

# Mid-Depth Retrieval Collapse in T5 Encoders: A Layerwise Case Study on Medieval Latin Witnesses

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## Abstract

Practitioners often read pre-trained encoders at intermediate layers, diagnose them with an anisotropy statistic, and correct them with a post-hoc projection. These steps assume that the statistic tracks retrieval quality. We test that assumption layer by layer on 1,705 medieval Latin canon-law witnesses, whose equivalence labels come from a scholarly edition. In the middle layers of three T5 encoders, cosine retrieval falls to or near chance (AUROC 0.50–0.65), while three embedding-trained models stay above 0.80 at every layer. Mean pairwise cosine gives no warning. Top-PC share, the first principal component’s variance share, flags all 26 collapsed layers with one false alarm in 100, but orders layers only weakly within each group. A top- $D$  projection fit on training vectors lifts every layer to AUROC 0.96–0.99 and cuts the cross-model spread in routing accuracy from 39.3 to 3.3 points. The repair matches, but does not exceed, a character n-gram reference. Contrastive fine-tuning of LaTa improves its last layer but not its collapsed middle layers. At the collapsed layers one direction holds most of the variance, but removing it alone restores only part of the ranking; removing two to five restores nearly all of it. We test whether a few massive, passage-varying residual coordinates cause the collapse. Zeroing up to ten of them restores none of the 26 collapsed layers to AUROC 0.90 or above, while per-coordinate standardization restores 9. A token audit finds token carriers in LaTa, where mean pooling without three token types (11 percent of test tokens) restores all 10 collapsed layers. For PhilTa only a token ablation supports carriers, and it needs 30 token types (a median 33 percent of test tokens) to restore all 9 collapsed layers; in mT5-base, dropping up to 100 token types ranked on training passages restores no layer. We also test whether T5 pretraining or the missing embedding objective produces the collapse [Pending: P2x2: one-sentence result]. We release the frozen testbed and split, the eval-

uation code, and lexical reference scores.

## 1 Introduction

Pre-trained encoders are a standard representation for semantic matching in text-reuse and digital-humanities pipelines (e.g., Manjavacas et al., 2019; Schelb et al., 2026; Reichbauer et al., 2026). Practitioners take a model off the shelf and often read an intermediate layer, since intermediate layers can outperform the final one (Skean et al., 2025). They diagnose the embedding space with an anisotropy statistic, usually the mean cosine between passages, adapted from the word-level statistic of Ethayarajh (2019). They then apply a post-hoc correction, such as removing the top principal directions (Mu and Viswanath, 2018) or whitening (Su et al., 2021). The last two steps assume that the statistic tracks retrieval quality: a layer the statistic calls healthy should retrieve well, and a layer it flags should gain from the correction. We test that assumption layer by layer.

Medieval canon law gives a controlled case for this test. Before the twelfth century it survives in many compilations, whose witnesses differ in spelling, wording, and fragment boundaries. Matching a witness to its source by hand is labor-intensive expert work, because scribes often omitted or misnamed the source. The Carolingian Canon Law project (CCL; Firey, 2009) has already made these decisions for 1,705 witnesses of 840 sources. Its labels are expert-assigned by the project and checked by collation in its parallel display, so they are independent of any model. A character n-gram reference retrieves at least as well as every pre-trained encoder configuration we test, so the equivalence signal is present in the surface text. A retrieval failure therefore points to the representation rather than to the data. Latin also has dedicated encoders in two architectures. The T5 models LaTa and PhilTa have encoder-only siblings, LaBERTa and PhilBERTa, trained by the same authors on

the same data (Riemenschneider and Frank, 2023). Latin BERT (Bamman and Burns, 2020) is a further encoder-only model. This allows a matched comparison of architectures (Section 6).

The layerwise view shows a failure that the usual diagnostic misses. In the middle layers of the three T5 encoders in our panel, cosine retrieval falls to or near chance: pairwise AUROC at the worst layer is 0.50 for LaTa, 0.54 for PhilTa, and 0.65 for mT5-base. The three embedding-trained models stay above 0.80 at every layer. Mean pairwise cosine gives no warning. It is highest in the models that retrieve best, and within the T5 encoders it is lower at the collapsed layers than at the healthy ones. Variance statistics do better. A top-PC share of 0.6 or more marks all 26 collapsed layers, with one false alarm in 100 model-layers. Within a regime, however, top-PC share and effective rank order layers only weakly (Section 4). A top- $D$  projection fit on training vectors (ABTT; Mu and Viswanath, 2018), with no parameter update, lifts every layer of every model to AUROC 0.96 to 0.99. It also compresses the spread in routing accuracy across models from 39.3 to 3.3 points. The repair matches the  $n$ -gram reference and does not exceed it. Contrastive fine-tuning raises LaTa’s last layer from 0.938 to 0.984 AUROC, yet its layers 2 to 11 stay at 0.50 to 0.57.

Prior work shows that a few residual coordinates of very large magnitude can dominate cosine similarity (Timkey and van Schijndel, 2021). In decoder language models they persist through the middle layers (Sun et al., 2024), and in T5 encoders they grow with depth (Zhao et al., 2025). We test whether mean pooling turns these coordinates into values that vary from passage to passage and so drown out content (Section 5). We also test whether T5 pretraining or the missing embedding objective produces the collapse (Section 6). A label-free check finds the same low-rank middle layers on English court opinions, including in a T5 encoder pretrained on English alone.

We measure the geometry on two tasks. Pairwise retrieval (Task A) scores every pair of witnesses and tests ranking directly. Open-set routing (Task B) files each witness under a known source or flags it as new, which adds a threshold decision. The embedding-trained models show a second regime. In them every passage shares a large mean vector, which raises every cosine (mean pairwise cosine 0.59 to 0.97), while variance stays spread over many directions. These models rank

well, yet the projection still raises the share of witnesses LaBSE files correctly as known or new from 83.3 to 90.8 percent. Section 7 shows that this gain holds without any threshold and does not come from the shared mean vector.

This paper makes four contributions. (1) We give the first per-layer retrieval profile of T5 encoder states in isolation that we know of: in our panel, T5 encoders collapse to or near chance at mid-depth. A  $2 \times 2$  control crosses T5 pretraining with an embedding objective [Pending: P2x2: raw encoder-only and embedding-trained T5 cells]. (2) We compare anisotropy statistics with retrieval layer by layer. Top-PC share and effective rank separate collapsed from healthy layers, whereas mean pairwise cosine is lower at collapsed layers than at healthy ones. A train-only projection restores every layer, which shows that the signal was present in the layer but masked. (3) We give a falsifiable test of the massive-coordinate account for mean-pooled passage retrieval (coordinate ablation, token audit, subspace split, and a  $D$  ablation), and find that the nuisance spans two to five directions, which rank below the retained directions in every pre-trained model. The account’s main prediction fails: zeroing up to ten top coordinates restores none of the 26 collapsed layers to AUROC 0.90 or above, while per-coordinate standardization restores 9 and a three-component projection all 26. The token audit finds that mean pooling without at most 30 token types restores every collapsed layer of LaTa and PhilTa (three types, 11 percent of test tokens, suffice in LaTa; PhilTa’s 30 are a median 33 percent), and finds no such token set in mT5-base. (4) We release the frozen testbed and split, the evaluation code, and lexical reference scores (Section 3).

## 2 Related Work

**Anisotropy, outlier dimensions, and post-hoc correction.** Contextual embeddings crowd into a narrow cone, usually measured by the mean cosine of random pairs (Gao et al., 2019; Ethayarajh, 2019). That statistic mainly reflects a shared mean offset (Rudman et al., 2022). A separate distortion comes from a few outlier dimensions of very large magnitude (Kovaleva et al., 2021; Puccetti et al., 2022), which dominate cosine similarity (Timkey and van Schijndel, 2021). In decoder language models they appear as massive activations that emerge early and persist through the mid-

dle layers (Sun et al., 2024). T5 encoders carry such outliers too, and their magnitude grows with depth (Zhao et al., 2025). Post-hoc corrections remove the mean and top principal directions (Mu and Viswanath, 2018), whiten the space (Su et al., 2021; Huang et al., 2021), or down-weight frequent tokens (Arora et al., 2017). Timkey and van Schijndel (2021) apply such corrections at every layer of four pre-trained language models and recover word-similarity quality. After task fine-tuning, the dominant directions carry task information, and removing them hurts (Rajaei and Pilehvar, 2021). For final-layer embeddings, top-dimension variance share predicts when cosine fails (Parupudi, 2026), and effective rank is the most informative geometric predictor of retrieval quality, though the prediction is moderate (Rottach et al., 2025). Kulkarni et al. (2026) find that geometric metrics of the final-layer unembeddings of OLMo-style language models do not reliably predict their downstream performance, and the cosine geometry of sentence embeddings does not predict their task performance (Nastase and Merlo, 2025). None of these studies compares the statistics with retrieval quality layer by layer in T5 encoders. We make that comparison and fit the correction on training data only.

**Layerwise similarity and retrieval quality.** Intermediate layers often outperform the final layer on embedding tasks (Skean et al., 2025). Layerwise geometry studies place the anisotropy peak in the middle layers of decoder models (Razzhigaev et al., 2024; Machina and Mercer, 2024) and report flatter profiles for encoders (Razzhigaev et al., 2024). Hämmerl et al. (2023) find a large outlier dimension in a middle layer of a multilingual encoder and improve cross-lingual retrieval by zeroing it. Evidence for T5 stops short of its encoder layers. Mean-pooled final-layer T5 embeddings trail fine-tuned encoders on semantic similarity (Ni et al., 2022a). Whitening gives T5 large similarity gains, reported for an average of its first and last layers (Huang et al., 2021). Layerwise anisotropy for T5 has been reported with encoder and decoder states concatenated (Godey et al., 2024). We report the first per-layer retrieval profile of T5 encoder states in isolation that we know of.

**Latin encoders and text reuse.** Latin has dedicated encoder-only models, Latin BERT (Bamman and Burns, 2020) and LaBERTa and PhilBERTa, and the encoder-decoder models LaTa and PhilTa

trained on the same data (Riemenschneider and Frank, 2023). Work on allusive text reuse in Latin finds that distributional semantics moderately improves retrieval over lexical matching (Manjavacas et al., 2019). Recent benchmarks target intertextuality in classical Latin literature (Schelb et al., 2026) and semantic search in Latin and Ancient Greek (Toyin et al., 2026). Reichbauer et al. (2026) mine parallel Latin and Greek sentences and find that whitening helps specialized encoders and slightly hurts LaBSE. Our testbed differs in its labels, which come from an edition of medieval canon-law witnesses, and in its question: where the retrieval signal sits inside each encoder.

### 3 Testbed and Protocol

**Corpus.** The Carolingian Canon Law project (CCL; Firey, 2009) is an open-access edition of early medieval, mostly ninth-century, canon-law manuscripts that keys each transcribed unit to the source it copies. A *directory* is one source, and a *witness* is one manuscript’s copy of it; the testbed holds 1,705 witnesses in 840 directories. The labels are expert-assigned by the project and checked by collation in its parallel display (Appendix A), so no model shaped them. The material is licensed CC BY-NC-SA 4.0 with attribution to the project. We will release the frozen testbed and split under the same license, in coordination with the project’s versioned exports, together with the evaluation code and lexical reference scores.

**Split.** We divide the witnesses into 847 training and 858 test files, and pairs form only within a split. Singleton directories (545 of 840) are divided file by file, doubletons go whole to one side, and larger directories are divided internally so that every size class yields training positives. This gives 565 positive training and 596 positive test pairs; 535 test files have a same-directory partner in test (*existing*) and 323 do not (*new*).

**Tasks and metrics.** Task A scores every pair of test witnesses by the cosine of their pooled embeddings, and we report AUROC with same-directory pairs as positives. Task B is open-set routing: file an unidentified witness under a known directory, or flag it as new. Each test file scores every test directory by its maximum cosine to a member. A pseudo-directory NEW scores the threshold  $\tau$ . The top-scoring entry is the decision. Directory accuracy at rank 1 (DirAcc@1) requires that entry to be

the labeled directory, or NEW for a new file. Assignment accuracy scores only the existing-versus-new decision. Labeling every file as existing already scores 62.4%, so we report assignment accuracy alongside DirAcc@1.

Per layer we also read five statistics: three label-free geometry statistics and two label-based separation indices. The *mean pairwise cosine* over all pairs of pooled test passages is the usual anisotropy statistic, adapted from the word-level statistic of [Ethayarajh \(2019\)](#), which uses random pairs. Let  $p_i = \lambda_i / \sum_j \lambda_j$  for the covariance eigenvalues  $\lambda_1 \geq \lambda_2 \geq \dots$  of the centered training embeddings. The *top-PC share* (the share of centered variance on the first principal component) is  $p_1$ . The *effective rank* is  $\exp(-\sum_i p_i \log p_i)$ , the entropy effective rank of [Roy and Vetterli \(2007\)](#) applied to the covariance eigenvalues; it is 1 when one direction holds all variance. The *cosine gap*  $g$  is the mean same-directory minus the mean different-directory test cosine, and the *standardized gap*  $g/s$  divides it by the standard deviation  $s$  of all pairwise test cosines, putting models with different cosine spreads on one scale. The two gaps need labels.

**Train-only rule.** A headline cell reads each model and setting at the layer with the highest training AUROC for Task A, or training DirAcc@1 for Task B (Table 11). The threshold  $\tau$  maximizes training F1 over same- versus different-directory pairs. SIF frequencies are fit on training texts, and ABTT components and  $D$  on training embeddings. Layerwise profiles show every layer and select nothing. The rule selects every layer, threshold, and projection from training data alone. Analysis quantities (the collapse cutoff and the worst layers) use test AUROC and select no reported headline cell. Appendix K uses a predeclared variant of the rule for attribution layers.

**Post-processing.** We mean-pool the hidden states of every transformer block, dropping only tokenizer artifacts without lexical content. All-but-the-Top (ABTT; [Mu and Viswanath, 2018](#)) subtracts the training mean and removes the top  $D$  principal components of the centered training embeddings from all vectors, with  $D \in \{1, 2, 3, 5, 7, 10\}$  chosen per layer by training DirAcc@1. We write ABTT-only when we contrast it with SIF+ABTT. SIF ([Arora et al., 2017](#)) instead reweights each token by  $a/(a + p(w))$  before pooling ( $a = 10^{-3}$ ), with  $p(w)$  estimated on training texts, and gives special tokens, such as the end-of-sequence token,

| Model                   | Arch. | Data       | Emb. obj.   | L  | Status |
|-------------------------|-------|------------|-------------|----|--------|
| LaTa                    | T5    | Latin      | none        | 12 | run    |
| PhilTa                  | T5    | Grc/La/En  | none        | 12 | run    |
| mT5-base                | T5    | 101 lang.  | none        | 12 | run    |
| T5-v1.1-base            | T5    | English    | none        | 12 | run    |
| LaBERTa                 | Enc.  | Latin      | none        | 12 | [P2x2] |
| PhilBERTa               | Enc.  | Grc/La/En  | none        | 12 | [P2x2] |
| Latin BERT              | Enc.  | Latin      | none        | 12 | [P2x2] |
| Sentence-T5 (or GTR-T5) | T5    | English    | contrastive | 12 | [P2x2] |
| LaBSE                   | Enc.  | 109 lang.  | contrastive | 12 | run    |
| Qwen3-0.6B              | Dec.  | multiling. | contrastive | 28 | run    |
| KaLM-mini               | Dec.  | multiling. | contrastive | 24 | run    |

Table 1: The model panel as a  $2 \times 2$  design. Arch.: a T5 encoder with span-corruption pretraining, or an encoder-only model (Enc.) with masked-language-model pretraining; LaBERTa and PhilBERTa are RoBERTa-style. Emb. obj.: the embedding objective trained after pretraining, if any. Blocks: raw T5 encoders; raw encoder-only models; embedding-trained T5 and encoder-only models; embedding-trained decoders (Dec.) outside the design. T5-v1.1-base is the matched raw partner of Sentence-T5. Data: pretraining languages (Grc/La/En: Ancient Greek, Latin, English). L: transformer blocks read (encoder blocks for T5). KaLM-mini runs its decoder with bidirectional attention. [P2x2]: not yet run.

zero weight. We use only this weighting step of SIF, not its common-component removal. SIF+ABTT applies both. Full-rank PCA whitening ([Su et al., 2021](#)) is ill-conditioned here, since it estimates 768 to 1,024 dimensions from 847 training vectors. It degenerates the learned threshold and also ranks below ABTT at every layer, at Task A AUROC 0.82–0.88 for the T5 encoders and LaBSE and 0.55–0.64 for the two decoders, so we exclude it (Appendix G). Whitening to a reduced dimension, as [Su et al. \(2021\)](#) also do, is the fairer comparison. Fit on training embeddings with  $k = 128$  or 256 components, it ranks on par with ABTT (Table 20, Appendix G): Task A AUROC 0.973–0.984 at each model’s train-selected layer against 0.971–0.987, and within 0.021 of ABTT at every one of the 100 model-layers, collapsed layers included;  $k = 64$  falls slightly below. On pairwise ranking the repair therefore does not depend on ABTT in particular. On routing it does: reduced-rank whitening does not degenerate the threshold as full rank does, but at each model’s train-selected Task B layer  $k = 128$  reaches DirAcc@1 1.9 to 4.5 points below ABTT, and  $k = 256$  up to 15 points below. We therefore keep ABTT as the reference.

**Model panel.** Table 1 crosses two factors. The first is the architecture with its pretraining objec-

tive: a T5 encoder with span-corruption pretraining, or an encoder-only model with masked-language-model pretraining. The second is whether a contrastive embedding objective follows pretraining. The T5 encoders we have run are the Latin LaTa, its trilingual sibling PhilTa (Riemenschneider and Frank, 2023), and mT5-base (Xue et al., 2021). The controls we have run are LaBSE (Feng et al., 2022), Qwen3-0.6B (Zhang et al., 2025), and KaLM-mini (Hu et al., 2025). In the models run so far the two factors are confounded: every T5 encoder is raw and every control is embedding-trained, so a T5-only collapse could reflect either. The pending rows break the confound. LaBERTa and PhilBERTa share authors and pretraining data with LaTa and PhilTa (Riemenschneider and Frank, 2023), and Latin BERT (Bamman and Burns, 2020) is a second raw Latin encoder. Sentence-T5 (Ni et al., 2022a) fills the embedding-trained T5 cell, and T5-v1.1-base (Raffel et al., 2020) is its matched raw partner.

**Lexical reference.** Character 3–5-gram TF-IDF, fit on training files, reaches 0.987 AUROC on Task A, tying the best pre-trained encoder cell, and 89.9 DirAcc@1 on Task B, level with the best pre-trained ones. In every tercile of n-gram overlap among positive test pairs (terciles cut on the TF-IDF score itself), it ranks a witness’s partner first at least as often as every baseline and ABTT encoder configuration at its train-selected layer. The equivalence signal is therefore present in the surface text. Our claims concern where the retrieval signal sits inside encoders and what masks it, not whether embeddings beat n-grams on this corpus. Table 28 (Appendix I) gives all three lexical scorers.

## 4 Two Geometries

This section asks whether the anisotropy statistic that practitioners read tracks retrieval layer by layer. We first give the retrieval profile of every model, then set the geometry statistics of Section 3 against it.

**Retrieval by depth.** Figure 1 plots test AUROC against depth for all six models. The three T5 encoders fall to or near chance in their middle layers. Baseline AUROC bottoms out at 0.496 for LaTa at layer 6, at 0.538 for PhilTa at layer 10, and at 0.654 for mT5-base at layer 5. We call a layer *collapsed* when its baseline AUROC is below 0.70. By this definition, 26 of the 36 T5 layers collapse: layers 2–

11 of LaTa, 3–11 of PhilTa, and 5–11 of mT5-base. We set this cutoff after inspecting test AUROC; it describes the profiles and selects no headline cell. Each T5 encoder recovers at its last layer, where LaTa reaches 0.938. The three embedding-trained models never fall below 0.806, and they improve broadly with depth to a best layer of 0.960 to 0.972. In our panel the collapse is thus confined to the T5 encoders, which are also the only models without an embedding objective (Section 6). ABTT, fit on training embeddings, restores every layer of every model (Section 7). The question is whether a geometry statistic would have flagged the collapsed layers.

**Mean pairwise cosine.** The usual diagnostic gives no warning of the collapse. Table 2 summarizes each model’s geometry over its layers. Mean pairwise cosine, averaged over all pairs of test passages, is highest in the embedding-trained models. Their mean cosine ranges from 0.585 to 0.974 across layers, while their retrieval stays above 0.80. At the collapsed T5 layers the statistic ranges from 0.23 to 0.62. LaTa (0.23–0.30) and mT5-base (0.30–0.36) sit below every embedding-trained layer, and PhilTa (0.53–0.62) overlaps the bottom of that range. Within the T5 encoders the statistic runs against the usual reading. Collapsed layers have lower mean cosine than healthy ones, so it correlates with AUROC at Spearman +0.67 (Table 3). Within the embedding-trained models it carries almost no information (Spearman +0.10, Pearson −0.06). Computing it on training passages instead gives the same picture (Spearman +0.72 within T5, +0.12 within the embedding-trained models). A practitioner who reads a high mean cosine as a warning would flag the healthy models and pass the collapsed layers.

**Variance statistics.** The variance statistics separate the two groups, with one exception (mT5-base layer 4). We compute top-PC share and effective rank on centered training embeddings, so the shared offset that drives mean cosine does not enter them. At the collapsed T5 layers the top principal component holds 76 to 100 percent of the variance, and effective rank lies between 1.0 and 4.6. The two transition layers at the onset of collapse, LaTa layer 2 and PhilTa layer 3, set the lower end of the share and the upper end of the rank. Without them the ranges narrow to 83–100 percent and 1.00–2.17. At mT5-base layers 5–11 one direction holds all the variance to three decimals. In the embedding-

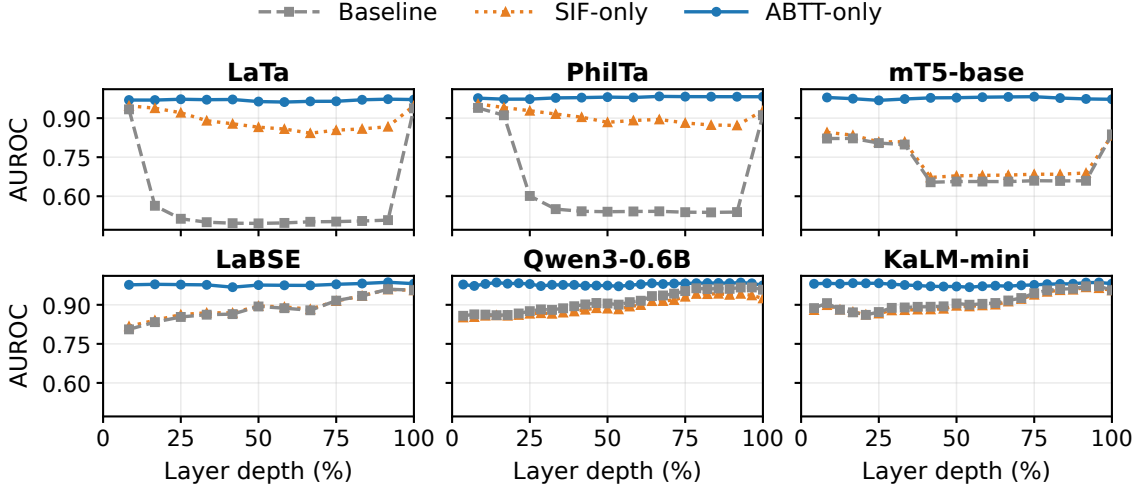


Figure 1: Pairwise retrieval AUROC (Task A, test split) of the cosine between mean-pooled passage vectors, against normalized layer depth (layer  $\ell$  of  $L$  is plotted at  $100\ell/L$  percent). Top row: the three T5 encoders. Bottom row: the three embedding-trained models. Gray: baseline mean pooling. Orange: SIF-weighted pooling. Blue: ABTT, with the number of removed components  $D$  chosen per layer on training data. AUROC 0.5 is chance. The T5 baselines fall to or near chance in their middle layers (LaTa 0.496 at layer 6, PhilTa 0.538 at layer 10, mT5-base 0.654 at layer 5) and recover at the last layer. The embedding-trained baselines stay above 0.80 and rise broadly with depth. ABTT holds every layer of every model between 0.96 and 0.99. SIF lifts LaTa and PhilTa and barely moves mT5-base.

| Model      | Type         | $L$ | Geometry (baseline) |           |         |          | Baseline test AUROC |             |             |
|------------|--------------|-----|---------------------|-----------|---------|----------|---------------------|-------------|-------------|
|            |              |     | Mean cos.           | Cos. SD   | Max PC1 | Min rank | $\ell^*$            | at $\ell^*$ | Min (layer) |
| LaTa       | T5 raw       | 12  | 0.23–0.58           | 0.10–0.63 | 0.952   | 1.36     | 4                   | 0.500       | 0.496 (6)   |
| PhilTa     | T5 raw       | 12  | 0.53–0.71           | 0.09–0.46 | 0.858   | 1.81     | 6                   | 0.541       | 0.538 (10)  |
| mT5-base   | T5 raw       | 12  | 0.30–0.95           | 0.05–0.76 | 1.000   | 1.00     | 5                   | 0.654       | 0.654 (5)   |
| LaBSE      | BERT emb.    | 12  | 0.73–0.93           | 0.04–0.10 | 0.524   | 16.19    | 8                   | 0.879       | 0.806 (1)   |
| Qwen3-0.6B | Decoder emb. | 28  | 0.85–0.97           | 0.01–0.04 | 0.183   | 71.60    | 16                  | 0.910       | 0.858 (1)   |
| KaLM-mini  | Decoder emb. | 24  | 0.58–0.93           | 0.02–0.07 | 0.419   | 23.83    | 5                   | 0.861       | 0.861 (5)   |

Table 2: Geometry and retrieval regimes of the six encoders under baseline mean pooling (no post-hoc correction). Type: T5 raw is a T5 encoder pretrained without a sentence-embedding objective; emb. marks a BERT encoder or a decoder trained with one.  $L$ : number of transformer layers. Mean cos. and Cos. SD: range over layers of the mean and the standard deviation of the cosine over all pairs of test passages. Max PC1: peak top-PC share over layers, the largest share of variance on the first principal component of the centered train embeddings. Min rank: smallest entropy effective rank of the same centered train embeddings over layers. All four geometry columns are label-free.  $\ell^*$ : layer where the PC1 share peaks. The last two columns give the baseline test pairwise AUROC of cosine at  $\ell^*$  and its minimum over layers, with the layer in parentheses. Layer 1 is the first transformer block.

trained models the largest top-PC share at any layer is 0.18 for Qwen3-0.6B, 0.42 for KaLM-mini, and 0.52 for LaBSE. Their effective rank ranges from 16 to 150. ABTT with  $D = 10$  removes the concentration. On the training vectors it is fit on, this follows by construction. On test passages at the collapsed layers, top-PC share falls to 0.03–0.05 and effective rank rises to 148–173.

**What the statistics predict.** Figure 2 sets both kinds of statistic against retrieval over all 100 model-layers, and Table 3 gives the correlations. Across the panel, top-PC share correlates with baseline AUROC at Spearman  $-0.77$  and effective rank

at  $+0.77$ , against  $+0.55$  for mean pairwise cosine. Both coefficients are inflated by the two clusters of collapsed and healthy layers visible in panel (a), and the Pearson value of  $-0.92$  most of all. Layers within a model are also not independent, so the 100 model-layers carry the evidence of roughly six clusters. Leaving out any one model keeps the top-PC Spearman between  $-0.70$  and  $-0.83$ ; this shows robustness to single models, not to the clustering. A threshold states the separation more directly. A top-PC share of 0.6 or more marks all 26 collapsed layers with one false alarm, mT5-base layer 4, and every cutoff from 0.53 to 0.76 gives the same split. Like the collapse cutoff, this

| Statistic                      | All (100) |        | T5 (36) |        | Non-T5 (64) |        | LOMO $\rho$ |       |
|--------------------------------|-----------|--------|---------|--------|-------------|--------|-------------|-------|
|                                | $r$       | $\rho$ | $r$     | $\rho$ | $r$         | $\rho$ | min         | max   |
| <i>Target: baseline AUROC</i>  |           |        |         |        |             |        |             |       |
| Mean pairwise cosine           | 0.83      | 0.55   | 0.61    | 0.67   | -0.06       | 0.10   | 0.42        | 0.67  |
| Top-PC share                   | -0.92     | -0.77  | -0.86   | -0.50  | -0.47       | -0.43  | -0.83       | -0.70 |
| Effective rank                 | 0.80      | 0.77   | 0.87    | 0.53   | 0.48        | 0.40   | 0.70        | 0.85  |
| Raw cosine gap                 | 0.24      | 0.22   | 0.63    | 0.72   | 0.15        | -0.03  | -0.10       | 0.48  |
| Standardized gap               | 0.86      | 0.97   | 0.87    | 0.96   | 0.89        | 0.93   | 0.96        | 0.98  |
| <i>Target: ABTT AUROC gain</i> |           |        |         |        |             |        |             |       |
| Mean pairwise cosine           | -0.83     | -0.54  | -0.60   | -0.66  | 0.07        | -0.07  | -0.66       | -0.41 |
| Top-PC share                   | 0.92      | 0.76   | 0.86    | 0.50   | 0.45        | 0.41   | 0.69        | 0.83  |
| Effective rank                 | -0.80     | -0.76  | -0.88   | -0.52  | -0.44       | -0.37  | -0.84       | -0.68 |
| Raw cosine gap                 | -0.23     | -0.23  | -0.61   | -0.71  | -0.16       | 0.00   | -0.48       | 0.09  |
| Standardized gap               | -0.85     | -0.97  | -0.87   | -0.95  | -0.88       | -0.92  | -0.98       | -0.96 |

Table 3: Correlation between per-layer geometry and retrieval, one observation per model-layer, baseline mean pooling.  $r$ : Pearson;  $\rho$ : Spearman. Columns pool all 100 model-layers, the 36 layers of the three T5 encoders, or the 64 layers of the three embedding-trained models; LOMO gives the minimum and maximum Spearman over the six leave-one-model-out subsets. Mean pairwise cosine is on test passages; top-PC share and effective rank are on centered train embeddings. These three statistics are label-free. The two gaps are label-based separation indices. Raw cosine gap is the mean test cosine of same-directory pairs minus that of different-directory pairs; standardized gap divides it by the standard deviation of all test pairwise cosines at that layer. Targets are the baseline test AUROC and the ABTT gain (AUROC with ABTT at its train-selected  $D$  minus baseline AUROC). The standardized gap is a signal-to-noise ratio of the kind AUROC summarizes, so it tracks AUROC by construction. ABTT AUROC varies little (0.962–0.987), so the gain correlations mirror the baseline ones. The pooled Pearson and Spearman values are both inflated by the two clusters of collapsed and healthy layers, and layers within a model are not independent.

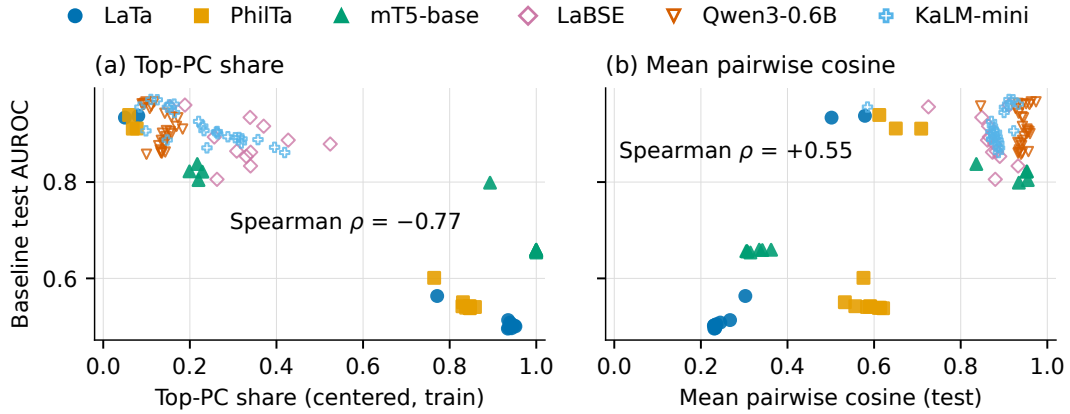


Figure 2: Per-layer geometry against baseline retrieval, one marker per model-layer (100 in all). Filled markers: T5 encoders (LaTa, PhilTa, mT5-base). Hollow markers: embedding-trained models (LaBSE, Qwen3-0.6B, KaLM-mini). Vertical axis: baseline Task A test AUROC. (a) Top-PC share, the share of centered training variance on the first principal component. (b) Mean pairwise cosine over all pairs of test passages, the usual anisotropy statistic. In (a) every collapsed layer (AUROC below 0.70) has a share of at least 0.76, and every other layer except mT5-base layer 4 (share 0.89, AUROC 0.80) lies below 0.53. In (b) the collapsed LaTa and mT5-base layers sit below every embedding-trained layer, and the collapsed PhilTa layers overlap the bottom of that range, so a high mean cosine gives no warning of collapse.

threshold is descriptive and was read off the data. Within a family, the variance statistics track AUROC only weakly. Top-PC share gives Spearman  $-0.50$  within the T5 encoders and  $-0.43$  within the embedding-trained models; effective rank gives  $+0.53$  and  $+0.40$ . Within single models, the top-PC Spearman runs from  $-0.28$  (LaBSE) to  $-0.94$  (mT5-base) over 12 to 28 layers. mT5-base layer 4

shows the limit. Its top-PC share is 0.893 and its effective rank 2.15, yet its AUROC of 0.799 is close to that of its healthy layers 1–3. The lower block of Table 3 adds no separate evidence. ABTT lifts every layer into a narrow band (0.962 to 0.987), so its gain is nearly a constant minus the baseline, and the gain correlations mirror the baseline ones. The variance statistics thus separate the collapsed

regime from the healthy one. They do not rank layers within a regime, and a high share alone does not guarantee collapse. This limit fits reports that geometric metrics of the final-layer unembeddings of OLMo-style language models do not reliably predict their downstream performance (Kulkarni et al., 2026) and that the cosine geometry of sentence embeddings does not predict their task performance (Nastase and Merlo, 2025).

**The standardized gap.** The raw cosine gap, the margin a threshold works with, tracks retrieval even less well. Across the panel it correlates with AUROC at Spearman +0.22. The standardized gap reaches +0.97, and for the AUROC target every Spearman column of Table 3 keeps it between +0.93 and +0.98. This agreement is expected by construction. The standardized gap is a label-based signal-to-noise ratio, close to the  $d'$  from which a binormal model computes AUROC, so we do not count it as a predictor. Its use here is to dissolve the mT5-base paradox. mT5-base shows why the two gaps differ (Figure 5, Appendix B). Its raw gap rises from about 0.03 to 0.21–0.25 exactly where ranking fails, and within the model it correlates with AUROC at −0.78. Over the same layers, the standard deviation of all pairwise cosines grows from 0.05–0.07 at layers 1–3 to about 0.75. Divided by that spread, the gap falls to 0.28–0.33, below the 0.41–0.49 of layers 1–3 and the 0.78 of layer 12. The within-model correlation then turns to +0.52. A gap that rises as ranking fails is thus no paradox: the collapsed layers spread all cosines widely, and same-directory pairs gain little relative to that spread. The resolution is partial. At mT5-base layer 4 the standardized gap is the lowest of the model (0.19), yet AUROC is 0.80. LaTa adds an anomaly that the standardized gap does not explain. Its raw gap is negative at layers 3–11, between −0.05 and −0.07, so same-directory pairs score slightly below different-directory pairs on average. We do not have an account of this: the token audit of Section 5 tests a length account and rejects it (E2).

**SIF at the collapsed layers.** SIF offers a first hint about what feeds the dominant direction. It down-weights frequent tokens and drops special tokens before pooling (Section 3), and its effect at the collapsed layers splits the T5 encoders. It lifts LaTa from 0.50–0.56 to 0.84–0.94 and PhilTa from 0.54–0.60 to 0.87–0.93. It moves mT5-base only from 0.65–0.66 to 0.67–0.69. ABTT reaches 0.96–0.98

at the same layers in all three. This pattern suggests that tokens SIF down-weights or drops carry the dominant direction in LaTa and PhilTa, and that tokens it keeps carry it in mT5-base. Because SIF differs from mean pooling in two ways, the token audit of Section 5 separates them (E2). It supports the first suggestion for LaTa: mean pooling without the comma, the period and the end-of-sequence token restores every collapsed layer. For PhilTa the support comes from the token ablation alone, which needs 30 token types (a median 33 percent of test tokens) to restore every collapsed layer. It does not support the second suggestion: in mT5-base no token group accounts for the direction.

**Two regimes.** The panel therefore shows two regimes. In the embedding-trained models every passage shares a large mean vector, which raises every cosine, and variance spreads over many directions. These models rank well but still gain from the projection. Their baseline AUROC is 0.806 to 0.972, and ABTT adds 0.007 to 0.031 at their respective train-selected layers, a gain within bootstrap noise for Qwen3-0.6B and KaLM-mini (Section 7). In the middle layers of the T5 encoders one direction takes most of the variance, and cosine ranking falls to or near chance. The cosines there are spread across both signs rather than moderate. At mT5-base’s collapsed layers the 5th percentile of test cosines lies between −0.83 and −0.77, the median between 0.84 and 0.90, and the standard deviation about 0.75. At LaTa’s collapsed layers the 5th percentile lies between −0.50 and −0.26, and the standard deviation reaches 0.63. Prior work already separates these two components. Timkey and van Schijndel (2021) distinguish a constant offset from a few high-variance dimensions that dominate cosine, and IsoScore (Rudman et al., 2022) shows that the mean cosine of random pairs mainly measures the offset. For final-layer embeddings, top-dimension variance share predicts cosine failure better than mean cosine (Parupudi, 2026), and effective rank is the most informative geometric predictor of retrieval quality across many models, though the prediction is moderate (Rottach et al., 2025). Our contribution is the layerwise version of this comparison within one testbed, including raw T5 encoder states, where the low-rank regime coincides with chance retrieval. Section 5 asks what produces that dominant direction.

## 5 Localizing the Collapse

Section 4 showed that collapsed T5 layers concentrate their centered variance on one direction. Their cosines also take both signs, so the pooled vectors lie on both sides of one axis. The projection on the dominant direction therefore changes sign across passages, which a shift shared by all passages cannot produce. Prior work on outlier dimensions offers a candidate account. In pre-trained transformers a few residual coordinates reach magnitudes far above the median (Kovaleva et al., 2021; Puccetti et al., 2022) and dominate cosine similarity (Timkey and van Schijndel, 2021). In decoder language models these massive activations take near-constant values on the tokens that carry them (Sun et al., 2024), and T5 encoders carry such outliers too (Zhao et al., 2025). Their prevalence tracks token frequency in pretraining (Puccetti et al., 2022). Mean pooling adds one step. It averages the per-token values over a token mix that differs from passage to passage, so the pooled coordinate varies across passages. If that variation is large, cosine ranks passages by one coordinate and ignores their content.

Pooled vectors at a collapsed layer do contain massive coordinates. At LaTa layer 7, the layer the attribution rule selects (Appendix K), the three coordinates of largest mean magnitude over training passages reach 805, 780 and 753 against a median of 9.5. Massive coordinates alone do not collapse retrieval, however. At mT5-base layer 1 the top three reach 135, 131 and 63 against a median of 1.9, yet AUROC is 0.82 and top-PC share is 0.23. The summed per-coordinate contribution of Timkey and van Schijndel (2021) gives the top three coordinates 0.86 of the mean pairwise cosine of training passages at mT5-base layer 1 (mean cosine 0.951), and 0.22 at LaTa layer 7 (mean cosine 0.228). That share measures a shared shift. Averaged over pairs, a coordinate’s contribution is approximately the square of its mean over normalized passages, so the share credits a coordinate for moving all passages together and not for varying between them. A share of 0.22 at the collapsed layer against 0.86 at the healthy one is consistent with a distinction between offset and variation; it is not evidence that the massive coordinates dominate similarity at LaTa layer 7. The refined hypothesis is that collapse occurs when the pooled massive coordinates vary across passages at least as much as they shift all passages together. We make this measurable with  $r$ , the

standard deviation of a pooled coordinate across training passages divided by the absolute value of its mean. The hypothesis predicts  $r \geq 1$  for the top coordinates at collapsed T5 layers, and  $r < 1$  at mT5-base layer 1 and in Qwen3-0.6B. A high centered top-PC share already means that one direction varies across passages. The testable content is that this direction is aligned with a few coordinates and has token carriers. The four tests below each state a prediction and its alternative. Every intervention is fit on training embeddings only.

**Coordinate ablation (E1).** Zeroing a single outlier dimension already improves cross-lingual retrieval in a multilingual encoder (Hämmerl et al., 2023). We test whether such coordinates explain a collapse to chance, and at which depths. At every layer of LaTa, PhilTa, mT5-base, LaBSE, Qwen3-0.6B and KaLM-mini, we rank residual coordinates two ways on training passages: by mean absolute value, and by variance across passages. For each ranking we zero the top  $k$  coordinates,  $k \in \{1, 3, 5, 10\}$ . We also standardize every coordinate by its training mean and standard deviation (Timkey and van Schijndel, 2021), and we center the vectors without removing any component ( $D = 0$ ). ABTT with  $D = 1$  serves as a reference, since it removes the dominant direction, and ABTT with  $D = 3$  as a second, since it recovers at least 80 percent of the  $D = 10$  gain at every collapsed T5 layer. If a few passage-varying coordinates cause the collapse, zeroing the top  $k \leq 5$  coordinates by variance restores AUROC at the collapsed T5 layers to at least 0.90. Zeroing by mean magnitude, which also picks coordinates that shift all passages together, should help less. Top-PC share then falls toward the range of the non-T5 models. For LaBSE and Qwen3-0.6B, which have no collapsed layer, zeroing should change AUROC by at most 0.03 at their train-selected layers, where ABTT itself adds 0.007 to 0.031, and should recover only a small part of the ABTT gain at their weakest layers. If standardization repairs a collapsed layer but zeroing  $k \leq 5$  does not, the nuisance sits on a few coordinates, but more than five. If neither repairs it and only a projection does, the dominant axis is spread over many coordinates of similar variance. The account then reduces to “a few dominant directions”, which is what Section 4 and the  $D$  ablation below already report. Centering alone tests whether the shared offset matters at all at these layers. Table 4 gives AUROC (top block) and top-PC

share (bottom block) for each model at its worst baseline layer. We find that the main prediction fails. Zeroing restores none of the 26 collapsed T5 layers to AUROC 0.90 or above under either ranking, for  $k \leq 5$  or for  $k = 10$ : the best cell is 0.736 for  $k \leq 5$  and 0.794 for  $k = 10$ , and top-PC share at  $k = 5$  stays at 0.568 to 0.998, above the whole range of the non-T5 models (at most 0.524). The two rankings nearly coincide at these layers (on average 2.4 of the top 3 and 9.1 of the top 10 coordinates are shared; median AUROC difference 0.000), so the prediction that ranking by mean magnitude helps less gets no verdict. Standardization partly works: it restores 9 of the 26 layers (1 of 10 in LaTa, 5 of 9 in PhilTa, 3 of 7 in mT5-base), reaches AUROC 0.798 to 0.946 over the 26, recovers a median 79 percent of the  $D = 10$  gain, and brings top-PC share below 0.2 at 14 of the 26. Centering alone restores none (AUROC change  $-0.044$  to  $+0.022$ ); ABTT restores 1 of 26 with  $D = 1$  and all 26 with  $D = 3$ . By the rule above, 9 layers fall under “a few coordinates, but more than five” and 17 under “only a projection repairs”, but the gloss of the second branch, an axis spread over many coordinates of similar variance, does not fit the data. A descriptive measure, added after we saw these results and carrying no prediction, shows a concentrated direction: at the collapsed layers the ten highest-variance coordinates hold a median 89 percent of the training variance (61 to 99.6 percent) and a median 92 percent of the first principal component’s squared loading (76 to 99.7 percent), and 90 percent of that loading sits on 4 to 22 coordinates. Yet after those ten are zeroed, one direction still holds 0.324 to 0.995 of the remaining variance (at least 0.6 at 24 of the 26 layers), and AUROC stays below 0.80. Most of the dominant direction’s loading thus sits on about ten coordinates, yet zeroing them leaves one direction dominant. A follow-up, designed after we saw the E1 results, shows that it is what remains of the same direction. The new top direction is the part of the first component outside those ten coordinates (median cosine 0.999 with that remainder). It holds a median 0.72 of the remaining variance, although zeroing leaves only 0.7 percent of the variance along the original first axis. Passage scores on the old and the new top direction agree linearly (median absolute Pearson correlation 0.99 on test passages) but less in rank: the absolute Spearman correlation ranges from 0.31 to 0.99 (median 0.87), with a median of 0.45 in mT5-base and 4 of the

26 layers below 0.5. A high Pearson with a low Spearman correlation points to a few extreme passages driving the agreement in mT5-base. The ten coordinates hold a median 0.50 of the second component’s squared loading and 0.18 of the third’s, so parts of those components survive zeroing, but they are not why it fails. Removing them while keeping the first restores 0 of the 26 layers. Zeroing followed by removal of one, two or three components fit on the zeroed vectors restores 9, 21 and 26 layers, against 1, 13 and 26 without zeroing; these intervention cells were added post hoc and carry no prediction. The first component’s direction is thus also written on the coordinates outside the top ten; its passage scores largely keep their rank order in LaTa and PhilTa (median absolute Spearman correlation 0.86 and 0.98) and only partly in mT5-base. LaBSE and Qwen3-0.6B behave as predicted, and so does KaLM-mini, which we added to this experiment after the predictions were written and hold to the same bound: at their train-selected layers the largest absolute AUROC change over the eight zeroings is 0.004 for LaBSE, 0.007 for Qwen3-0.6B and 0.002 for KaLM-mini, and at their weakest layers zeroing recovers  $-18$  to  $18$  percent of the  $D = 10$  gain, against 36 to 80 percent for centering alone. The prediction for  $r$  fails as stated: the three largest-magnitude coordinates all have  $r \geq 1$  at 18 of the 26 collapsed layers (10 of 10 in LaTa, 7 of 7 in mT5-base, 1 of 9 in PhilTa). In PhilTa one of the three lies between 0.81 and 0.96 at layers 5 to 10 while the other two exceed 1, and the lowest values at layers 3 and 4 are 0.21 and 0.61. Chosen by variance, the top three all have  $r \geq 1$  at all 26 layers. As predicted, the three largest-magnitude coordinates have  $r < 1$  at mT5-base layer 1 (0.05 to 0.12) and at every layer of Qwen3-0.6B (at most 0.20); the same holds for LaBSE and KaLM-mini (at most 0.39).

**Token audit (E2).** The account names a carrier: some tokens hold the massive values. We revised this test after E1 and before reading any E2 number. E1 shows that zeroing coordinates restores no collapsed layer while removing three principal components restores all 26. The audit therefore targets the directions that projection removes, the first three principal components of the training vectors, and not only the coordinates. The ratio  $r$  and its test on Qwen3-0.6B left the audit, because E1 reports them. The audit runs at every layer of the six models, with the three embedding-trained mod-

| Model                                       | L  | Base  | Zeroed, ranked by mean $ x $ |       |       |       | Zeroed, ranked by variance |       |       |       | Std.  | $D=0$ | ABTT $_{D=1}$ | ABTT $_{D=3}$ |
|---------------------------------------------|----|-------|------------------------------|-------|-------|-------|----------------------------|-------|-------|-------|-------|-------|---------------|---------------|
|                                             |    |       | 1                            | 3     | 5     | 10    | 1                          | 3     | 5     | 10    |       |       |               |               |
| <i>Task A test AUROC</i>                    |    |       |                              |       |       |       |                            |       |       |       |       |       |               |               |
| LaTa                                        | 6  | 0.496 | 0.498                        | 0.514 | 0.512 | 0.574 | 0.498                      | 0.514 | 0.516 | 0.574 | 0.822 | 0.482 | 0.671         | 0.973         |
| PhilTa                                      | 10 | 0.538 | 0.537                        | 0.540 | 0.540 | 0.557 | 0.538                      | 0.538 | 0.544 | 0.558 | 0.907 | 0.501 | 0.602         | 0.964         |
| mT5-base                                    | 5  | 0.654 | 0.655                        | 0.660 | 0.694 | 0.716 | 0.655                      | 0.660 | 0.694 | 0.716 | 0.806 | 0.670 | 0.835         | 0.914         |
| LaBSE                                       | 1  | 0.806 | 0.804                        | 0.793 | 0.794 | 0.788 | 0.802                      | 0.788 | 0.775 | 0.790 | 0.907 | 0.867 | 0.886         | 0.929         |
| Qwen3-0.6B                                  | 1  | 0.858 | 0.866                        | 0.874 | 0.877 | 0.880 | 0.866                      | 0.870 | 0.873 | 0.874 | 0.963 | 0.954 | 0.959         | 0.970         |
| KaLM-mini                                   | 5  | 0.861 | 0.862                        | 0.857 | 0.857 | 0.862 | 0.865                      | 0.865 | 0.869 | 0.863 | 0.933 | 0.917 | 0.926         | 0.969         |
| <i>Top-PC share of the training vectors</i> |    |       |                              |       |       |       |                            |       |       |       |       |       |               |               |
| LaTa                                        | 6  | 0.934 | 0.921                        | 0.885 | 0.859 | 0.712 | 0.921                      | 0.885 | 0.851 | 0.712 | 0.211 | 0.934 | 0.712         | 0.071         |
| PhilTa                                      | 10 | 0.847 | 0.827                        | 0.796 | 0.760 | 0.580 | 0.828                      | 0.777 | 0.733 | 0.608 | 0.143 | 0.847 | 0.763         | 0.088         |
| mT5-base                                    | 5  | 1.000 | 1.000                        | 1.000 | 0.998 | 0.995 | 1.000                      | 1.000 | 0.998 | 0.995 | 0.910 | 1.000 | 0.458         | 0.138         |
| LaBSE                                       | 1  | 0.263 | 0.263                        | 0.267 | 0.258 | 0.199 | 0.202                      | 0.206 | 0.205 | 0.213 | 0.189 | 0.263 | 0.138         | 0.077         |
| Qwen3-0.6B                                  | 1  | 0.100 | 0.101                        | 0.096 | 0.095 | 0.095 | 0.101                      | 0.096 | 0.097 | 0.098 | 0.102 | 0.100 | 0.077         | 0.049         |
| KaLM-mini                                   | 5  | 0.419 | 0.298                        | 0.301 | 0.306 | 0.192 | 0.284                      | 0.195 | 0.193 | 0.198 | 0.196 | 0.419 | 0.210         | 0.067         |

Table 4: Coordinate ablation at each model’s worst baseline layer (L). Top block: Task A test AUROC for the unmodified mean-pooled vectors (Base); after zeroing the  $k$  residual coordinates with the largest mean absolute training value, or with the largest variance across training passages; after per-coordinate standardization with training statistics (Std.); after centering alone ( $D=0$ ); and after ABTT with one or three components (ABTT $_{D=1}$ , ABTT $_{D=3}$ ), the two reference repairs. Bottom block: top-PC share of the training vectors after the same interventions, the share of their centered variance on the first principal component; centering leaves it unchanged by definition. In the three T5 rows, AUROC stays at or below 0.716 after zeroing ten coordinates under either ranking, while standardization reaches 0.806 to 0.907 and ABTT with three components 0.914 to 0.973. At the collapsed T5 layers (baseline AUROC below 0.70, 26 layers), one component recovers a median 45 percent of the AUROC gain of  $D=10$ , and three recover at least 80 percent at every layer (Table 24). All statistics are fit on training embeddings only.

els as the healthy contrast, and has four parts: a pooling control, the token audit proper, a token ablation, and a length check. SIF rescues LaTa and PhilTa but not mT5-base (Section 4), and it differs from mean pooling in two ways: it down-weights frequent tokens, and it drops special tokens such as the end-of-sequence token. Three pooling arms separate the two. The first is mean pooling without special tokens. The second uses SIF’s frequency weights but keeps special tokens at weight one, which about doubles their share of the pooling weight (a median 0.040 at the collapsed layers, against 0.020 under mean pooling). The third is mean pooling without the 100 most frequent training tokens. An arm rescues a collapsed layer if it recovers at least 80 percent of SIF’s AUROC gain over mean pooling, judged only where that gain is at least 0.05. For the audit, a passage’s score on a component is the mean of its tokens’ contributions. We fit one mean contribution per token type on training tokens and measure how much of the variance of the test passages’ scores this token mix explains. The token mix carries the first component if it explains at least half. We also split that variance into the shares of special tokens, of the 100 most frequent tokens and of all other tokens, and list the token types with the largest shares. A share is a covariance share: the covariance of a group’s

or a type’s part of the score with the whole score, divided by the score’s variance. The shares of the three groups sum to one, but a single share can fall below zero or exceed one. We repeat each quantity on 20 random directions, drawn orthogonal to the top ten principal components, as a matched control, and on the joint span of the components. For the ablation, we rank token types on training passages by their share of the first component’s score variance, or by the mean of their shares on the first three components, and mean-pool without the top  $m$  types,  $m \in \{1, 3, 10, 30, 100\}$ . The ablation restores a layer if AUROC reaches 0.90 for some  $m$ . Its control is mass-matched: it drops random other token types that hold at least as many training tokens as the dropped ones, in five draws. We added this control on review after a first run, whose control matched the number of dropped types and their training counts but dropped far fewer tokens. The length check addresses LaTa’s negative raw gap (Section 4): the length account holds if the first component’s score correlates with log token count at  $|\rho| \geq 0.5$  and same-directory pairs differ more in log length than different-directory pairs. We expected frequent tokens to hold the largest share in LaTa and PhilTa, so that the frequency weights alone rescue them and dropping special tokens does not. For mT5-base we expected no arm

to rescue it and the tokens SIF keeps, taken as those outside the 100 most frequent, to hold the largest share: if the end-of-sequence token carried the direction, SIF would delete that token and rescue the model. Table 5 gives the audit for each model at its worst baseline layer. We find token carriers in LaTa. The frequency weights alone rescue all 10 of its collapsed layers (median AUROC 0.861; 96 percent of the gain of our SIF arm, 99 percent of the gain of the Section 4 cells). Dropping the 100 most frequent tokens also rescues all 10 (median 0.901), and dropping special tokens alone rescues none (median 0.498). The token mix explains a median 0.985 of the first component’s test score variance, against a median 0.18 for random directions. At layer 6 the comma alone holds 0.962 of that component’s score variance over training passages, the period 0.916 of the second component’s, and the comma 0.897 of the third’s. Mean pooling without three token types, the comma, the period and the end-of-sequence token, restores all 10 layers (AUROC 0.912 to 0.929, median 0.919); they are the top three of the three-component ranking at every one of these layers and 11 percent of the test tokens. The mass-matched control at the same setting leaves the median AUROC at 0.500 while dropping 10 percent of the test tokens, and it restores none of the 26 collapsed layers at any  $m$  under either ranking. The ranking by the first component alone restores at most 2 of the 10 LaTa layers at any  $m$ . In PhilTa the same three token types lead at layer 10. Frequent tokens hold the largest share of the first component at 8 of its 9 collapsed layers (median 0.740, against 0.235 for special tokens), not at all 9 as expected: at layer 3 special tokens lead (0.515 against 0.455). Neither single-factor arm rescues any of the 9 layers (median AUROC 0.653 with the frequency weights, 0.500 with special tokens dropped), so our expectation for the frequency weights is not met. Given how that arm weights special tokens, we conclude only that neither change alone suffices. The ablation restores all 9 PhilTa layers, but it needs more token types: under the three-component ranking, dropping the top three restores 1 of the 9 (AUROC 0.847 to 0.904), the top 10 restore 8, and the top 30 all 9 (AUROC 0.905 to 0.930, median 0.925). At  $m = 30$  the dropped types are a median 33 percent of the test tokens (32 to 35 percent), and the mass-matched control, which drops a median 32 percent, stays at a median AUROC of 0.543. PhilTa’s token mix explains a median 0.42 of the

first component’s score and less than one half at all 9 layers, so the token-mix rule fails there, and the case for token carriers in PhilTa rests on the ablation. Our reading, made after seeing this result, is that the values of PhilTa’s leading token types depend on context. It rests on one statistic: at layer 10 the end-of-sequence token has a mean contribution of  $-1.1 \times 10^5$  against a mean magnitude of  $1.6 \times 10^5$ , so its sign changes across passages and one mean per token type cannot predict it. In mT5-base the account fails. The token mix explains 0.066 to 0.067 of the first component’s score, below the random-direction mean at 6 of the 7 collapsed layers. The group shares (0.53 frequent, 0.47 other) stay close to the groups’ shares of tokens (0.48 and 0.50), and the largest single-type share is 0.08, although some rare types make very large contributions per token. Top-PC share stays at 1.000 under every pooling arm. Dropping up to 100 token types ranked on training passages restores none of the 7 layers: at  $m = 100$  under the three-component ranking, AUROC is 0.724 to 0.748. SIF gains less than 0.05 there, so the rescue rule cannot be evaluated; the median raw AUROC changes of the three arms are 0.000, +0.027 and +0.057. The expectation that the tokens SIF keeps hold the largest share is not met at any layer. The length account fails at all 26 collapsed layers. The first component’s score correlates with log length at  $|\rho| < 0.5$  everywhere, and same-directory pairs differ less in log length than different-directory pairs (mean absolute difference 0.32 against 0.78 on LaTa’s test pairs). LaTa’s negative gap is therefore not a length effect and stays unexplained. The score’s correlation with a passage’s share of the 100 most frequent tokens is weak as well ( $|\rho| \leq 0.36$  at all 26 layers). The healthy contrast shows that a token mix that explains a component does not by itself mark a collapse. At Qwen3-0.6B layer 1 it explains 0.99 of the first component’s score, with the first token contributing 0.69, yet that component holds 0.10 of the variance and AUROC is 0.858. Random directions also reach 0.71 there and 0.73 at LaBSE layer 1, so the statistic must be read against its control. One reproduction check failed: our SIF arm, recomputed with training-only token frequencies, equals the SIF pooling of our extraction code exactly but differs from the SIF cells reported in Section 4 by up to 0.015 AUROC (LaTa; 0.007 for PhilTa, 0.001 for mT5-base). We had planned to withhold the pooling-control conclusions in that case. After reading the results we

report them under both references instead: all 78 rescue verdicts (26 layers, three arms) are the same with the Section 4 cells as the reference. With E1, these results place LaTa’s nuisance on a few token types and not on a few coordinates: dropping three token types restores every collapsed layer, as removing three components does, while the direction they feed extends beyond its ten top coordinates. For PhilTa the same reading rests on the ablation alone.

**Subspace split (E3).** Our account says that the top  $D$  directions ABTT removes hold nuisance in pre-trained spaces. We rank passages by their projection onto PC1 alone, onto PCs 2 to  $D$ , and onto the retained subspace, and we compare the removed with the retained subspace. At collapsed layers the full vector already ranks at chance and the removed subspace dominates it, so the test informs mainly where the full vector is healthy. The cleanest contrast is LaTa’s layer 12, which the pre-trained and fine-tuned models share. ABTT raises the pre-trained layer from 0.938 to 0.971 and lowers the fine-tuned one from 0.984 to 0.970 in point estimate. If the removed subspace holds nuisance in pre-trained spaces, it ranks well below the retained subspace there, although PCs 2 to  $D$  may still carry some topical signal. Fine-tuning is reported to concentrate task signal in the dominant directions (Rajae and Pilehvar, 2021). If that holds here, the removed subspace of fine-tuned LaTa, Qwen3-0.6B and KaLM-mini ranks well above chance and close to the retained one. This would explain the drop in fine-tuned LaTa’s AUROC under ABTT. If the removed subspace ranks no better after fine-tuning than before, that drop needs another cause. We find that the removed subspace ranks below the retained one in every pre-trained model (Table 25, Appendix H). At the 26 collapsed T5 layers it ranks at chance (median AUROC 0.489; PC1 alone 0.463), while the retained subspace reaches 0.977. Where the full vector is healthy the removed subspace still carries retrieval signal, mostly in PCs 2 to  $D$ , but the retained subspace leads by at least 0.08 in the T5 encoders and 0.04 in the other models at every pre-trained layer. The third outcome that would falsify the account (Scope, below) therefore does not occur. At the selected layers of the pre-trained models the top  $D$  directions also rank below the next  $D$  (LaTa layer 12: 0.868 against 0.907), so per dimension they carry less signal than what follows them; after fine-tuning

they rank above it (0.953 against 0.900 for LaTa). At LaTa’s layer 12 the pre-trained gain from ABTT comes from centering alone (0.972), so removing the top components changes AUROC by  $-0.001$  before fine-tuning and  $-0.015$  after it, in point estimate. In fine-tuned Qwen3-0.6B and KaLM-mini the removed subspace also comes within 0.02 of the retained one (0.983 against 0.992, and 0.978 against 0.994, at  $D = 10$ ). This is consistent with fine-tuning moving task signal into the dominant directions (Rajae and Pilehvar, 2021), and with the point-estimate drop of fine-tuned LaTa’s AUROC under ABTT from 0.984 to 0.970.

**How many directions (H1).** ABTT selects  $D = 10$ , the top of the searched grid, in 87 of 100 model-layers (81 of 100 when SIF precedes it). Selection by training DirAcc@1 never picks  $D \leq 2$ : it picks  $D = 3$  once,  $D = 5$  twice, and  $D = 7$  ten times. This already sits uneasily with a single nuisance axis, at least for routing. We therefore sweep  $D \in \{0, 1, 2, 3, 5, 7, 10, 15, 20, 30, 50\}$  at all 100 model-layers, where  $D = 0$  is centering alone, and read Task A AUROC. If the nuisance is rank-1,  $D = 1$  recovers most of the AUROC gain at the collapsed T5 layers. We set the threshold at 80 percent of the  $D = 10$  gain. The curve then stays flat beyond  $D = 1$ , and centering alone recovers little. If  $D = 1$  falls below that threshold, the nuisance is a low-dimensional subspace, and we drop the one-direction wording. Figure 3 shows the sweep. We find that the nuisance is not rank-1. At the 26 collapsed T5 layers,  $D = 1$  recovers a median 45 percent of the  $D = 10$  AUROC gain (13 to 89 percent) and reaches the 80 percent threshold at only one layer, LaTa layer 2. Centering alone recovers nothing (median  $-4$  percent). Two components recover a median 78 percent, and three recover at least 80 percent at every collapsed layer (median 96 percent, AUROC 0.964 against 0.977 at  $D = 10$ ; Table 24, Appendix H). This happens although the first component holds a median 94 percent of the centered training variance at these layers: one direction dominates the variance, but the nuisance that breaks cosine ranking spans a few directions, two to three in LaTa and PhilTa and more in mT5-base.  $D = 1$  recovers the most in mT5-base (median 64 percent) and the least in PhilTa (14 percent), but mT5-base needs the most components overall: LaTa reaches the full gain with two (median 98 percent), PhilTa with three (95 percent), while mT5-base still lacks 15 percent

| Model      | L  | AUROC by pooling |          |           |       |          | Token ablation |       | Token-mix EV |       | Share by token group |       |       | $ \rho $ |
|------------|----|------------------|----------|-----------|-------|----------|----------------|-------|--------------|-------|----------------------|-------|-------|----------|
|            |    | Mean             | No spec. | SIF+spec. | SIF   | No freq. | Drop           | Rand. | PC1          | Rand. | Spec.                | Freq. | Other |          |
| LaTa       | 6  | 0.496            | 0.493    | 0.860     | 0.878 | 0.904    | 0.922          | 0.483 | 0.98         | 0.19  | 0.01                 | 0.97  | 0.02  | 0.38     |
| PhilTa     | 10 | 0.538            | 0.496    | 0.677     | 0.879 | 0.692    | 0.917          | 0.540 | 0.44         | 0.14  | 0.23                 | 0.75  | 0.02  | 0.11     |
| mT5-base   | 5  | 0.654            | 0.653    | 0.675     | 0.673 | 0.726    | 0.724          | 0.656 | 0.07         | 0.24  | 0.00                 | 0.53  | 0.47  | 0.20     |
| LaBSE      | 1  | 0.806            | 0.809    | 0.812     | 0.818 | 0.816    | 0.827          | 0.800 | 0.36         | 0.73  | 0.09                 | 0.37  | 0.53  | 0.95     |
| Qwen3-0.6B | 1  | 0.858            | 0.855    | 0.856     | 0.852 | 0.869    | 0.885          | 0.836 | 0.99         | 0.71  | 0.74                 | 0.08  | 0.18  | 0.58     |
| KaLM-mini  | 5  | 0.861            | 0.859    | 0.867     | 0.863 | 0.872    | 0.879          | 0.854 | 0.98         | 0.47  | 0.01                 | 0.02  | 0.97  | 0.51     |

Table 5: Token audit at each model’s worst baseline layer (L). Pooling: Task A test AUROC of mean pooling (Mean); mean pooling without special tokens (No spec.); SIF frequency weights with special tokens kept (SIF+spec.); SIF pooling, which also drops special tokens, recomputed in this experiment’s forward pass with training-only token frequencies (SIF; its AUROC differs from the SIF cells of Section 4 by up to 0.015 over all models and layers); and mean pooling without the 100 most frequent training tokens (No freq.). Token ablation: test AUROC of mean pooling after dropping the  $m$  token types that contribute most to the top principal components (Drop), where  $m$  between 1 and 100 and one of two rankings are chosen by training AUROC, and the mean test AUROC of five random sets of other token types that hold at least as many training tokens as the dropped types (Rand.). Token-mix EV: the share of the variance of the first principal component’s score across test passages that is explained by which token types a passage contains, with one mean per token type fit on training tokens (PC1), next to the mean of the same quantity over 20 random directions orthogonal to the top ten principal components (Rand.). Share by token group: the covariance share of that variance contributed by special tokens, by the 100 most frequent tokens and by all other tokens; the three sum to one, but a single share can fall below zero or exceed one.  $|\rho|$ : absolute Spearman correlation between the PC1 score and the log token count of test passages (the sign of a principal component is a convention). Components, token frequencies and token rankings are fit on training passages only.

at  $D = 3$  and keeps rising until  $D = 10$ . The failure of  $D = 1$  is not an estimation artifact: at LaTa layer 6, PhilTa layer 10 and mT5-base layer 8 the first component fit on training and on test passages agree to  $|\cos| \geq 0.995$ , removing a test-fitted first component also leaves AUROC at 0.684, 0.666 and 0.869, and after the training-fitted removal the next direction still holds 66 to 74 percent of the remaining test variance in LaTa and PhilTa. Beyond  $D = 10$  the curve is flat to within about 0.02, and extending the selection grid to  $D = 50$  would move training selection to  $D \geq 15$  at 99 of 100 model-layers while changing test AUROC by at most 0.023. We therefore drop the one-direction wording for what breaks ranking.

**Scope.** We do not ask where these coordinates come from in pretraining, or why they peak at mid-depth in the T5 encoders of our panel. We ask only whether they explain the retrieval failure. Three outcomes would falsify the account. First, neither zeroing  $k \leq 5$  coordinates nor standardization repairs the collapsed layers while ABTT does, which makes the nuisance a distributed direction. Second, Qwen3-0.6B shows passage-varying massive coordinates ( $r \geq 1$ ) of T5 scale without collapse. Third, the removed subspace ranks as well as the retained one in the pre-trained T5s, which means ABTT discards signal along with nuisance. Under any of these, the paper’s claim reduces to the geometric

description of Section 4 and the repair. E1 decides the first two. The first occurs at 17 of the 26 collapsed layers; at the other 9, standardization repairs and zeroing  $k \leq 5$  does not. The second does not occur: the three largest-magnitude coordinates of Qwen3-0.6B have  $r \leq 0.20$  at every layer. For the coordinate account, our claim therefore reduces to the geometric description and the repair. One qualification applies to the first outcome: the dominant direction is not evenly distributed, since about ten coordinates hold most of its loading, yet zeroing them leaves the rest of that direction dominant (E1 and its follow-up). E2 adds token carriers in LaTa, where the token mix explains the dominant direction’s score and mean pooling without three token types (11 percent of test tokens) restores every collapsed layer. For PhilTa the evidence is the ablation alone: mean pooling without 30 token types (a median 33 percent of test tokens) restores all 9 collapsed layers, while the token-mix rule fails at all 9. In mT5-base, dropping up to 100 token types ranked on training passages restores no layer, so the claim there stays the geometric description and the repair. Why those token types take such values is outside our scope. Section 6 turns from the coordinates to a second question: whether T5 pretraining or the missing embedding objective produces them.

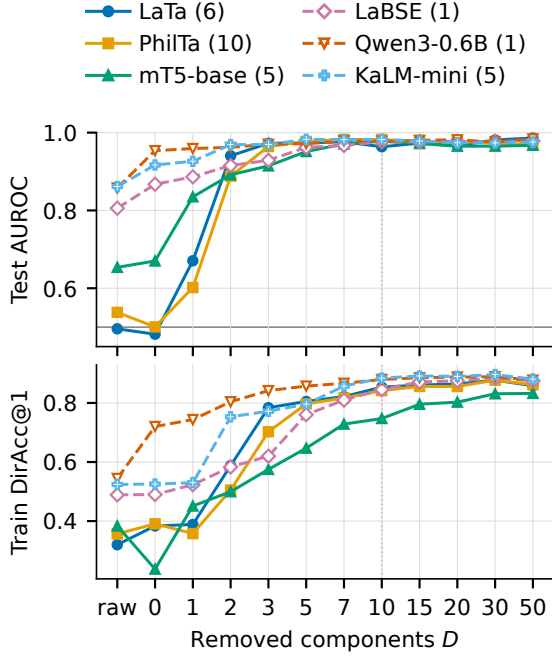


Figure 3: Task A test AUROC (top) and training DirAcc@1 (bottom) against the number  $D$  of removed principal components, one line per model at its lowest-AUROC baseline layer (in parentheses in the legend). The horizontal axis is categorical: raw is the uncorrected vector,  $D = 0$  is centering on the training mean alone, and the ticks are not evenly spaced in  $D$ . ABTT is fit on training embeddings only; the dotted vertical line marks  $D = 10$ , the top of the selection grid, and the gray horizontal line chance (AUROC 0.5). Filled markers and solid lines: T5 encoders; hollow markers and dashed lines: embedding-trained models. At the collapsed T5 layers one component restores part of the ranking, and two to five restore nearly all of it.

## 6 Objective or Architecture?

Our panel confounds two factors. Every T5 we have run is a raw pre-trained encoder. Every model that does not collapse is embedding-trained: LaBSE, Qwen3-0.6B and KaLM-mini. The collapse may follow T5 pretraining, or it may follow the absence of an embedding objective. The panel so far cannot separate the two, so the claim that collapse is T5-specific holds only in our panel.

Table 1 gives the design. The LaTa–LaBERTa and PhilTa–PhilBERTa pairs hold data and authors fixed and vary the architecture factor. The embedding-trained T5 cell holds Sentence-T5 (or GTR-T5) (Ni et al., 2022a,b). Its matched raw partner is T5-v1.1-base on the Latin corpus, since Sentence-T5 starts from an English T5 checkpoint rather than from LaTa.

The raw encoder-only cell is the discriminat-

| Cell       | Models                         | AUROC <sub>min</sub> | PC1 <sub>max</sub> |
|------------|--------------------------------|----------------------|--------------------|
| T5, raw    | LaTa, PhilTa, mT5-base         | 0.50–0.65            | 0.86–1.00          |
| T5, raw    | T5-v1.1-base                   | 0.489                | 0.975              |
| T5, raw+FT | LaTa (fine-tuned)              | 0.50–0.57            | 0.945              |
| T5, emb.   | Sentence-T5                    | [P2x2]               | [P2x2]             |
| Enc., raw  | LaBERTa, PhilBERTa, Latin BERT | [P2x2]               | [P2x2]             |
| Enc., emb. | LaBSE                          | 0.806                | 0.524              |

Table 6: Architecture-by-objective panel. AUROC<sub>min</sub> is the lowest Task A test AUROC over layers of the unmodified mean-pooled vectors (a range when the cell holds several models; for fine-tuned LaTa, the range over layers 2–11). PC1<sub>max</sub> is the peak top-PC share over layers, the largest share of centered training variance on the first principal component. Enc. is encoder-only, emb. is embedding-trained, and FT is contrastive fine-tuning on training pairs. T5-v1.1-base is read on the Latin corpus as the raw partner of Sentence-T5.

ing test. Both accounts predict that the raw T5s collapse and that LaBSE does not. If LaBERTa and PhilBERTa keep retrieval at every depth, the collapse follows T5 pretraining. If they collapse, it follows the missing embedding objective. The embedding-trained T5 cell is weaker evidence, because its loss, like our fine-tuning below, attaches at the output. A Sentence-T5 that collapses at mid-depth is then predicted by both accounts, while one that stays healthy favors the objective account. We read this cell against LaBSE’s healthy middle layers and against raw T5-v1.1-base. An English model may also retrieve Latin poorly at every layer for reasons unrelated to collapse, so for this cell the label-free top-PC share and effective rank are the primary readout. Table 6 reports the lowest baseline AUROC over layers and the peak top-PC share of each cell. We find [Pending: P2x2: lowest baseline AUROC and peak top-PC share over layers for LaBERTa, PhilBERTa, Latin BERT and Sentence-T5; expected under the architecture account: the raw encoder-only models stay above 0.80 AUROC at every layer, while Sentence-T5 and T5-v1.1-base reach a top-PC share above 0.8 at mid-depth].

Fine-tuning gives a first piece of evidence. Contrastive fine-tuning on 499 of the training pairs (Appendix I) raises LaTa’s last layer from 0.938 to 0.984 AUROC (Table 26). Its layers 2 to 11 still collapse, at AUROC 0.50 to 0.57. Their geometry barely moves: the peak top-PC share is 0.945 after fine-tuning against 0.952 before (Table 27, Appendix I). Supervision attached to the output thus leaves the intermediate layers unrepaired. A similar output-only pattern is reported for a multi-

lingual encoder: after sentence-level tuning it has no large outlier dimension at the output layer, while the base model’s top outlier persists, reduced, at layer 8 (Hämmerl et al., 2023). This result does not separate the two accounts: small-scale supervision at the output may leave intermediate layers untouched under either. Our fine-tuning runs at most eight epochs on 499 pairs, far less than the large-scale contrastive training behind Sentence-T5, so the raw encoder-only cell remains the test.

All retrieval evidence comes from one labeled corpus. The geometry, however, needs no labels, so we can cross model with text. We compute top-PC share and effective rank per layer for mT5-base and PhilTa, whose pretraining covers English, on an English sample. We compute the same for T5-v1.1-base (Raffel et al., 2020) on both the Latin corpus and the English sample. If the collapse is a property of the model, each model shows the same profile on both texts: top-PC share above 0.8 and effective rank near 1 at mid-depth. If mT5-base stays high-rank on English, the collapse depends on the input text. If T5-v1.1-base stays high-rank on English while mT5-base collapses there, it depends on the pretraining data. Every cell shows the same profile (Table 10 and Figure 6, Appendix B). The English sample is 1,705 public-domain passages of US court opinions from the Caselaw Access Project (Harvard Law School Library Innovation Lab, 2024), as redistributed in the Common Pile (Kandpal et al., 2025), one per Latin passage and matched to it in mT5 token length; as on Latin, we read the statistics on the 847 passages paired with training passages. On English, mT5-base reaches a top-PC share of at least 0.999 with effective rank 1.0 at layers 5 to 11, as on Latin, and PhilTa reaches 0.85 to 0.89 at layers 4 to 11 (0.83 to 0.86 on Latin). PhilTa’s effective rank is 1.8 to 2.3 on both texts, as on Latin, rather than near 1. T5-v1.1-base, pre-trained on English alone, is low-rank from layer 2 to layer 11 on both texts, at top-PC share 0.93 to 0.95 on English and 0.96 to 0.97 on Latin. In every cell the first and last layers stay high-rank. Neither fail condition occurs: in these cells the low-rank middle layers appear on both texts, whether or not the input is in the model’s pretraining language. This check speaks to geometry only. Whether retrieval also collapses on English text stays outside what this testbed measures.

## 7 Repair and Its Limits

Sections 5 and 6 ask what masks the retrieval signal. This section asks whether a train-only projection can remove the mask, and where it stops helping. ABTT (Section 3) subtracts the training mean and removes the top  $D$  training components; no model parameter changes.

**ABTT restores every layer.** ABTT lifts every layer of every model to a pairwise AUROC between 0.96 and 0.99, including the T5 layers that sit at chance. At the train-selected layers of Table 7, baseline AUROC spans 0.838 to 0.972, and ABTT moves every model into a band from 0.971 to 0.987. The gain is largest where the baseline is weakest: mT5-base gains 0.137 (95% CI 0.109 to 0.166), while KaLM-mini gains 0.009 (−0.002 to 0.021), within noise. At each T5 encoder’s worst baseline layer, ABTT recovers AUROC to within 1.0 point of that model’s best ABTT layer. Because ABTT is a linear projection fit on training vectors, this shows that the retrieval signal was linearly present in every layer. The character 3–5-gram TF-IDF reference reaches 0.987 on the same split (Table 28). The repair therefore matches the lexical reference but does not exceed it.

**Routing converges.** Task B shows the same pattern at the directory level (Table 8). Under the baseline, DirAcc@1 spans 39.3 points across models, so model choice dominates. ABTT compresses that spread to 3.3 points, from 86.1 to 89.4. Assignment accuracy moves from a 40.3-point spread to a band from 88.5 to 91.7. Over five random reassignments of query and reference files, the baseline spread of 36.6 points falls to 1.7 under SIF+ABTT, with standard deviations of at most 1.0. The lexical reference routes at 89.9 on the single split, level with the best of the ABTT cells, which span 86.1 to 89.4, and at 91.1 over five seeds, against 88.9 to 90.6 for the SIF+ABTT cells. Surface overlap routes as well as the best repaired encoder on this corpus. Under SIF+ABTT every model places the correct directory in a two-item shortlist for more than 95 percent of queries (Table 19, Appendix G). Directory-level bootstrap intervals (Appendix J, Table 29) span about  $\pm 0.02$  AUROC and  $\pm 3$  points of routing accuracy per cell. Every zero-shot routing gain from ABTT excludes zero, as do the AUROC gains of the three T5 encoders and LaBSE; those of Qwen3-0.6B (0.966 to 0.973; 95% CI of the gain −0.010 to 0.019) and KaLM-mini (0.972 to 0.981;

| Model                   | Task A AUROC |       |              |              | Task A cosine gap |       |              |              |
|-------------------------|--------------|-------|--------------|--------------|-------------------|-------|--------------|--------------|
|                         | Base         | SIF   | ABTT         | SIF+ABTT     | Base              | SIF   | ABTT         | SIF+ABTT     |
| LaTa                    | 0.938        | 0.948 | <b>0.971</b> | 0.966        | 0.237             | 0.366 | 0.525        | <b>0.580</b> |
| PhilTa                  | 0.939        | 0.955 | <b>0.983</b> | 0.982        | 0.231             | 0.319 | 0.540        | <b>0.573</b> |
| mT5-base                | 0.838        | 0.845 | <b>0.975</b> | 0.974        | 0.081             | 0.036 | <b>0.536</b> | 0.519        |
| LaBSE                   | 0.956        | 0.956 | <b>0.987</b> | 0.976        | 0.183             | 0.184 | <b>0.611</b> | 0.580        |
| Qwen3-0.6B              | 0.966        | 0.942 | <b>0.973</b> | 0.972        | 0.024             | 0.053 | <b>0.553</b> | 0.540        |
| KaLM-mini               | 0.972        | 0.964 | 0.981        | <b>0.981</b> | 0.056             | 0.056 | <b>0.568</b> | 0.535        |
| LaTa (fine-tuned)       | <b>0.984</b> | –     | 0.970        | –            | 0.387             | –     | <b>0.548</b> | –            |
| Qwen3-0.6B (fine-tuned) | <b>0.996</b> | –     | 0.994        | –            | 0.684             | –     | <b>0.716</b> | –            |
| KaLM-mini (fine-tuned)  | <b>0.997</b> | –     | 0.994        | –            | 0.636             | –     | <b>0.651</b> | –            |
| TF-IDF char 3–5         |              |       | 0.987        |              |                   |       | 0.555        |              |

Table 7: Task A pairwise duplicate detection for all six models under four post-processing settings. Each cell is a test-set score at the layer chosen by training-set AUROC, listed in Table 11; ABTT uses  $D$  tuned on train. Baseline AUROC spans 0.838 to 0.972, and ABTT lifts every model into a 0.971 to 0.987 band. Cosine gap is defined in Section 3. Per-layer grids: Appendix Tables 12, 16, and 21. Below the rule, reference systems on the same split with the same evaluation code: LaTa, Qwen3-0.6B, and KaLM-mini fine-tuned contrastively on 499 of the 565 positive train pairs (a 28-directory dev carve takes the rest), a ceiling at this training budget, and the character 3–5-gram TF-IDF reference (TF-IDF char 3–5), fitted on train files (both setups: Appendix I). TF-IDF char 3–5 is level with the best zero-shot ABTT AUROC (0.987 against 0.987), and ABTT moves the fine-tuned encoders’ AUROC 0.984 to 0.970 for LaTa (gap 0.387 to 0.548), 0.996 to 0.994 for Qwen3-0.6B (gap 0.684 to 0.716), and 0.997 to 0.994 for KaLM-mini (gap 0.636 to 0.651). The zero-shot rows diagnose where the retrieval signal sits inside encoders; they are not offered as retrievers that beat surface overlap.

–0.002 to 0.021) do not (Table 30). The spread in DirAcc@1 falls from 39.3 (34.6 to 43.7) to 3.3 (1.8 to 6.0) points, and the lexical reference is not distinguishable from the best ABTT cells (+0.5; 95% CI –1.4 to 2.5 against Qwen3-0.6B, –1.3 to 2.3 against KaLM-mini); only LaTa’s ABTT cell is significantly below it.

### The offset regime and the open-set decision.

The embedding-trained models do not collapse, yet ABTT still changes their routing. At their respective train-selected layers, their baseline AUROC was already 0.96 to 0.97. Under ABTT, assignment accuracy rises from 83.3 to 90.8 for LaBSE, from 82.3 to 91.5 for Qwen3-0.6B, and from 87.5 to 91.7 for KaLM-mini. ABTT also improves LaBSE’s ranking, from 0.956 to 0.987 AUROC, while the AUROC gains of Qwen3-0.6B and KaLM-mini are within noise; DirAcc@1 gains 3.5 to 9.1 points across the three. Centering alone, which removes the shared offset, does not account for this (below).

If an offset only added a constant to every cosine, order would be preserved and a threshold fit on training files would absorb it. A shared offset vector need not act as such a constant. Its effect on each cosine depends on the passage’s norm and its projection onto the offset, so it can reorder pairs. LaBSE’s AUROC gain under ABTT does not come from the mean: centering alone leaves its AUROC unchanged (–0.002, –0.009 to 0.004), and the gain arrives with the ten removed com-

ponents (+0.029, 0.013 to 0.048). A cost to the open-set decision beyond ranking would need off-sets that differ by query, such as hub vectors that sit close to many queries (Radovanović et al., 2010), or a shift between the training and test score distributions. The threshold grid offers a mundane alternative. We fit  $\tau$  over 200 evenly spaced values in  $[0, 1]$ , a step of about 0.005, while at layer 27 the test cosines of Qwen3-0.6B spread with a standard deviation of only 0.007. At such a layer one grid step spans most of a standard deviation, and ABTT could help routing simply by spreading the cosines. Five checks on cached embeddings test these readings (Table 31, Appendix J). The gain does not depend on the threshold. With no threshold at all, the AUROC of each file’s best-match cosine for existing against new files rises from the baseline cell to the ABTT cell by 0.037 (95% CI 0.021 to 0.055) for LaBSE, 0.055 (0.034 to 0.077) for Qwen3-0.6B, and 0.027 (0.012 to 0.043) for KaLM-mini. At their baseline cells the train-fit  $\tau$  gives assignment accuracy within 1.0 point of the best test threshold (upper bounds 1.1 to 3.5), and the gain survives when both settings use their best test thresholds (7.0, 8.9, and 4.4 points). Refitting  $\tau$  as the exact best-F1 cut leaves the three ABTT cells unchanged and moves the baselines by –1.3 to +0.3 points. Layer 27 is not a routing cell: at Qwen3-0.6B’s routing cells the test cosines spread with a standard deviation of 0.043 (layer 28, raw) and 0.086

| Model                   | Assignment accuracy |      |             |             | Directory accuracy @1 |      |             |             |
|-------------------------|---------------------|------|-------------|-------------|-----------------------|------|-------------|-------------|
|                         | Base                | SIF  | ABTT        | SIF+ABTT    | Base                  | SIF  | ABTT        | SIF+ABTT    |
| LaTa                    | 73.8                | 83.2 | 88.5        | <b>90.2</b> | 72.1                  | 81.8 | 86.1        | <b>88.2</b> |
| PhilTa                  | 70.9                | 79.0 | 91.1        | <b>91.4</b> | 69.3                  | 77.7 | 88.3        | <b>88.9</b> |
| mT5-base                | 47.2                | 54.5 | <b>90.3</b> | <b>90.3</b> | 46.6                  | 54.1 | <b>88.5</b> | 87.5        |
| LaBSE                   | 83.3                | 83.4 | <b>90.8</b> | 89.6        | 81.5                  | 81.9 | <b>88.5</b> | 86.9        |
| Qwen3-0.6B              | 82.3                | 76.0 | <b>91.5</b> | 90.8        | 80.3                  | 74.2 | <b>89.4</b> | 88.6        |
| KaLM-mini               | 87.5                | 85.7 | <b>91.7</b> | 90.8        | 85.9                  | 84.0 | <b>89.4</b> | 88.0        |
| LaTa (fine-tuned)       | 83.4                | –    | <b>87.8</b> | –           | 81.6                  | –    | <b>85.2</b> | –           |
| Qwen3-0.6B (fine-tuned) | 91.4                | –    | <b>92.1</b> | –           | 90.8                  | –    | <b>91.4</b> | –           |
| KaLM-mini (fine-tuned)  | 92.4                | –    | <b>93.4</b> | –           | 91.7                  | –    | <b>92.5</b> | –           |
| TF-IDF char 3–5         |                     |      | 91.8        |             |                       |      | 89.9        |             |

Table 8: Task B autonomous routing for all six models under four post-processing settings, in percent. Baseline directory accuracy at rank 1 spans 46.6 to 85.9 across models (39.3 points), and ABTT narrows it to 86.1 to 89.4 (3.3 points). Each cell is a test-set score at the layer chosen by training-set directory accuracy at rank 1, listed in Table 11. Assignment accuracy scores the existing-versus-new decision alone, comparing each file’s maximum cosine against the train-fit threshold  $\tau$ ; a degenerate threshold that routes everything as existing already attains the 62.4 percent class prior. Directory accuracy at rank 1 requires the correct directory, so read the two columns together. Per-layer grids: Appendix Tables 13, 17, and 22. Below the rule, reference systems on the same split with the same evaluation code: LaTa, Qwen3-0.6B, and KaLM-mini fine-tuned contrastively on 499 of the 565 positive train pairs (a 28-directory dev carve takes the rest), a ceiling at this training budget, and the character 3–5-gram TF-IDF reference (TF-IDF char 3–5), fitted on train files (both setups: Appendix I). TF-IDF char 3–5 is level with the best zero-shot ABTT cell (91.8 against 91.7 assignment accuracy, 89.9 against 89.4 directory accuracy at rank 1), and the fine-tuned encoders with ABTT sit below every zero-shot ABTT cell for LaTa (87.8 and 85.2), above every zero-shot ABTT cell for Qwen3-0.6B (92.1 and 91.4), and above every zero-shot ABTT cell for KaLM-mini (93.4 and 92.5), against cells spanning 88.5 to 91.7 and 86.1 to 89.4; these are point comparisons, and Appendix J gives their bootstrap intervals. The zero-shot rows diagnose where the retrieval signal sits inside encoders; they are not offered as retrievers that beat surface overlap.

after ABTT (layer 5; 0.034 raw). Nor does the shared offset produce the gain. At each baseline cell’s layer, centering alone ( $D = 0$ ) changes assignment accuracy by  $-4.7$  points ( $-7.4$  to  $-2.0$ ) for LaBSE,  $+2.6$  ( $-0.4$  to  $5.5$ ) for Qwen3-0.6B, and  $-1.9$  ( $-3.7$  to  $-0.1$ ) for KaLM-mini, and at the ABTT layer of each of the three the removed components add 12.0 to 25.9 points on top of centering. From the baseline cell to the ABTT cell hubness also falls (Radovanović et al., 2010): the skewness of the 10-occurrence distribution drops from 2.0–2.5 to 0.6–1.1, while centering alone at the ABTT layer leaves it near 1.8. We therefore read the routing gain as better separation of known from new witnesses by their best match, carried by the top components beyond the mean; whether the reduced hubness causes it remains untested.

**SIF.** SIF never matches ABTT on any model or task; we use it only as the diagnostic of Section 4 (full sweeps in Appendix G).

**Where the repair stops helping.** ABTT does not help fine-tuned ranking: AUROC falls by 0.002 to 0.014 in point estimate, and every 95% interval includes zero. Fine-tuned LaTa falls from 0.984 to 0.970 ( $-0.014$ ,  $-0.032$  to  $+0.001$ ), and

Qwen3-0.6B and KaLM-mini fall by 0.002 and 0.003 ( $-0.005$  to  $+0.001$ ;  $-0.007$  to  $+0.001$ ); for Qwen3-0.6B the two rows also sit at different train-selected layers (28 and 27). The same projection raises fine-tuned LaTa’s DirAcc@1 from 81.6 to 85.2 ( $+3.6$ , 1.0 to 6.3); the rises for Qwen3-0.6B (90.8 to 91.4) and KaLM-mini (91.7 to 92.5) are within noise (Table 26). The direction of the ranking loss is consistent with Rajaei and Pilehvar (2021), who found that removing top directions lowers the STS performance of BERT after Siamese fine-tuning. Reichbauer et al. (2026) find, by contrast, that whitening still helps fine-tuned Latin and Greek encoders. Whether the projection costs ranking in fine-tuned models is the subspace question of Section 5: if fine-tuning moves task signal into the top components, then removing them removes signal (E3). E3 finds that fine-tuning does move signal there: at layer 12 the removed subspace of fine-tuned LaTa ranks at 0.953, against 0.868 before fine-tuning and 0.900 for the next ten components.

Fine-tuning on 499 training pairs also beats the training-free projection for the embedding-trained models. Over five random reassignments of query and reference files, fine-tuned Qwen3-0.6B with

ABTT routes at 92.3 against 90.5 for its zero-shot ABTT row, and KaLM-mini at 93.0 against 90.9; for LaTa the two tie at 87.7 (Appendix I). On the single split the DirAcc@1 differences are +2.0 (0.0 to 4.0) and +3.1 (1.2 to 5.2).

**Practitioner rule.** On this testbed, a top- $D$  projection fit on training vectors helped every pre-trained encoder at every layer and did not help ranking in contrastively fine-tuned ones. A high top-PC share (0.76 or more here) flags a collapsed layer; mean pairwise cosine did not. Neither statistic chooses among healthy layers, and a character n-gram reference routes as well as the best pre-trained configuration on this corpus.

## 8 Discussion

The analysis describes one failure precisely. In the middle layers of the T5 encoders in our panel, a low-dimensional subspace takes over the pooled vectors, and cosine ranking falls to or near chance. One direction holds most of its variance, but removing it alone restores only part of the ranking; removing two to five restores nearly all of it. Top-PC share separates these layers from healthy ones; mean pairwise cosine does not. In the pre-trained models the removed directions rank below the retained ones, and at the selected layers below the next ten as well; in the fine-tuned models they rank above the next ten, a reversal for LaTa (E3). Massive coordinates do not explain the subspace in the way we predicted (E1). Zeroing up to ten of them repairs none of the 26 collapsed layers, although the ten highest-variance coordinates hold a median 92 percent of the dominant direction’s squared loading; per-coordinate standardization repairs 9. In LaTa, token types localize the nuisance where coordinates do not (E2): the comma alone holds 0.96 of the dominant direction’s score variance over training passages at layer 6, and mean pooling without three token types (11 percent of test tokens) restores all 10 collapsed layers. For PhilTa the evidence is weaker and rests on the ablation: mean pooling without 30 token types (a median 33 percent of test tokens) restores all 9 collapsed layers, but the token mix explains less than half of the dominant direction’s score at every one. In mT5-base neither the token mix nor an ablation of up to 100 token types accounts for the direction. Qwen3-0.6B and KaLM-mini are decoders of the kind in which massive activations are reported (Sun et al., 2024), yet they never collapse, and their

three largest-magnitude coordinates have  $r < 1$  at every layer (E1). At mT5-base layer 1, three massive coordinates carry 0.86 of the mean pairwise cosine, yet the layer still ranks at AUROC 0.82; their  $r$  is 0.05 to 0.12, so they shift all passages together. A projection fit on training vectors removes these few directions and restores ranking at every depth. Choosing among healthy layers still needs a train-only task signal.

Several observations remain open. In LaTa’s middle layers the raw cosine gap is negative ( $-0.05$  to  $-0.07$ ), so same-directory pairs score slightly below different-directory pairs on average, while AUROC stays at chance. Section 4 shows that the standardized gap does not account for it, and the token audit rejects a length account (E2). Architecture and training objective are confounded in the models we have run, so the T5 attribution stays provisional until the controlled panel of Section 6 lands.

For the editor who needs to route new witnesses, the repair narrows the choice of model (Section 7). Under SIF+ABTT every model places the correct directory in a two-item shortlist for more than 95 percent of queries. The existing-versus-new call remains wrong for roughly one file in ten, so the open-set decision stays with the scholar. A character n-gram reference routes as well as any repaired pre-trained encoder on this corpus, and a practical tool should start from it. We release the frozen testbed and split, the evaluation code, and lexical reference scores so that others can test these accounts on their own encoders. Token attributions at the train-selected layers are reported in Appendix K; ABTT yields leave-one-out rank faithfulness gains there, with mixed ERASER-style rationale metrics.

## Limitations

The study uses one labeled corpus in one language and one domain: medieval Latin canon law. A label-free check extends the geometry, but not the retrieval result, to one English register (Section 6). The 1,705-witness corpus is large for this field but small by current NLP standards. We test six models, and others may behave differently. In the models we have run, architecture and training objective are confounded, so we cannot yet say whether the mid-depth collapse follows from T5 pretraining or from the absence of an embedding objective; the controlled panel of Section 6 is designed to separate them (P2x2). The bootstrap intervals resample test directories with the layer, threshold, and projections held fixed, so they cover test-set sampling but not the choice of split; they span about  $\pm 0.02$  AUROC and  $\pm 3$  routing points per cell, and differences whose paired interval includes zero, among them the ABTT drops on the fine-tuned encoders and the lexical reference’s lead over the best ABTT cells, are not established effects. The printed routing accuracies are precise to a few files: refitting the threshold on a finer grid moves 45 of the 66 printed routing numbers, by a median of 3 files (0.35 points); ABTT, SIF+ABTT, and fine-tuned cells move by at most 0.8, and Base and SIF cells by up to 2.6 (Appendix J). ABTT needs enough text to estimate principal components, so smaller archives need separate study.

Task B measures ranking quality and a single threshold decision. It does not measure the full judgment a working scholar brings to deciding whether a witness belongs to an existing directory. The attribution analysis in the appendix is local to fixed query-candidate scores and failed its pre-registered threshold; its consistent effect is a leave-one-out rank faithfulness gain, and the ERASER-style rationale metrics are mixed.

The labels are expert-assigned, but by one project. Its director or supervised graduate students assigned the keys, and verification is by collation in the project’s parallel display, not by independent double annotation, so no inter-annotator agreement figure exists for this testbed. Collation compares the witnesses of one key, so it has nothing to compare when a key has a single witness, and 545 of the 840 directories hold one witness in this testbed.

## Ethical Considerations

The benchmark holds early medieval legal texts, transcribed from manuscripts and early printings, and contains no personal data. The Carolingian Canon Law project publishes its material under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 licence (CCL, 2026a). The project has confirmed in writing that the licence covers the transcriptions, annotations and markup in the exports we use and that, in the view of the project’s Advisory Board, attribution to the project suffices; the project’s interface credits the transcriber and proofreader of each unit. The project’s corpus keeps growing and being corrected, so benchmark version 1 is a dated snapshot, fixed by a sha256 manifest of its 1,705 labelled files, and every number in this paper comes from it. We will release it under the same licence, in coordination with the project’s periodic versioned exports, and later versions will follow those exports. The intended use is research on retrieval and on assisted source identification: the system ranks candidates for an expert to judge and does not identify sources on its own.

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## A Corpus Provenance and Derivation

This appendix records where the testbed text comes from and how our files were derived from it, in the detail that Section 3 leaves out.

**The project.** The Carolingian Canon Law project (CCL; [Firey, 2009](#)) is an open-access digital edition founded in 2009; [Eichbauer \(2014\)](#) reviews it.<sup>1</sup> Its texts are early medieval canon-law manuscripts, most of them ninth-century and never edited to modern standards. Each manuscript is identified by its siglum (Place-Library-Shelfmark), and a witness is one manuscript’s copy of one unit (one transcribed canon or excerpt).

**Transcription.** Transcription is manual, from manuscript images or, for some texts, eighteenth-century printings, with one exception: HTR has been attempted on one manuscript (Paris, BnF, lat. 1454; 142 of the 1,705 witnesses), and its transcription is still in proofreading. Before 2019 transcribers worked from purchased microfilm into word-processed files, which CCL contributors converted to TEI-P5 (the Text Encoding Initiative’s XML standard for scholarly editions). Since 2019 they transcribe in the CCL Transcription Desk beside a IIIF image of the folio (a zoomable web-served scan of the manuscript leaf); an approved proofreader on the Proofreading Desk then checks

<sup>1</sup>Founding year, pre-2019 workflow, proofreading standard, the one manuscript transcribed by handwritten text recognition (HTR), the design of the identifiers, the assignment and verification of keys, and the extent of unidentified material: unpublished documentation and correspondence supplied by the project, 2026.

the text against the image to the letter and adds TEI markup ([CCL, 2026b](#)). That proofreading stage belongs to the post-2019 workflow. The interface credits the transcriber and proofreader of each unit ([CCL, 2026a](#)).

**Identifiers.** The project has two identifier systems, both designed at its launch in 2009 by its director in consultation with a mathematician, a software developer and a second scholar of early medieval canon law. The first is a location identifier. Every manuscript is a unique object, so every unit has exactly one place in one manuscript, and the identifier records it as the manuscript siglum, the folium, and the unit’s position on that folium. Our file names are these identifiers. The second is the source key. Many units in different manuscripts copy the same prescription issued by the same authority, and the key names that common source. It encodes the type of source (a church council, a papal decretal, a royal capitulary, a theologian, a penitential), the point of origin (a particular council, pope or author), the date of origin, and the item’s reference in modern scholarship, such as a canon, letter or register number. The key is designed so that a researcher can find the modern authoritative edition of the source where one exists.

**Labels.** CCL keys a unit to its original source, such as a fourth-century council or a papal decretal, once that source is identified, and our directories reuse that key as supplied. In a label, a four-letter code carries the type of source and its point of origin, followed by the year and the canon or chapter number, as in CT0U.567.16; some labels carry a trailing annotation, as in CT0U.567.16(15) or CARL.501?.18. Counting a directory as key-named when its label begins with a four-letter code, a period, a year and a period, whatever follows, 689 of the 840 directory labels are source keys. The other 151 are modern edition citations, the Apostolic Canons (Can.apost.N), references to other collections such as the Capitula Martini, biblical references, and a handful of unidentified sources; each unidentified source is a singleton directory and contributes no positive pairs.

**Assignment and verification.** Keys were assigned by the project’s director, a specialist in early medieval canon law, or by graduate students working under the director’s supervision, who had enough Latin to check a manuscript version against the modern authoritative version of the source. Ver-

ification, and occasional correction, runs through the project’s parallel display, which sets side by side all units that share a key and shows how far their wording agrees. A misidentified unit stands out there as a text that cannot be a version of the others. This is verification by collation (see Limitations). The project supplied two label corrections during this study, and both are logged with the frozen testbed. We assigned no label ourselves and changed only those two. What remains unidentified is large: across the several hundred manuscripts that contain the project’s texts, thousands of units still carry no key, and they are the queries that Task B is meant to serve.

**Derivation.** We hold the witnesses as plain text derived from CCL TEI-P5 exports. The rule is one file per unit: the directory is the source key from the unit’s `scholSource` division, which is stripped from the text so that no file carries its own label; the text is the manuscript rubric followed by the canon text, with inline markup flattened (editorial notes kept, deletions dropped) and layout whitespace collapsed; a unit without a source key receives no directory. The rule is scripted; checked against a current export of one manuscript (Paris, BnF, lat. 2123), it reproduces all 279 units present in both the export and the corpus, 100 percent after whitespace normalization with no character-level disagreement, and places every one of them in the same directory.

## B Layer Diagnostics and Attribution Layer Selection

This appendix complements the layerwise view of Section 4. Figure 4 adds the raw cosine gap against depth for all six models to the AUROC profile of Figure 1; the gap rises in mT5-base’s middle layers while its ranking fails. Figure 5 plots the raw gap, the standardized gap, and baseline AUROC side by side. It also separates the most anisotropic layers, where geometry collapses, from the operational attribution layers, which are selected by training-set retrieval accuracy. Table 9 lists both for the three T5 encoders, together with the top-PC share and effective rank before and after ABTT.

Top-PC share and effective rank correlate with the ABTT gain across all 100 model-layers (Spearman +0.76 and −0.76, Table 3), but only because ABTT lifts every layer into a narrow band, so the gain mirrors the baseline. Within a family the correlations are weak. The statistics locate where ABTT

helps; they do not select layers, which the train-only rule does.

**Geometry on English text.** Table 10 and Figure 6 give the label-free check of Section 6. The English sample comes from the public-domain US court opinions of the Caselaw Access Project (Harvard Law School Library Innovation Lab, 2024), in the redistribution of the Common Pile (Kandpal et al., 2025); every sampled document is marked public domain. Legal prose is the nearest openly licensed English register to canon law. Each non-empty Latin passage receives one English passage from a distinct opinion (the two empty Latin files receive an empty one): a run of whole words from a random sentence start in the body text, whose mT5 token count is within  $\max(2, 3\%)$  tokens of the Latin passage’s, and the English passage inherits its partner’s split. The two texts then have the same mT5 length distribution (median 98 tokens). The match holds only under the mT5 tokenizer: under PhilTa’s tokenizer the English passages are about 40 percent longer, and under T5-v1.1’s English-only tokenizer the Latin passages are about 55 percent longer. Re-pairing the training passages on each model’s own token lengths (653 to 847 pairs) moves top-PC share at the collapsed layers by about 0.01 to 0.02, depending on the matching rule, and leaves every cell collapsed on the same layers.

## C Score Distributions at the Collapsed Retrieval Layer

Figure 7 restates the collapse and the repair as score distributions rather than as summary metrics, for the model whose collapse is easiest to read.

## D Expanded Per-Layer Results for T5 Encoders

The main text compresses the headline retrieval results into Tables 7 and 8, which report one train-selected layer per model and method; Table 11 lists those layers. This appendix provides the expanded per-layer Task A duplicate-detection, Task B autonomous-routing, and Task B top- $k$  ranking tables for LaTa, PhilTa, and mT5-base; Appendix F gives the same view for the other three models.

## E Last-Token vs. Mean Pooling

Decoder-only embedding models often use last-token pooling. We compare last-token pooling

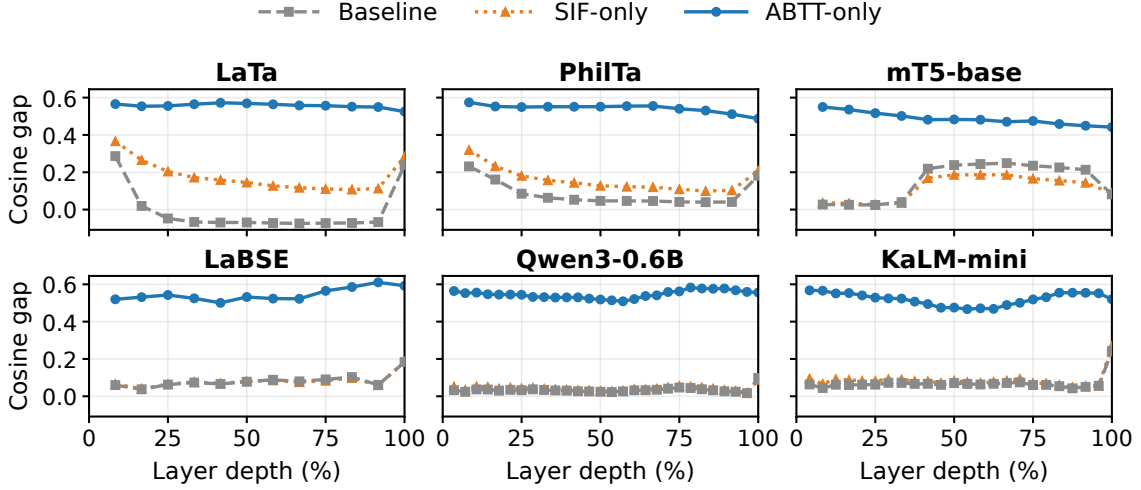


Figure 4: Raw cosine gap against normalized layer depth for all six models, test split, same panels as Figure 1. The gap is the mean cosine of same-directory pairs minus that of different-directory pairs, the margin a threshold works with. Gray: baseline mean pooling. Orange: SIF-weighted pooling. Blue: ABTT, with  $D$  chosen per layer on training data. Baseline gaps stay below 0.29 at every layer of every model. In the middle layers, LaTa’s gap turns slightly negative and PhilTa’s falls toward zero, while mT5-base’s gap rises as its ranking fails (compare Figure 1). Figure 5 resolves the mT5-base case by dividing the gap by the spread of all cosines. ABTT raises every model’s gap to between 0.44 and 0.61 at every depth. SIF raises the LaTa and PhilTa gaps and leaves the other four near or below baseline.

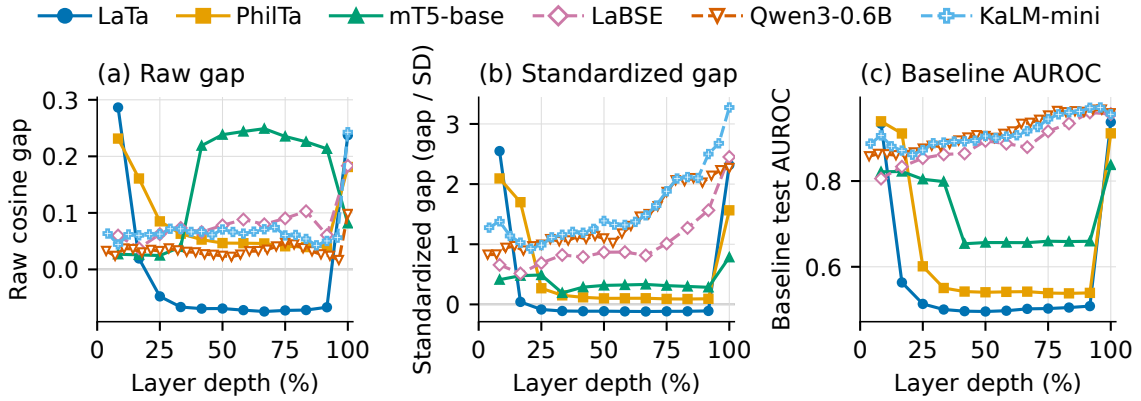


Figure 5: Cosine gap and retrieval against normalized layer depth for the six models, baseline mean pooling, test split. Solid lines with filled markers: T5 encoders. Dashed lines with hollow markers: embedding-trained models. (a) Raw gap: mean cosine of same-directory pairs minus that of different-directory pairs. (b) Standardized gap: the raw gap divided by the standard deviation of all pairwise test cosines at that layer. (c) Baseline Task A test AUROC. In the middle layers of mT5-base the raw gap rises to 0.21–0.25 while AUROC falls to 0.65, and the standardized gap falls with AUROC. LaTa’s raw gap is negative at layers 3–11. Across all 100 model-layers the standardized gap tracks AUROC at Spearman 0.97, and the raw gap at 0.22.

with mean pooling under the same ABTT pipeline, train/test split, and ABTT choices. Table 15 reports, for each model and pooling, the layer chosen on the train split by DirAcc@1, the rule of Table 8, with test-set Task B assignment accuracy, DirAcc@1, and cosine gap at that layer. Mean pooling performs better on assignment accuracy for every model in this experiment, including the decoder-only models.

## F Per-Layer Results for Non-T5 Models

The main text focuses on LaTa, PhilTa, and mT5-base. This appendix reports the same per-layer view for LaBSE, Qwen3-0.6B, and KaLM-mini under the same comparison: baseline mean pooling versus ABTT with  $D$  tuned per layer on the train split.

| Model    | Operational layer | Most anisotropic layer | PC1 before | PC1 after | Eff. rank before→after |
|----------|-------------------|------------------------|------------|-----------|------------------------|
| LaTa     | 7                 | 8                      | 0.956      | 0.036     | 1.34→168.40            |
| PhilTa   | 1                 | 6                      | 0.862      | 0.039     | 1.83→155.02            |
| mT5-base | 1                 | 5                      | 1.000      | 0.044     | 1.00→151.74            |

Table 9: Most anisotropic layers, by top-PC share on the test split, and operational attribution layers for the three T5 encoders. PC1: top-PC share. PC1 and effective-rank values compare baseline geometry with ABTT-D10 at the most anisotropic layer, on the test split. The main text (Table 2) reads top-PC share on the training split, where LaTa peaks at layer 4 instead of 8; LaTa’s training curve is flat, with a top-PC share of 0.93 to 0.95 over layers 3–11. Operational layers come from the train-only retrieval rule of Appendix K.

| Model        | Text    | PC1 <sub>max</sub> ( $\ell$ ) | Layers with PC1 $\geq 0.6$ |             |           |
|--------------|---------|-------------------------------|----------------------------|-------------|-----------|
|              |         |                               | Layers                     | PC1         | Eff. rank |
| mT5-base     | Latin   | 1.000 (5)                     | 4–11                       | 0.893–1.000 | 1.00–2.15 |
|              | English | 1.000 (5)                     | 4–11                       | 0.822–1.000 | 1.00–3.10 |
| PhilTa       | Latin   | 0.858 (6)                     | 3–11                       | 0.764–0.858 | 1.81–4.34 |
|              | English | 0.894 (6)                     | 4–11                       | 0.846–0.894 | 1.82–2.30 |
| T5-v1.1-base | Latin   | 0.975 (9)                     | 2–11                       | 0.958–0.975 | 1.14–1.20 |
|              | English | 0.950 (9)                     | 2–11                       | 0.931–0.950 | 1.28–1.36 |

Table 10: Label-free layer geometry crossed by model and text. Each cell reads the mean-pooled hidden states of 847 passages: the Latin training passages, or the English passages paired with them one to one and matched in mT5 token length (US court opinions, Caselaw Access Project). PC1<sub>max</sub>: peak top-PC share over the 12 layers, the share of centered variance on the first principal component, with its layer  $\ell$ . The last three columns cover the layers whose top-PC share is at least 0.6, the threshold that marks every collapsed Latin layer (one false alarm, mT5-base layer 4), and give the range of top-PC share and of entropy effective rank over them. No labels are used, so the English rows say nothing about retrieval.

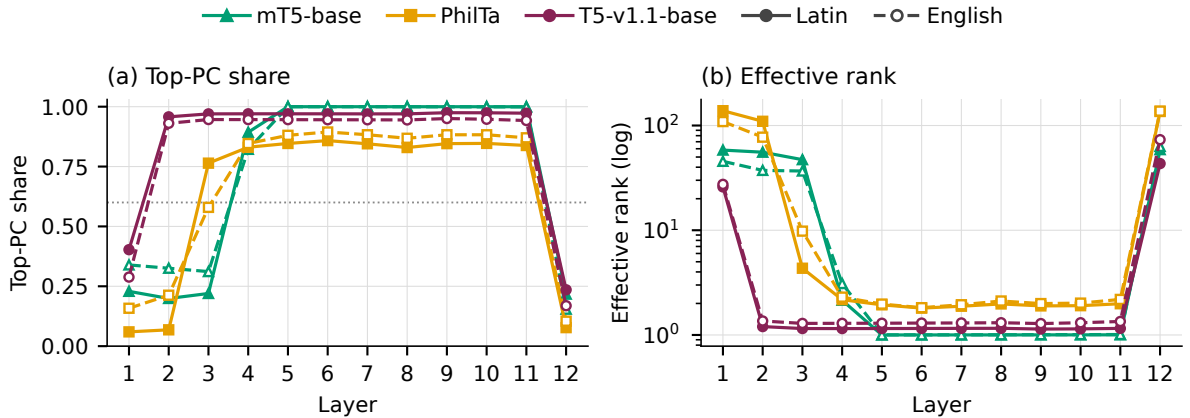


Figure 6: Label-free layer geometry of three T5 encoders on the Latin corpus (solid, filled markers) and on English US court opinions matched to the Latin passages in mT5 token length (dashed, hollow markers), read on 847 passages per cell. (a) Top-PC share, the share of centered variance on the first principal component; the dotted line is the 0.6 threshold that marks every collapsed Latin layer (one false alarm, mT5-base layer 4). (b) Entropy effective rank, log scale. Every model is low-rank from layer 2 to 4 through layer 11 on both texts and recovers at layer 12.

| Model                   | Task A layer |     |      |          | Task B layer |     |      |          |
|-------------------------|--------------|-----|------|----------|--------------|-----|------|----------|
|                         | Base         | SIF | ABTT | SIF+ABTT | Base         | SIF | ABTT | SIF+ABTT |
| LaTa                    | 12           | 1   | 12   | 7        | 1            | 1   | 8    | 12       |
| PhilTa                  | 1            | 1   | 9    | 8        | 1            | 1   | 1    | 1        |
| mT5-base                | 12           | 1   | 2    | 1        | 12           | 1   | 1    | 2        |
| LaBSE                   | 12           | 12  | 11   | 10       | 11           | 12  | 11   | 11       |
| Qwen3-0.6B              | 26           | 22  | 2    | 2        | 28           | 27  | 5    | 5        |
| KaLM-mini               | 23           | 23  | 1    | 2        | 23           | 22  | 3    | 1        |
| LaTa (fine-tuned)       | 12           | –   | 12   | –        | 12           | –   | 12   | –        |
| Qwen3-0.6B (fine-tuned) | 28           | –   | 27   | –        | 28           | –   | 27   | –        |
| KaLM-mini (fine-tuned)  | 24           | –   | 24   | –        | 24           | –   | 24   | –        |

Table 11: Train-selected layer behind every cell of Tables 7 and 8. For each model and post-processing setting, the Task A layer has the highest training-set AUROC and the Task B layer the highest training-set directory accuracy at rank 1; the test scores in those tables are read at these layers, so no layer is chosen on test. Layer 1 is the first transformer block. The fine-tuned rows below the rule have no SIF setting. Table 26 reads its pre-trained rows at the Base and ABTT layers above the rule and its fine-tuned rows at the layers below it.

| Task A: Pairwise Duplicate Detection (main) |       |              |              |                   |                   |
|---------------------------------------------|-------|--------------|--------------|-------------------|-------------------|
| Model                                       | Layer | AUROC (base) | AUROC (ABTT) | Cosine gap (base) | Cosine gap (ABTT) |
| LaTa                                        | 1     | 0.934        | 0.970        | 0.286             | 0.566             |
|                                             | 2     | 0.563        | 0.969        | 0.019             | 0.554             |
|                                             | 3     | 0.513        | 0.973        | -0.048            | 0.555             |
|                                             | 4     | 0.500        | 0.971        | -0.067            | 0.565             |
|                                             | 5     | 0.496        | 0.972        | -0.069            | 0.572             |
|                                             | 6     | 0.496        | 0.964        | -0.069            | 0.569             |
|                                             | 7     | 0.498        | 0.962        | -0.072            | 0.564             |
|                                             | 8     | 0.502        | 0.965        | -0.074            | 0.558             |
|                                             | 9     | 0.503        | 0.965        | -0.073            | 0.557             |
|                                             | 10    | 0.505        | 0.971        | -0.072            | 0.551             |
|                                             | 11    | 0.508        | 0.973        | -0.067            | 0.550             |
|                                             | 12    | <b>0.938</b> | <b>0.971</b> | <b>0.237</b>      | <b>0.525</b>      |
| PhilTa                                      | 1     | 0.939        | 0.977        | 0.231             | 0.574             |
|                                             | 2     | 0.911        | 0.973        | 0.161             | 0.553             |
|                                             | 3     | 0.601        | 0.973        | 0.085             | 0.549             |
|                                             | 4     | 0.551        | 0.978        | 0.063             | 0.551             |
|                                             | 5     | 0.542        | 0.979        | 0.053             | 0.551             |
|                                             | 6     | 0.541        | 0.981        | 0.047             | 0.551             |
|                                             | 7     | 0.542        | 0.979        | 0.046             | 0.554             |
|                                             | 8     | 0.542        | 0.984        | 0.046             | 0.555             |
|                                             | 9     | <b>0.539</b> | <b>0.983</b> | <b>0.041</b>      | <b>0.540</b>      |
|                                             | 10    | 0.538        | 0.982        | 0.040             | 0.531             |
|                                             | 11    | 0.539        | 0.982        | 0.041             | 0.511             |
|                                             | 12    | 0.911        | 0.982        | 0.181             | 0.487             |
| mT5-base                                    | 1     | 0.822        | 0.979        | 0.027             | 0.550             |
|                                             | 2     | <b>0.822</b> | <b>0.975</b> | <b>0.026</b>      | <b>0.536</b>      |
|                                             | 3     | 0.804        | 0.968        | 0.025             | 0.517             |
|                                             | 4     | 0.799        | 0.974        | 0.040             | 0.502             |
|                                             | 5     | 0.654        | 0.978        | 0.219             | 0.482             |
|                                             | 6     | 0.657        | 0.978        | 0.238             | 0.483             |
|                                             | 7     | 0.656        | 0.980        | 0.244             | 0.482             |
|                                             | 8     | 0.656        | 0.982        | 0.249             | 0.471             |
|                                             | 9     | 0.660        | 0.982        | 0.235             | 0.475             |
|                                             | 10    | 0.659        | 0.977        | 0.226             | 0.459             |
|                                             | 11    | 0.659        | 0.974        | 0.213             | 0.449             |
|                                             | 12    | 0.838        | 0.972        | 0.081             | 0.442             |

Table 12: Per-layer Task A pairwise duplicate-detection metrics for the three T5 encoders (LaTa, PhilTa, mT5-base). baseline is mean pooling without correction; abtt\_optimal applies ABTT (no SIF weighting) with  $D$  tuned per layer on the train split. Both AUROC and cosine gap are computed over the test  $n \times n$  cosine matrix with no threshold and no directory routing: AUROC treats same-directory test pairs as positives, and cosine gap is the difference between mean same-directory and mean different-directory cosine. Rows in bold mark the layer chosen on the train split, the layer with the highest training-set AUROC under ABTT. It is the layer behind the ABTT cell of Table 7 (listed in Table 11), and not always the layer with the highest test score in this table.

| Task B: Autonomous Routing (main, file-level, $\tau$ -thresholded) |          |                    |                    |               |               |                   |                   |
|--------------------------------------------------------------------|----------|--------------------|--------------------|---------------|---------------|-------------------|-------------------|
| Model                                                              | Layer    | Existing<br>(base) | Existing<br>(ABTT) | New<br>(base) | New<br>(ABTT) | Overall<br>(base) | Overall<br>(ABTT) |
| LaTa                                                               | 1        | 0.630              | 0.910              | 0.916         | 0.907         | 0.738             | 0.909             |
|                                                                    | 2        | 0.153              | 0.905              | 0.913         | 0.892         | 0.439             | 0.900             |
|                                                                    | 3        | 0.256              | 0.882              | 0.842         | 0.950         | 0.477             | 0.908             |
|                                                                    | 4        | 0.389              | 0.893              | 0.718         | 0.920         | 0.513             | 0.903             |
|                                                                    | 5        | 0.374              | 0.892              | 0.715         | 0.913         | 0.502             | 0.900             |
|                                                                    | 6        | 0.378              | 0.884              | 0.712         | 0.913         | 0.503             | 0.895             |
|                                                                    | 7        | 0.372              | 0.875              | 0.715         | 0.916         | 0.501             | 0.890             |
|                                                                    | <b>8</b> | <b>0.379</b>       | <b>0.884</b>       | <b>0.706</b>  | <b>0.885</b>  | <b>0.502</b>      | <b>0.885</b>      |
|                                                                    | 9        | 0.366              | 0.886              | 0.724         | 0.873         | 0.501             | 0.881             |
|                                                                    | 10       | 0.344              | 0.873              | 0.752         | 0.882         | 0.498             | 0.876             |
|                                                                    | 11       | 0.310              | 0.867              | 0.786         | 0.901         | 0.490             | 0.880             |
|                                                                    | 12       | 0.576              | 0.890              | 0.932         | 0.833         | 0.710             | 0.868             |
| PhilTa                                                             | <b>1</b> | <b>0.591</b>       | <b>0.918</b>       | <b>0.904</b>  | <b>0.901</b>  | <b>0.709</b>      | <b>0.911</b>      |
|                                                                    | 2        | 0.479              | 0.895              | 0.867         | 0.885         | 0.625             | 0.892             |
|                                                                    | 3        | 0.193              | 0.897              | 0.889         | 0.879         | 0.455             | 0.890             |
|                                                                    | 4        | 0.148              | 0.886              | 0.923         | 0.873         | 0.439             | 0.881             |
|                                                                    | 5        | 0.252              | 0.879              | 0.796         | 0.873         | 0.457             | 0.876             |
|                                                                    | 6        | 0.338              | 0.905              | 0.700         | 0.817         | 0.474             | 0.872             |
|                                                                    | 7        | 0.301              | 0.901              | 0.746         | 0.830         | 0.469             | 0.874             |
|                                                                    | 8        | 0.262              | 0.875              | 0.808         | 0.885         | 0.467             | 0.879             |
|                                                                    | 9        | 0.303              | 0.890              | 0.762         | 0.870         | 0.476             | 0.882             |
|                                                                    | 10       | 0.299              | 0.908              | 0.780         | 0.793         | 0.480             | 0.865             |
|                                                                    | 11       | 0.256              | 0.869              | 0.824         | 0.836         | 0.470             | 0.857             |
|                                                                    | 12       | 0.338              | 0.880              | 0.898         | 0.876         | 0.549             | 0.879             |
| mT5-base                                                           | <b>1</b> | <b>0.135</b>       | <b>0.877</b>       | <b>0.981</b>  | <b>0.947</b>  | <b>0.453</b>      | <b>0.903</b>      |
|                                                                    | 2        | 0.110              | 0.890              | 0.975         | 0.910         | 0.436             | 0.897             |
|                                                                    | 3        | 0.161              | 0.794              | 0.926         | 0.941         | 0.449             | 0.850             |
|                                                                    | 4        | 0.269              | 0.787              | 0.858         | 0.882         | 0.491             | 0.823             |
|                                                                    | 5        | 0.422              | 0.746              | 0.659         | 0.864         | 0.512             | 0.790             |
|                                                                    | 6        | 0.636              | 0.766              | 0.489         | 0.876         | 0.580             | 0.808             |
|                                                                    | 7        | 0.594              | 0.793              | 0.517         | 0.851         | 0.565             | 0.815             |
|                                                                    | 8        | 0.557              | 0.809              | 0.560         | 0.842         | 0.558             | 0.822             |
|                                                                    | 9        | 0.364              | 0.841              | 0.746         | 0.814         | 0.508             | 0.831             |
|                                                                    | 10       | 0.314              | 0.794              | 0.783         | 0.876         | 0.491             | 0.825             |
|                                                                    | 11       | 0.297              | 0.763              | 0.780         | 0.854         | 0.479             | 0.797             |
|                                                                    | 12       | 0.198              | 0.834              | 0.926         | 0.799         | 0.472             | 0.821             |

Table 13: Per-layer Task B autonomous routing accuracy for the three T5 encoders. baseline is mean pooling without correction; abtt\_optimal applies ABTT (no SIF weighting) with  $D$  tuned per layer on the train split. For every test file we compute  $s_i = \max_{j \neq i} \cos(e_i, e_j)$  against the rest of the test set and compare to the learned threshold  $\tau$  (fit on train via  $F_1$ -optimal cut on same-directory vs. different-directory pairs). The decision is binary first (existing if  $s_i \geq \tau$ , else new); only files predicted existing are then routed to their top-1 neighbour’s directory. We report **existing accuracy** (restricted to files whose true partner is in the test set), **new accuracy** (files that should be flagged as novel), and **overall assignment accuracy**. Rows in bold mark the layer chosen on the train split, the layer with the highest training-set directory accuracy at rank 1 under ABTT. It is the layer behind the ABTT cell of Table 8 (listed in Table 11), and not always the layer with the highest test score in this table.

| Task B: Top-K Ranking (main, single-seed) |          |                 |                 |                    |                    |               |               |
|-------------------------------------------|----------|-----------------|-----------------|--------------------|--------------------|---------------|---------------|
| Model                                     | Layer    | Acc@1<br>(base) | Acc@1<br>(ABTT) | Existing<br>(base) | Existing<br>(ABTT) | New<br>(base) | New<br>(ABTT) |
| LaTa                                      | 1        | 0.721           | 0.880           | 0.630              | 0.910              | 0.916         | 0.907         |
|                                           | 2        | 0.383           | 0.869           | 0.153              | 0.905              | 0.913         | 0.892         |
|                                           | 3        | 0.350           | 0.882           | 0.256              | 0.882              | 0.842         | 0.950         |
|                                           | 4        | 0.303           | 0.881           | 0.389              | 0.893              | 0.718         | 0.920         |
|                                           | 5        | 0.297           | 0.875           | 0.374              | 0.892              | 0.715         | 0.913         |
|                                           | 6        | 0.297           | 0.874           | 0.378              | 0.884              | 0.712         | 0.913         |
|                                           | 7        | 0.302           | 0.868           | 0.372              | 0.875              | 0.715         | 0.916         |
|                                           | <b>8</b> | <b>0.302</b>    | <b>0.861</b>    | <b>0.379</b>       | <b>0.884</b>       | <b>0.706</b>  | <b>0.885</b>  |
|                                           | 9        | 0.309           | 0.858           | 0.366              | 0.886              | 0.724         | 0.873         |
|                                           | 10       | 0.317           | 0.852           | 0.344              | 0.873              | 0.752         | 0.882         |
|                                           | 11       | 0.331           | 0.857           | 0.310              | 0.867              | 0.786         | 0.901         |
|                                           | 12       | 0.689           | 0.841           | 0.576              | 0.890              | 0.932         | 0.833         |
| PhilTa                                    | <b>1</b> | <b>0.693</b>    | <b>0.883</b>    | <b>0.591</b>       | <b>0.918</b>       | <b>0.904</b>  | <b>0.901</b>  |
|                                           | 2        | 0.603           | 0.869           | 0.479              | 0.895              | 0.867         | 0.885         |
|                                           | 3        | 0.389           | 0.865           | 0.193              | 0.897              | 0.889         | 0.879         |
|                                           | 4        | 0.369           | 0.859           | 0.148              | 0.886              | 0.923         | 0.873         |
|                                           | 5        | 0.334           | 0.854           | 0.252              | 0.879              | 0.796         | 0.873         |
|                                           | 6        | 0.310           | 0.845           | 0.338              | 0.905              | 0.700         | 0.817         |
|                                           | 7        | 0.325           | 0.847           | 0.301              | 0.901              | 0.746         | 0.830         |
|                                           | 8        | 0.350           | 0.857           | 0.262              | 0.875              | 0.808         | 0.885         |
|                                           | 9        | 0.337           | 0.857           | 0.303              | 0.890              | 0.762         | 0.870         |
|                                           | 10       | 0.344           | 0.833           | 0.299              | 0.908              | 0.780         | 0.793         |
|                                           | 11       | 0.355           | 0.829           | 0.256              | 0.869              | 0.824         | 0.836         |
|                                           | 12       | 0.528           | 0.854           | 0.338              | 0.880              | 0.898         | 0.876         |
| mT5-base                                  | <b>1</b> | <b>0.451</b>    | <b>0.885</b>    | <b>0.135</b>       | <b>0.877</b>       | <b>0.981</b>  | <b>0.947</b>  |
|                                           | 2        | 0.435           | 0.874           | 0.110              | 0.890              | 0.975         | 0.910         |
|                                           | 3        | 0.445           | 0.830           | 0.161              | 0.794              | 0.926         | 0.941         |
|                                           | 4        | 0.467           | 0.798           | 0.269              | 0.787              | 0.858         | 0.882         |
|                                           | 5        | 0.352           | 0.756           | 0.422              | 0.746              | 0.659         | 0.864         |
|                                           | 6        | 0.346           | 0.775           | 0.636              | 0.766              | 0.489         | 0.876         |
|                                           | 7        | 0.352           | 0.775           | 0.594              | 0.793              | 0.517         | 0.851         |
|                                           | 8        | 0.366           | 0.784           | 0.557              | 0.809              | 0.560         | 0.842         |
|                                           | 9        | 0.394           | 0.787           | 0.364              | 0.841              | 0.746         | 0.814         |
|                                           | 10       | 0.387           | 0.788           | 0.314              | 0.794              | 0.783         | 0.876         |
|                                           | 11       | 0.378           | 0.765           | 0.297              | 0.763              | 0.780         | 0.854         |
|                                           | 12       | 0.466           | 0.779           | 0.198              | 0.834              | 0.926         | 0.799         |

Table 14: Per-layer Task B top- $k$  ranking metrics for the three T5 encoders. baseline is mean pooling without correction; abtt\_optimal applies ABTT (no SIF weighting) with  $D$  tuned per layer on the train split. This table is computed on the single-seed split (no  $M$ -seed averaging, since the mseed sweep was run only for SIF-conditioned variants; see Table 23 for the multi-seed SIF+ABTT view). **Acc@1** is directory accuracy at rank 1, with an existing/new decomposition. Rows in bold mark the layer chosen on the train split, the layer with the highest training-set directory accuracy at rank 1 under ABTT. It is the layer behind the ABTT cell of Table 8 (listed in Table 11), and not always the layer with the highest test score in this table.

| Model      | Pooling    | Layer | $D$ | Assignment acc. | Dir. acc. @1 | Cosine gap |
|------------|------------|-------|-----|-----------------|--------------|------------|
| LaTa       | Mean       | 8     | 10  | 0.885           | 0.861        | 0.558      |
|            | Last token | 7     | 10  | 0.664           | 0.466        | 0.308      |
| PhilTa     | Mean       | 1     | 10  | 0.911           | 0.883        | 0.573      |
|            | Last token | 11    | 10  | 0.589           | 0.406        | 0.205      |
| LaBSE      | Mean       | 11    | 10  | 0.908           | 0.885        | 0.610      |
|            | Last token | 12    | 10  | 0.895           | 0.871        | 0.586      |
| Qwen3-0.6B | Mean       | 5     | 10  | 0.915           | 0.894        | 0.545      |
|            | Last token | 26    | 5   | 0.881           | 0.864        | 0.580      |
| KaLM-mini  | Mean       | 3     | 10  | 0.917           | 0.894        | 0.552      |
|            | Last token | 24    | 10  | 0.875           | 0.847        | 0.483      |

Table 15: Mean versus last-token pooling under ABTT with  $D$  tuned per layer on the train split, for the five paper models whose last-token embeddings were extracted (mT5-base was not part of this run). For each model and pooling the layer is the one with the highest training-set directory accuracy at rank 1, the rule behind the ABTT cells of Table 8 (their layers are listed in Table 11);  $D$  is the number of principal components removed at that layer. The three score columns are test-set Task B assignment accuracy, Task B directory accuracy at rank 1, and Task A cosine gap at that layer.

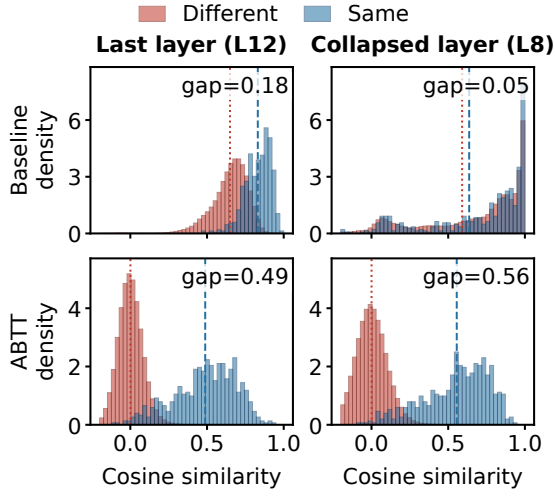


Figure 7: Cosine similarity distributions for PhilTa at the last layer and at its collapsed retrieval layer, before and after ABTT. Equivalent pairs share a directory (Same in the legend); non-equivalent pairs do not (Different). The collapsed retrieval layer is layer 8, chosen by a train-only rule (highest training-set AUROC within 30 to 70 percent of encoder depth, layers 4 to 8), so the panel understates the collapse rather than selecting its worst case; its test AUROC is 0.542, close to chance. Two other PhilTa layers carry other meanings and are not shown here: the most anisotropic layer under the top-PC rule of Appendix B is layer 6, and the lowest baseline test AUROC, 0.538, falls at layer 10. Before ABTT the two distributions sit almost on top of each other, and ABTT separates them.

Table 16: Per-layer Task A pairwise metrics for the non-T5 models (LaBSE, Qwen3-0.6B, KaLM-mini), under the same two-method comparison as the main paper. baseline is mean pooling without correction; abtt\_optimal applies ABTT (no SIF weighting) with  $D$  tuned per layer on the train split. Both AUROC and cosine gap are computed over the test  $n \times n$  cosine matrix with no threshold and no directory routing: AUROC treats same-directory test pairs as positives, and cosine gap is the difference between mean same-directory and mean different-directory cosine. Rows in bold mark the layer chosen on the train split, the layer with the highest training-set AUROC under ABTT. It is the layer behind the ABTT cell of Table 7 (listed in Table 11), and not always the layer with the highest test score in this table.

| Task A: Pairwise Duplicate Detection (appendix models) |           |                 |                 |                      |                      |
|--------------------------------------------------------|-----------|-----------------|-----------------|----------------------|----------------------|
| Model                                                  | Layer     | AUROC<br>(base) | AUROC<br>(ABTT) | Cosine gap<br>(base) | Cosine gap<br>(ABTT) |
| LaBSE                                                  | 1         | 0.806           | 0.977           | 0.060                | 0.520                |
|                                                        | 2         | 0.833           | 0.979           | 0.037                | 0.531                |
|                                                        | 3         | 0.853           | 0.978           | 0.062                | 0.543                |
|                                                        | 4         | 0.862           | 0.977           | 0.073                | 0.525                |
|                                                        | 5         | 0.864           | 0.968           | 0.066                | 0.501                |
|                                                        | 6         | 0.894           | 0.976           | 0.078                | 0.532                |
|                                                        | 7         | 0.887           | 0.975           | 0.088                | 0.524                |
|                                                        | 8         | 0.879           | 0.975           | 0.080                | 0.522                |
|                                                        | 9         | 0.916           | 0.979           | 0.090                | 0.565                |
|                                                        | 10        | 0.934           | 0.982           | 0.103                | 0.586                |
|                                                        | <b>11</b> | <b>0.960</b>    | <b>0.987</b>    | <b>0.061</b>         | <b>0.611</b>         |
|                                                        | 12        | 0.956           | 0.981           | 0.183                | 0.592                |
| Qwen3-0.6B                                             | 1         | 0.858           | 0.978           | 0.033                | 0.564                |
|                                                        | <b>2</b>  | <b>0.863</b>    | <b>0.973</b>    | <b>0.024</b>         | <b>0.553</b>         |
|                                                        | 3         | 0.862           | 0.980           | 0.037                | 0.556                |
|                                                        | 4         | 0.860           | 0.986           | 0.036                | 0.547                |
|                                                        | 5         | 0.861           | 0.981           | 0.030                | 0.545                |
|                                                        | 6         | 0.867           | 0.983           | 0.035                | 0.545                |
|                                                        | 7         | 0.876           | 0.980           | 0.033                | 0.544                |
|                                                        | 8         | 0.882           | 0.972           | 0.037                | 0.532                |
|                                                        | 9         | 0.881           | 0.977           | 0.034                | 0.532                |
|                                                        | 10        | 0.887           | 0.976           | 0.031                | 0.529                |
|                                                        | 11        | 0.895           | 0.977           | 0.030                | 0.530                |
|                                                        | 12        | 0.901           | 0.973           | 0.027                | 0.530                |
|                                                        | 13        | 0.907           | 0.974           | 0.026                | 0.523                |
|                                                        | 14        | 0.905           | 0.974           | 0.024                | 0.519                |
|                                                        | 15        | 0.901           | 0.971           | 0.022                | 0.514                |
|                                                        | 16        | 0.910           | 0.975           | 0.026                | 0.509                |
|                                                        | 17        | 0.916           | 0.979           | 0.032                | 0.521                |
|                                                        | 18        | 0.933           | 0.983           | 0.032                | 0.537                |
|                                                        | 19        | 0.936           | 0.980           | 0.035                | 0.541                |
|                                                        | 20        | 0.943           | 0.982           | 0.041                | 0.559                |
|                                                        | 21        | 0.953           | 0.982           | 0.047                | 0.562                |
|                                                        | 22        | 0.963           | 0.983           | 0.044                | 0.583                |
|                                                        | 23        | 0.961           | 0.984           | 0.038                | 0.578                |
|                                                        | 24        | 0.963           | 0.984           | 0.032                | 0.576                |
|                                                        | 25        | 0.961           | 0.983           | 0.027                | 0.578                |
|                                                        | 26        | 0.966           | 0.985           | 0.024                | 0.568                |
|                                                        | 27        | 0.967           | 0.982           | 0.017                | 0.559                |
|                                                        | 28        | 0.958           | 0.975           | 0.098                | 0.556                |
| KaLM-mini                                              | <b>1</b>  | <b>0.887</b>    | <b>0.981</b>    | <b>0.063</b>         | <b>0.568</b>         |
|                                                        | 2         | 0.906           | 0.983           | 0.044                | 0.566                |
|                                                        | 3         | 0.881           | 0.981           | 0.062                | 0.551                |
|                                                        | 4         | 0.871           | 0.983           | 0.060                | 0.553                |
|                                                        | 5         | 0.861           | 0.983           | 0.062                | 0.541                |
|                                                        | 6         | 0.872           | 0.983           | 0.063                | 0.529                |
|                                                        | 7         | 0.888           | 0.979           | 0.072                | 0.524                |
|                                                        | 8         | 0.890           | 0.976           | 0.072                | 0.524                |
|                                                        | 9         | 0.892           | 0.974           | 0.066                | 0.507                |
|                                                        | 10        | 0.893           | 0.971           | 0.067                | 0.494                |
|                                                        | 11        | 0.894           | 0.970           | 0.061                | 0.475                |
|                                                        | 12        | 0.905           | 0.970           | 0.071                | 0.475                |
|                                                        | 13        | 0.900           | 0.968           | 0.066                | 0.467                |

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| Task A: Pairwise Duplicate Detection (appendix models) |       |                 |                 |                      |                      |
|--------------------------------------------------------|-------|-----------------|-----------------|----------------------|----------------------|
| Model                                                  | Layer | AUROC<br>(base) | AUROC<br>(ABTT) | Cosine gap<br>(base) | Cosine gap<br>(ABTT) |
|                                                        | 14    | 0.904           | 0.971           | 0.063                | 0.471                |
|                                                        | 15    | 0.907           | 0.973           | 0.067                | 0.468                |
|                                                        | 16    | 0.917           | 0.972           | 0.071                | 0.489                |
|                                                        | 17    | 0.926           | 0.975           | 0.075                | 0.501                |
|                                                        | 18    | 0.945           | 0.977           | 0.060                | 0.519                |
|                                                        | 19    | 0.956           | 0.979           | 0.061                | 0.531                |
|                                                        | 20    | 0.960           | 0.980           | 0.054                | 0.555                |
|                                                        | 21    | 0.962           | 0.981           | 0.043                | 0.555                |
|                                                        | 22    | 0.971           | 0.985           | 0.050                | 0.555                |
|                                                        | 23    | 0.972           | 0.986           | 0.056                | 0.552                |
|                                                        | 24    | 0.956           | 0.981           | 0.243                | 0.520                |

Table 17: Per-layer Task B autonomous routing accuracy for the non-T5 models. baseline is mean pooling without correction; abtt\_optimal applies ABTT (no SIF weighting) with  $D$  tuned per layer on the train split. For every test file we compute  $s_i = \max_{j \neq i} \cos(e_i, e_j)$  against the rest of the test set and compare to the learned threshold  $\tau$  (fit on train via  $F_1$ -optimal cut on same-directory vs. different-directory pairs). The decision is binary first (existing if  $s_i \geq \tau$ , else new); only files predicted existing are then routed to their top-1 neighbour’s directory. We report **existing accuracy** (restricted to files whose true partner is in the test set), **new accuracy** (files that should be flagged as novel), and **overall assignment accuracy**. Rows in bold mark the layer chosen on the train split, the layer with the highest training-set directory accuracy at rank 1 under ABTT. It is the layer behind the ABTT cell of Table 8 (listed in Table 11), and not always the layer with the highest test score in this table.

| Task B: Autonomous Routing (appendix models) |           |                    |                    |               |               |                   |                   |
|----------------------------------------------|-----------|--------------------|--------------------|---------------|---------------|-------------------|-------------------|
| Model                                        | Layer     | Existing<br>(base) | Existing<br>(ABTT) | New<br>(base) | New<br>(ABTT) | Overall<br>(base) | Overall<br>(ABTT) |
| LaBSE                                        | 1         | 0.222              | 0.879              | 0.898         | 0.873         | 0.477             | 0.876             |
|                                              | 2         | 0.381              | 0.875              | 0.811         | 0.827         | 0.543             | 0.857             |
|                                              | 3         | 0.277              | 0.886              | 0.907         | 0.867         | 0.514             | 0.879             |
|                                              | 4         | 0.372              | 0.897              | 0.885         | 0.836         | 0.565             | 0.874             |
|                                              | 5         | 0.327              | 0.845              | 0.901         | 0.851         | 0.543             | 0.847             |
|                                              | 6         | 0.548              | 0.860              | 0.882         | 0.851         | 0.674             | 0.857             |
|                                              | 7         | 0.398              | 0.849              | 0.892         | 0.830         | 0.584             | 0.841             |
|                                              | 8         | 0.385              | 0.869              | 0.870         | 0.743         | 0.568             | 0.822             |
|                                              | 9         | 0.421              | 0.871              | 0.929         | 0.817         | 0.612             | 0.851             |
|                                              | 10        | 0.606              | 0.892              | 0.892         | 0.805         | 0.713             | 0.859             |
|                                              | <b>11</b> | <b>0.785</b>       | <b>0.923</b>       | <b>0.913</b>  | <b>0.882</b>  | <b>0.833</b>      | <b>0.908</b>      |
|                                              | 12        | 0.729              | 0.914              | 0.938         | 0.864         | 0.808             | 0.895             |
| Qwen3-0.6B                                   | 1         | 0.264              | 0.921              | 0.960         | 0.885         | 0.526             | 0.908             |
|                                              | 2         | 0.437              | 0.869              | 0.814         | 0.916         | 0.579             | 0.887             |
|                                              | 3         | 0.250              | 0.914              | 0.947         | 0.907         | 0.513             | 0.911             |
|                                              | 4         | 0.411              | 0.936              | 0.848         | 0.842         | 0.576             | 0.901             |
|                                              | <b>5</b>  | <b>0.181</b>       | <b>0.910</b>       | <b>0.932</b>  | <b>0.923</b>  | <b>0.464</b>      | <b>0.915</b>      |
|                                              | 6         | 0.422              | 0.935              | 0.830         | 0.796         | 0.576             | 0.882             |
|                                              | 7         | 0.250              | 0.890              | 0.947         | 0.876         | 0.513             | 0.885             |
|                                              | 8         | 0.409              | 0.931              | 0.895         | 0.799         | 0.592             | 0.881             |
|                                              | 9         | 0.460              | 0.901              | 0.851         | 0.851         | 0.607             | 0.882             |
|                                              | 10        | 0.533              | 0.892              | 0.836         | 0.864         | 0.647             | 0.881             |
|                                              | 11        | 0.293              | 0.903              | 0.978         | 0.808         | 0.551             | 0.867             |
|                                              | 12        | 0.407              | 0.912              | 0.938         | 0.817         | 0.607             | 0.876             |
|                                              | 13        | 0.432              | 0.908              | 0.944         | 0.836         | 0.625             | 0.881             |
|                                              | 14        | 0.456              | 0.907              | 0.926         | 0.811         | 0.633             | 0.871             |
|                                              | 15        | 0.469              | 0.901              | 0.920         | 0.845         | 0.639             | 0.880             |
|                                              | 16        | 0.639              | 0.897              | 0.830         | 0.808         | 0.711             | 0.864             |
|                                              | 17        | 0.480              | 0.886              | 0.935         | 0.851         | 0.652             | 0.873             |
|                                              | 18        | 0.553              | 0.882              | 0.944         | 0.839         | 0.700             | 0.866             |
|                                              | 19        | 0.514              | 0.871              | 0.966         | 0.858         | 0.684             | 0.866             |
|                                              | 20        | 0.654              | 0.865              | 0.898         | 0.892         | 0.746             | 0.875             |
|                                              | 21        | 0.647              | 0.912              | 0.950         | 0.898         | 0.761             | 0.907             |
|                                              | 22        | 0.731              | 0.888              | 0.938         | 0.916         | 0.809             | 0.899             |
|                                              | 23        | 0.622              | 0.914              | 0.966         | 0.895         | 0.752             | 0.907             |

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| Task B: Autonomous Routing (appendix models) |          |                    |                    |               |               |                   |                   |
|----------------------------------------------|----------|--------------------|--------------------|---------------|---------------|-------------------|-------------------|
| Model                                        | Layer    | Existing<br>(base) | Existing<br>(ABTT) | New<br>(base) | New<br>(ABTT) | Overall<br>(base) | Overall<br>(ABTT) |
|                                              | 24       | 0.748              | 0.901              | 0.935         | 0.935         | 0.818             | 0.914             |
|                                              | 25       | 0.574              | 0.893              | 0.975         | 0.923         | 0.725             | 0.904             |
|                                              | 26       | 0.613              | 0.899              | 0.975         | 0.895         | 0.749             | 0.897             |
|                                              | 27       | 0.793              | 0.877              | 0.833         | 0.923         | 0.808             | 0.894             |
|                                              | 28       | 0.796              | 0.877              | 0.867         | 0.913         | 0.823             | 0.890             |
| KaLM-mini                                    | 1        | 0.391              | 0.921              | 0.907         | 0.920         | 0.585             | 0.921             |
|                                              | 2        | 0.493              | 0.916              | 0.848         | 0.879         | 0.627             | 0.902             |
|                                              | <b>3</b> | <b>0.443</b>       | <b>0.923</b>       | <b>0.870</b>  | <b>0.907</b>  | <b>0.604</b>      | <b>0.917</b>      |
|                                              | 4        | 0.308              | 0.931              | 0.935         | 0.895         | 0.544             | 0.917             |
|                                              | 5        | 0.338              | 0.905              | 0.882         | 0.923         | 0.543             | 0.911             |
|                                              | 6        | 0.256              | 0.895              | 0.910         | 0.920         | 0.502             | 0.904             |
|                                              | 7        | 0.475              | 0.864              | 0.854         | 0.944         | 0.618             | 0.894             |
|                                              | 8        | 0.467              | 0.862              | 0.870         | 0.913         | 0.619             | 0.881             |
|                                              | 9        | 0.469              | 0.873              | 0.885         | 0.870         | 0.626             | 0.872             |
|                                              | 10       | 0.449              | 0.875              | 0.885         | 0.870         | 0.613             | 0.873             |
|                                              | 11       | 0.458              | 0.862              | 0.885         | 0.864         | 0.619             | 0.862             |
|                                              | 12       | 0.507              | 0.862              | 0.895         | 0.904         | 0.653             | 0.878             |
|                                              | 13       | 0.540              | 0.860              | 0.895         | 0.870         | 0.674             | 0.864             |
|                                              | 14       | 0.593              | 0.836              | 0.845         | 0.885         | 0.688             | 0.854             |
|                                              | 15       | 0.566              | 0.864              | 0.898         | 0.827         | 0.691             | 0.850             |
|                                              | 16       | 0.550              | 0.849              | 0.935         | 0.830         | 0.695             | 0.841             |
|                                              | 17       | 0.675              | 0.864              | 0.920         | 0.889         | 0.767             | 0.873             |
|                                              | 18       | 0.697              | 0.875              | 0.960         | 0.898         | 0.796             | 0.883             |
|                                              | 19       | 0.721              | 0.873              | 0.966         | 0.892         | 0.814             | 0.880             |
|                                              | 20       | 0.705              | 0.791              | 0.966         | 0.913         | 0.803             | 0.837             |
|                                              | 21       | 0.727              | 0.811              | 0.966         | 0.926         | 0.817             | 0.854             |
|                                              | 22       | 0.850              | 0.879              | 0.916         | 0.923         | 0.875             | 0.895             |
|                                              | 23       | 0.849              | 0.884              | 0.920         | 0.898         | 0.875             | 0.889             |
|                                              | 24       | 0.787              | 0.892              | 0.944         | 0.867         | 0.846             | 0.882             |

Table 18: Per-layer Task B top- $k$  ranking metrics for the non-T5 models. baseline is mean pooling without correction; abtt\_optimal applies ABTT (no SIF weighting) with  $D$  tuned per layer on the train split. This table is computed on the single-seed split (no  $M$ -seed averaging, since the mseed sweep was run only for SIF-conditioned variants; see Table 23 for the multi-seed SIF+ABTT view). **Acc@1** is directory accuracy at rank 1, with an existing/new decomposition. Rows in bold mark the layer chosen on the train split, the layer with the highest training-set directory accuracy at rank 1 under ABTT. It is the layer behind the ABTT cell of Table 8 (listed in Table 11), and not always the layer with the highest test score in this table.

| Task B: Top-K Ranking (appendix models, single-seed) |           |                 |                 |                    |                    |               |               |
|------------------------------------------------------|-----------|-----------------|-----------------|--------------------|--------------------|---------------|---------------|
| Model                                                | Layer     | Acc@1<br>(base) | Acc@1<br>(ABTT) | Existing<br>(base) | Existing<br>(ABTT) | New<br>(base) | New<br>(ABTT) |
| LaBSE                                                | 1         | 0.465           | 0.851           | 0.222              | 0.879              | 0.898         | 0.873         |
|                                                      | 2         | 0.494           | 0.830           | 0.381              | 0.875              | 0.811         | 0.827         |
|                                                      | 3         | 0.500           | 0.848           | 0.277              | 0.886              | 0.907         | 0.867         |
|                                                      | 4         | 0.542           | 0.841           | 0.372              | 0.897              | 0.885         | 0.836         |
|                                                      | 5         | 0.521           | 0.811           | 0.327              | 0.845              | 0.901         | 0.851         |
|                                                      | 6         | 0.640           | 0.831           | 0.548              | 0.860              | 0.882         | 0.851         |
|                                                      | 7         | 0.561           | 0.810           | 0.398              | 0.849              | 0.892         | 0.830         |
|                                                      | 8         | 0.544           | 0.779           | 0.385              | 0.869              | 0.870         | 0.743         |
|                                                      | 9         | 0.608           | 0.823           | 0.421              | 0.871              | 0.929         | 0.817         |
|                                                      | 10        | 0.693           | 0.832           | 0.606              | 0.892              | 0.892         | 0.805         |
|                                                      | <b>11</b> | <b>0.815</b>    | <b>0.885</b>    | <b>0.785</b>       | <b>0.923</b>       | <b>0.913</b>  | <b>0.882</b>  |
|                                                      | 12        | 0.791           | 0.868           | 0.729              | 0.914              | 0.938         | 0.864         |
| Qwen3-0.6B                                           | 1         | 0.521           | 0.875           | 0.264              | 0.921              | 0.960         | 0.885         |
|                                                      | 2         | 0.558           | 0.871           | 0.437              | 0.869              | 0.814         | 0.916         |
|                                                      | 3         | 0.508           | 0.890           | 0.250              | 0.914              | 0.947         | 0.907         |
|                                                      | 4         | 0.559           | 0.874           | 0.411              | 0.936              | 0.848         | 0.842         |
|                                                      | <b>5</b>  | <b>0.450</b>    | <b>0.894</b>    | <b>0.181</b>       | <b>0.910</b>       | <b>0.932</b>  | <b>0.923</b>  |
|                                                      | 6         | 0.551           | 0.858           | 0.422              | 0.935              | 0.830         | 0.796         |
|                                                      | 7         | 0.507           | 0.859           | 0.250              | 0.890              | 0.947         | 0.876         |

(continued on next page)

| Task B: Top-K Ranking (appendix models, single-seed) |       |                 |                 |                    |                    |               |               |
|------------------------------------------------------|-------|-----------------|-----------------|--------------------|--------------------|---------------|---------------|
| Model                                                | Layer | Acc@1<br>(base) | Acc@1<br>(ABTT) | Existing<br>(base) | Existing<br>(ABTT) | New<br>(base) | New<br>(ABTT) |
|                                                      | 8     | 0.580           | 0.852           | 0.409              | 0.931              | 0.895         | 0.799         |
|                                                      | 9     | 0.587           | 0.852           | 0.460              | 0.901              | 0.851         | 0.851         |
|                                                      | 10    | 0.622           | 0.848           | 0.533              | 0.892              | 0.836         | 0.864         |
|                                                      | 11    | 0.548           | 0.830           | 0.293              | 0.903              | 0.978         | 0.808         |
|                                                      | 12    | 0.603           | 0.848           | 0.407              | 0.912              | 0.938         | 0.817         |
|                                                      | 13    | 0.618           | 0.851           | 0.432              | 0.908              | 0.944         | 0.836         |
|                                                      | 14    | 0.621           | 0.840           | 0.456              | 0.907              | 0.926         | 0.811         |
|                                                      | 15    | 0.627           | 0.848           | 0.469              | 0.901              | 0.920         | 0.845         |
|                                                      | 16    | 0.682           | 0.830           | 0.639              | 0.897              | 0.830         | 0.808         |
|                                                      | 17    | 0.641           | 0.840           | 0.480              | 0.886              | 0.935         | 0.851         |
|                                                      | 18    | 0.691           | 0.837           | 0.553              | 0.882              | 0.944         | 0.839         |
|                                                      | 19    | 0.676           | 0.837           | 0.514              | 0.871              | 0.966         | 0.858         |
|                                                      | 20    | 0.733           | 0.852           | 0.654              | 0.865              | 0.898         | 0.892         |
|                                                      | 21    | 0.752           | 0.881           | 0.647              | 0.912              | 0.950         | 0.898         |
|                                                      | 22    | 0.795           | 0.876           | 0.731              | 0.888              | 0.938         | 0.916         |
|                                                      | 23    | 0.742           | 0.886           | 0.622              | 0.914              | 0.966         | 0.895         |
|                                                      | 24    | 0.808           | 0.896           | 0.748              | 0.901              | 0.935         | 0.935         |
|                                                      | 25    | 0.718           | 0.886           | 0.574              | 0.893              | 0.975         | 0.923         |
|                                                      | 26    | 0.740           | 0.881           | 0.613              | 0.899              | 0.975         | 0.895         |
|                                                      | 27    | 0.787           | 0.876           | 0.793              | 0.877              | 0.833         | 0.923         |
|                                                      | 28    | 0.803           | 0.872           | 0.796              | 0.877              | 0.867         | 0.913         |
| KaLM-mini                                            | 1     | 0.577           | 0.902           | 0.391              | 0.921              | 0.907         | 0.920         |
|                                                      | 2     | 0.607           | 0.879           | 0.493              | 0.916              | 0.848         | 0.879         |
|                                                      | 3     | <b>0.587</b>    | <b>0.894</b>    | <b>0.443</b>       | <b>0.923</b>       | <b>0.870</b>  | <b>0.907</b>  |
|                                                      | 4     | 0.538           | 0.887           | 0.308              | 0.931              | 0.935         | 0.895         |
|                                                      | 5     | 0.524           | 0.882           | 0.338              | 0.905              | 0.882         | 0.923         |
|                                                      | 6     | 0.490           | 0.876           | 0.256              | 0.895              | 0.910         | 0.920         |
|                                                      | 7     | 0.592           | 0.873           | 0.475              | 0.864              | 0.854         | 0.944         |
|                                                      | 8     | 0.597           | 0.860           | 0.467              | 0.862              | 0.870         | 0.913         |
|                                                      | 9     | 0.601           | 0.843           | 0.469              | 0.873              | 0.885         | 0.870         |
|                                                      | 10    | 0.584           | 0.844           | 0.449              | 0.875              | 0.885         | 0.870         |
|                                                      | 11    | 0.586           | 0.836           | 0.458              | 0.862              | 0.885         | 0.864         |
|                                                      | 12    | 0.622           | 0.850           | 0.507              | 0.862              | 0.895         | 0.904         |
|                                                      | 13    | 0.633           | 0.840           | 0.540              | 0.860              | 0.895         | 0.870         |
|                                                      | 14    | 0.636           | 0.829           | 0.593              | 0.836              | 0.845         | 0.885         |
|                                                      | 15    | 0.650           | 0.817           | 0.566              | 0.864              | 0.898         | 0.827         |
|                                                      | 16    | 0.676           | 0.819           | 0.550              | 0.849              | 0.935         | 0.830         |
|                                                      | 17    | 0.747           | 0.852           | 0.675              | 0.864              | 0.920         | 0.889         |
|                                                      | 18    | 0.777           | 0.865           | 0.697              | 0.875              | 0.960         | 0.898         |
|                                                      | 19    | 0.801           | 0.861           | 0.721              | 0.873              | 0.966         | 0.892         |
|                                                      | 20    | 0.789           | 0.817           | 0.705              | 0.791              | 0.966         | 0.913         |
|                                                      | 21    | 0.802           | 0.831           | 0.727              | 0.811              | 0.966         | 0.926         |
|                                                      | 22    | 0.857           | 0.875           | 0.850              | 0.879              | 0.916         | 0.923         |
|                                                      | 23    | 0.859           | 0.872           | 0.849              | 0.884              | 0.920         | 0.898         |
|                                                      | 24    | 0.833           | 0.858           | 0.787              | 0.892              | 0.944         | 0.867         |

## G SIF and Whitening Post-Processing Sweeps

The main text uses ABTT-only as the paper-facing repair. This appendix reports SIF-only, SIF plus fixed- $D$  ABTT, and SIF plus tuned- $D$  ABTT, and gives the full account of the full-rank PCA whitening variant that Section 3 excludes, with the reduced-rank variant it compares against ABTT (Table 20). Table 19 first gives the five-seed cumulative top- $K$  result discussed in the main text, with SIF plus ABTT at each model’s train-selected layer. The per-layer tables then keep one primary metric per task so the four-method grid stays readable.

**Why whitening is excluded.** Full-rank PCA whitening (Su et al., 2021) leaves the different-directory cosine mean at or below 0.015, so the learned threshold degenerates: across all 100 whitening cells every file is routed to an existing directory, existing accuracy is 1.000, new accuracy never exceeds one file in 323, and assignment accuracy sits at the 62.4 percent class prior. That prior beats baseline in 60 of 100 cells while DirAcc@1 never exceeds 41.5 percent, so the apparent gain is a thresholding artifact. SIF and full-rank whitening fall short of ABTT for different reasons. SIF acts before pooling, so it can only attenuate the frequent tokens that feed the dominant direction, and whatever survives that reweighting is still pooled into the passage vector. Full-rank whitening also rescales the near-null directions that 847 training vectors cannot estimate (condition numbers  $3 \times 10^5$  to  $10^{16}$ ), which amplifies noise and degenerates the routing threshold; reduced-rank whitening drops them and does not degenerate it (Table 20).

**Top- $k$  ranking.** In the top- $k$  ranking form of Task B, each directory receives its best document match score, a pseudo-option NEW receives score  $\tau$ , and the simulated expert succeeds when the labeled directory appears in the shortlist. Cumulative top- $K$  accuracy is the fraction of test queries whose labeled directory is among the top  $K$  options. Table 19 gives the shortlist view (per-layer: Appendix D): every model exceeds 95% by Top-2 and gains little after Top-3.

| Model      | Top-1          | Top-2          | Top-3          | Top-4          | Top-5          |
|------------|----------------|----------------|----------------|----------------|----------------|
| LaTa       | 88.9 $\pm$ 0.5 | 96.1 $\pm$ 0.5 | 97.4 $\pm$ 0.6 | 97.9 $\pm$ 0.3 | 98.2 $\pm$ 0.2 |
| PhilTa     | 89.8 $\pm$ 0.8 | 95.8 $\pm$ 0.6 | 97.4 $\pm$ 0.3 | 97.7 $\pm$ 0.3 | 97.8 $\pm$ 0.4 |
| mT5-base   | 89.3 $\pm$ 0.5 | 95.1 $\pm$ 0.2 | 96.7 $\pm$ 0.7 | 97.1 $\pm$ 0.4 | 97.2 $\pm$ 0.4 |
| LaBSE      | 90.6 $\pm$ 1.0 | 95.5 $\pm$ 0.5 | 96.4 $\pm$ 0.5 | 97.2 $\pm$ 0.6 | 97.8 $\pm$ 0.5 |
| Qwen3-0.6B | 89.3 $\pm$ 0.7 | 95.7 $\pm$ 0.4 | 97.1 $\pm$ 0.3 | 97.5 $\pm$ 0.4 | 97.6 $\pm$ 0.4 |
| KaLM-mini  | 89.3 $\pm$ 0.5 | 95.1 $\pm$ 0.4 | 96.6 $\pm$ 0.7 | 97.2 $\pm$ 0.7 | 97.6 $\pm$ 0.5 |

Table 19: Cumulative top- $K$  accuracy for Task B under SIF+ABTT, as mean  $\pm$  standard deviation over five query/reference reseeds at a fixed split. For each model the reported layer is chosen on the train split: the layer with the highest training-set directory accuracy at rank 1 for SIF+ABTT in the single-seed run, the layer behind the SIF+ABTT cell of Table 8, listed in Table 11.  $D$  is tuned per layer on the train split. This is not always the layer with the highest five-seed mean. Each row is the bold row of Table 23. Each cell is the percentage of test queries whose labelled directory appears within the top  $K$  options. Per-layer top- $k$  rankings: Appendix Tables 14, 18, and 23.

| Model      | Task A AUROC |       |                |       |       |       | Task B DirAcc@1 |                |      |      |      | Task B assignment |                |      |      |      |
|------------|--------------|-------|----------------|-------|-------|-------|-----------------|----------------|------|------|------|-------------------|----------------|------|------|------|
|            | Base         | ABTT  | whitening, $k$ |       |       |       | ABTT            | whitening, $k$ |      |      |      | ABTT              | whitening, $k$ |      |      |      |
|            |              |       | 64             | 128   | 256   | full  |                 | 64             | 128  | 256  | full |                   | 64             | 128  | 256  | full |
| LaTa       | 0.938        | 0.971 | 0.969          | 0.973 | 0.983 | 0.872 | 86.1            | 81.2           | 83.1 | 77.0 | 41.1 | 88.5              | 84.7           | 86.2 | 80.7 | 62.5 |
| PhilTa     | 0.939        | 0.983 | 0.974          | 0.977 | 0.979 | 0.854 | 88.3            | 83.0           | 86.4 | 83.7 | 41.0 | 91.1              | 86.5           | 88.8 | 86.8 | 62.5 |
| mT5-base   | 0.838        | 0.975 | 0.971          | 0.979 | 0.974 | 0.845 | 88.5            | 78.9           | 85.8 | 83.8 | 41.0 | 90.3              | 82.6           | 88.6 | 87.2 | 62.5 |
| LaBSE      | 0.956        | 0.987 | 0.979          | 0.984 | 0.976 | 0.866 | 88.5            | 86.1           | 84.0 | 73.5 | 40.3 | 90.8              | 88.8           | 86.7 | 78.0 | 62.5 |
| Qwen3-0.6B | 0.966        | 0.973 | 0.973          | 0.978 | 0.983 | 0.594 | 89.4            | 84.8           | 85.0 | 75.9 | 20.5 | 91.5              | 87.2           | 88.0 | 79.3 | 62.4 |
| KaLM-mini  | 0.972        | 0.981 | 0.976          | 0.981 | 0.981 | 0.562 | 89.4            | 82.6           | 86.7 | 84.5 | 16.8 | 91.7              | 85.4           | 89.2 | 87.8 | 62.4 |

Table 20: PCA whitening to  $k$  components against the baseline and ABTT. Whitening keeps the top  $k$  principal components of the centered training embeddings and rescales each to unit variance. Full keeps every component: 768 for the T5 encoders and LaBSE, and 847, the number of training passages, for Qwen3-0.6B and KaLM-mini, whose last component then has numerically zero variance. Task A cells are read at each setting’s train-selected layer (highest training AUROC), Task B cells at the layer with the highest training DirAcc@1. Task B values are single-split test DirAcc@1 and assignment accuracy in percent, with the threshold learned on training pairs; they are not the five-seed values of Table 8. Every transform is fit on training embeddings only. Whitening ranks on par with ABTT, but routes below it.

Table 21: Per-layer Task A AUROC across the SIF-conditioned post-processing suite for all six models. `sif_only` replaces mean pooling with SIF-weighted pooling; `sif_abtt_fixed` adds ABTT with fixed  $D=10$ ; `sif_abtt_optimal` tunes  $D$  per layer on the train split. Gap is omitted to keep the table narrow; the pure-ABTT comparison (not SIF-conditioned) is in Tables 12 and 16. Rows in bold mark the layer chosen on the train split, the layer with the highest training-set AUROC under SIF+ABTT. It is the layer behind the SIF+ABTT cell of Table 7 (listed in Table 11), and not always the layer with the highest test score in this table.

| Task A: SIF-suite AUROC (appendix) |           |                 |                |                                    |                     |
|------------------------------------|-----------|-----------------|----------------|------------------------------------|---------------------|
| Model                              | Layer     | AUROC<br>(base) | AUROC<br>(SIF) | AUROC<br>(SIF+ABTT <sub>10</sub> ) | AUROC<br>(SIF+ABTT) |
| LaTa                               | 1         | 0.934           | 0.948          | 0.972                              | 0.976               |
|                                    | 2         | 0.563           | 0.938          | 0.964                              | 0.971               |
|                                    | 3         | 0.513           | 0.921          | 0.967                              | 0.966               |
|                                    | 4         | 0.500           | 0.890          | 0.970                              | 0.968               |
|                                    | 5         | 0.496           | 0.878          | 0.970                              | 0.970               |
|                                    | 6         | 0.496           | 0.865          | 0.965                              | 0.965               |
|                                    | <b>7</b>  | <b>0.498</b>    | <b>0.858</b>   | <b>0.966</b>                       | <b>0.966</b>        |
|                                    | 8         | 0.502           | 0.842          | 0.964                              | 0.964               |
|                                    | 9         | 0.503           | 0.854          | 0.964                              | 0.964               |
|                                    | 10        | 0.505           | 0.859          | 0.966                              | 0.966               |
|                                    | 11        | 0.508           | 0.867          | 0.965                              | 0.965               |
|                                    | 12        | 0.938           | 0.948          | 0.963                              | 0.963               |
| PhilTa                             | 1         | 0.939           | 0.955          | 0.975                              | 0.975               |
|                                    | 2         | 0.911           | 0.941          | 0.973                              | 0.973               |
|                                    | 3         | 0.601           | 0.928          | 0.968                              | 0.968               |
|                                    | 4         | 0.551           | 0.916          | 0.972                              | 0.972               |
|                                    | 5         | 0.542           | 0.904          | 0.972                              | 0.972               |
|                                    | 6         | 0.541           | 0.884          | 0.981                              | 0.981               |
|                                    | 7         | 0.542           | 0.891          | 0.981                              | 0.981               |
|                                    | <b>8</b>  | <b>0.542</b>    | <b>0.895</b>   | <b>0.982</b>                       | <b>0.982</b>        |
|                                    | 9         | 0.539           | 0.881          | 0.977                              | 0.977               |
|                                    | 10        | 0.538           | 0.874          | 0.977                              | 0.977               |
|                                    | 11        | 0.539           | 0.872          | 0.979                              | 0.979               |
|                                    | 12        | 0.911           | 0.929          | 0.969                              | 0.969               |
| mT5-base                           | <b>1</b>  | <b>0.822</b>    | <b>0.845</b>   | <b>0.974</b>                       | <b>0.974</b>        |
|                                    | 2         | 0.822           | 0.835          | 0.966                              | 0.966               |
|                                    | 3         | 0.804           | 0.808          | 0.958                              | 0.958               |
|                                    | 4         | 0.799           | 0.809          | 0.966                              | 0.966               |
|                                    | 5         | 0.654           | 0.672          | 0.974                              | 0.974               |
|                                    | 6         | 0.657           | 0.678          | 0.971                              | 0.971               |
|                                    | 7         | 0.656           | 0.680          | 0.967                              | 0.967               |
|                                    | 8         | 0.656           | 0.681          | 0.968                              | 0.968               |
|                                    | 9         | 0.660           | 0.685          | 0.967                              | 0.967               |
|                                    | 10        | 0.659           | 0.684          | 0.966                              | 0.966               |
|                                    | 11        | 0.659           | 0.689          | 0.966                              | 0.966               |
|                                    | 12        | 0.838           | 0.833          | 0.970                              | 0.970               |
| LaBSE                              | 1         | 0.806           | 0.818          | 0.974                              | 0.974               |
|                                    | 2         | 0.833           | 0.840          | 0.975                              | 0.975               |
|                                    | 3         | 0.853           | 0.862          | 0.968                              | 0.968               |
|                                    | 4         | 0.862           | 0.869          | 0.966                              | 0.966               |
|                                    | 5         | 0.864           | 0.869          | 0.964                              | 0.964               |
|                                    | 6         | 0.894           | 0.895          | 0.965                              | 0.965               |
|                                    | 7         | 0.887           | 0.892          | 0.964                              | 0.964               |
|                                    | 8         | 0.879           | 0.883          | 0.968                              | 0.968               |
|                                    | 9         | 0.916           | 0.915          | 0.973                              | 0.973               |
|                                    | <b>10</b> | <b>0.934</b>    | <b>0.934</b>   | <b>0.976</b>                       | <b>0.976</b>        |
|                                    | 11        | 0.960           | 0.959          | 0.983                              | 0.983               |
|                                    | 12        | 0.956           | 0.956          | 0.980                              | 0.984               |
| Qwen3-0.6B                         | 1         | 0.858           | 0.852          | 0.982                              | 0.981               |
|                                    | <b>2</b>  | <b>0.863</b>    | <b>0.853</b>   | <b>0.972</b>                       | <b>0.972</b>        |
|                                    | 3         | 0.862           | 0.858          | 0.977                              | 0.975               |
|                                    | 4         | 0.860           | 0.860          | 0.976                              | 0.977               |
|                                    | 5         | 0.861           | 0.858          | 0.971                              | 0.971               |
|                                    | 6         | 0.867           | 0.861          | 0.974                              | 0.974               |
|                                    | 7         | 0.876           | 0.866          | 0.971                              | 0.971               |

(continued on next page)

| Task A: SIF-suite AUROC (appendix) |          |                 |                |                                    |                     |
|------------------------------------|----------|-----------------|----------------|------------------------------------|---------------------|
| Model                              | Layer    | AUROC<br>(base) | AUROC<br>(SIF) | AUROC<br>(SIF+ABTT <sub>10</sub> ) | AUROC<br>(SIF+ABTT) |
|                                    | 8        | 0.882           | 0.868          | 0.969                              | 0.969               |
|                                    | 9        | 0.881           | 0.866          | 0.968                              | 0.968               |
|                                    | 10       | 0.887           | 0.870          | 0.966                              | 0.966               |
|                                    | 11       | 0.895           | 0.875          | 0.964                              | 0.964               |
|                                    | 12       | 0.901           | 0.881          | 0.966                              | 0.966               |
|                                    | 13       | 0.907           | 0.887          | 0.967                              | 0.967               |
|                                    | 14       | 0.905           | 0.885          | 0.966                              | 0.966               |
|                                    | 15       | 0.901           | 0.883          | 0.968                              | 0.968               |
|                                    | 16       | 0.910           | 0.894          | 0.965                              | 0.965               |
|                                    | 17       | 0.916           | 0.899          | 0.967                              | 0.974               |
|                                    | 18       | 0.933           | 0.914          | 0.966                              | 0.973               |
|                                    | 19       | 0.936           | 0.915          | 0.966                              | 0.976               |
|                                    | 20       | 0.943           | 0.918          | 0.971                              | 0.977               |
|                                    | 21       | 0.953           | 0.931          | 0.971                              | 0.971               |
|                                    | 22       | 0.963           | 0.942          | 0.973                              | 0.973               |
|                                    | 23       | 0.961           | 0.941          | 0.973                              | 0.973               |
|                                    | 24       | 0.963           | 0.943          | 0.976                              | 0.974               |
|                                    | 25       | 0.961           | 0.938          | 0.978                              | 0.975               |
|                                    | 26       | 0.966           | 0.942          | 0.980                              | 0.980               |
|                                    | 27       | 0.967           | 0.936          | 0.978                              | 0.978               |
|                                    | 28       | 0.958           | 0.925          | 0.968                              | 0.968               |
| KaLM-mini                          | 1        | 0.887           | 0.880          | 0.982                              | 0.980               |
|                                    | <b>2</b> | <b>0.906</b>    | <b>0.899</b>   | <b>0.981</b>                       | <b>0.981</b>        |
|                                    | 3        | 0.881           | 0.882          | 0.977                              | 0.977               |
|                                    | 4        | 0.871           | 0.873          | 0.974                              | 0.974               |
|                                    | 5        | 0.861           | 0.862          | 0.971                              | 0.971               |
|                                    | 6        | 0.872           | 0.867          | 0.967                              | 0.967               |
|                                    | 7        | 0.888           | 0.879          | 0.972                              | 0.972               |
|                                    | 8        | 0.890           | 0.880          | 0.973                              | 0.975               |
|                                    | 9        | 0.892           | 0.882          | 0.968                              | 0.968               |
|                                    | 10       | 0.893           | 0.882          | 0.964                              | 0.964               |
|                                    | 11       | 0.894           | 0.885          | 0.969                              | 0.969               |
|                                    | 12       | 0.905           | 0.896          | 0.966                              | 0.966               |
|                                    | 13       | 0.900           | 0.894          | 0.964                              | 0.964               |
|                                    | 14       | 0.904           | 0.898          | 0.970                              | 0.970               |
|                                    | 15       | 0.907           | 0.901          | 0.970                              | 0.970               |
|                                    | 16       | 0.917           | 0.913          | 0.967                              | 0.967               |
|                                    | 17       | 0.926           | 0.923          | 0.968                              | 0.968               |
|                                    | 18       | 0.945           | 0.939          | 0.977                              | 0.977               |
|                                    | 19       | 0.956           | 0.950          | 0.978                              | 0.978               |
|                                    | 20       | 0.960           | 0.956          | 0.978                              | 0.978               |
|                                    | 21       | 0.962           | 0.958          | 0.980                              | 0.983               |
|                                    | 22       | 0.971           | 0.967          | 0.983                              | 0.984               |
|                                    | 23       | 0.972           | 0.964          | 0.986                              | 0.985               |
|                                    | 24       | 0.956           | 0.956          | 0.976                              | 0.976               |

Table 22: Per-layer Task B overall assignment accuracy across the SIF-conditioned suite for all six models. Restricted to overall routing accuracy (existing/new decomposition omitted) to keep the table compact; see Tables 13 and 17 for the pure-ABTT pairwise comparison with existing/new broken out. Rows in bold mark the layer chosen on the train split, the layer with the highest training-set directory accuracy at rank 1 under SIF+ABTT. It is the layer behind the SIF+ABTT cell of Table 8 (listed in Table 11), and not always the layer with the highest test score in this table.

| Task B: SIF-suite routing (appendix) |       |                   |                  |                                      |                       |
|--------------------------------------|-------|-------------------|------------------|--------------------------------------|-----------------------|
| Model                                | Layer | Overall<br>(base) | Overall<br>(SIF) | Overall<br>(SIF+ABTT <sub>10</sub> ) | Overall<br>(SIF+ABTT) |
| LaTa                                 | 1     | 0.738             | 0.832            | 0.920                                | 0.899                 |
|                                      | 2     | 0.439             | 0.791            | 0.904                                | 0.900                 |
|                                      | 3     | 0.477             | 0.720            | 0.910                                | 0.907                 |
|                                      | 4     | 0.513             | 0.668            | 0.900                                | 0.902                 |
|                                      | 5     | 0.502             | 0.648            | 0.906                                | 0.906                 |
|                                      | 6     | 0.503             | 0.668            | 0.895                                | 0.895                 |

(continued on next page)

| Task B: SIF-suite routing (appendix) |       |                   |                  |                                      |                       |
|--------------------------------------|-------|-------------------|------------------|--------------------------------------|-----------------------|
| Model                                | Layer | Overall<br>(base) | Overall<br>(SIF) | Overall<br>(SIF+ABTT <sub>10</sub> ) | Overall<br>(SIF+ABTT) |
|                                      | 7     | 0.501             | 0.642            | 0.903                                | 0.903                 |
|                                      | 8     | 0.502             | 0.614            | 0.892                                | 0.892                 |
|                                      | 9     | 0.501             | 0.618            | 0.889                                | 0.889                 |
|                                      | 10    | 0.498             | 0.655            | 0.890                                | 0.890                 |
|                                      | 11    | 0.490             | 0.621            | 0.880                                | 0.880                 |
|                                      | 12    | <b>0.710</b>      | <b>0.804</b>     | <b>0.902</b>                         | <b>0.902</b>          |
| PhilTa                               | 1     | <b>0.709</b>      | <b>0.790</b>     | <b>0.914</b>                         | <b>0.914</b>          |
|                                      | 2     | 0.625             | 0.738            | 0.892                                | 0.892                 |
|                                      | 3     | 0.455             | 0.713            | 0.892                                | 0.892                 |
|                                      | 4     | 0.439             | 0.663            | 0.886                                | 0.886                 |
|                                      | 5     | 0.457             | 0.626            | 0.899                                | 0.899                 |
|                                      | 6     | 0.474             | 0.612            | 0.906                                | 0.906                 |
|                                      | 7     | 0.469             | 0.625            | 0.906                                | 0.906                 |
|                                      | 8     | 0.467             | 0.614            | 0.886                                | 0.886                 |
|                                      | 9     | 0.476             | 0.590            | 0.895                                | 0.895                 |
|                                      | 10    | 0.480             | 0.580            | 0.888                                | 0.888                 |
|                                      | 11    | 0.470             | 0.565            | 0.886                                | 0.886                 |
|                                      | 12    | 0.549             | 0.667            | 0.899                                | 0.899                 |
| mT5-base                             | 1     | 0.453             | 0.545            | 0.907                                | 0.907                 |
|                                      | 2     | <b>0.436</b>      | <b>0.537</b>     | <b>0.903</b>                         | <b>0.903</b>          |
|                                      | 3     | 0.449             | 0.568            | 0.885                                | 0.885                 |
|                                      | 4     | 0.491             | 0.488            | 0.857                                | 0.857                 |
|                                      | 5     | 0.512             | 0.483            | 0.826                                | 0.826                 |
|                                      | 6     | 0.580             | 0.558            | 0.817                                | 0.817                 |
|                                      | 7     | 0.565             | 0.554            | 0.832                                | 0.832                 |
|                                      | 8     | 0.558             | 0.548            | 0.834                                | 0.834                 |
|                                      | 9     | 0.508             | 0.495            | 0.845                                | 0.845                 |
|                                      | 10    | 0.491             | 0.495            | 0.832                                | 0.832                 |
|                                      | 11    | 0.479             | 0.484            | 0.816                                | 0.816                 |
|                                      | 12    | 0.472             | 0.541            | 0.858                                | 0.858                 |
| LaBSE                                | 1     | 0.477             | 0.521            | 0.886                                | 0.886                 |
|                                      | 2     | 0.543             | 0.508            | 0.875                                | 0.875                 |
|                                      | 3     | 0.514             | 0.577            | 0.860                                | 0.860                 |
|                                      | 4     | 0.565             | 0.587            | 0.869                                | 0.869                 |
|                                      | 5     | 0.543             | 0.598            | 0.857                                | 0.857                 |
|                                      | 6     | 0.674             | 0.648            | 0.872                                | 0.872                 |
|                                      | 7     | 0.584             | 0.634            | 0.858                                | 0.858                 |
|                                      | 8     | 0.568             | 0.571            | 0.854                                | 0.854                 |
|                                      | 9     | 0.612             | 0.613            | 0.859                                | 0.859                 |
|                                      | 10    | 0.713             | 0.698            | 0.864                                | 0.864                 |
|                                      | 11    | <b>0.833</b>      | <b>0.807</b>     | <b>0.896</b>                         | <b>0.896</b>          |
|                                      | 12    | 0.808             | 0.834            | 0.890                                | 0.880                 |
| Qwen3-0.6B                           | 1     | 0.526             | 0.566            | 0.914                                | 0.909                 |
|                                      | 2     | 0.579             | 0.524            | 0.894                                | 0.894                 |
|                                      | 3     | 0.513             | 0.537            | 0.904                                | 0.901                 |
|                                      | 4     | 0.576             | 0.531            | 0.899                                | 0.904                 |
|                                      | 5     | <b>0.464</b>      | <b>0.537</b>     | <b>0.908</b>                         | <b>0.908</b>          |
|                                      | 6     | 0.576             | 0.526            | 0.906                                | 0.906                 |
|                                      | 7     | 0.513             | 0.573            | 0.911                                | 0.911                 |
|                                      | 8     | 0.592             | 0.627            | 0.911                                | 0.911                 |
|                                      | 9     | 0.607             | 0.576            | 0.910                                | 0.910                 |
|                                      | 10    | 0.647             | 0.599            | 0.900                                | 0.900                 |
|                                      | 11    | 0.551             | 0.606            | 0.893                                | 0.893                 |
|                                      | 12    | 0.607             | 0.663            | 0.900                                | 0.900                 |
|                                      | 13    | 0.625             | 0.675            | 0.897                                | 0.897                 |
|                                      | 14    | 0.633             | 0.571            | 0.893                                | 0.893                 |
|                                      | 15    | 0.639             | 0.600            | 0.893                                | 0.893                 |
|                                      | 16    | 0.711             | 0.679            | 0.880                                | 0.880                 |
|                                      | 17    | 0.652             | 0.686            | 0.900                                | 0.876                 |
|                                      | 18    | 0.700             | 0.731            | 0.899                                | 0.882                 |
|                                      | 19    | 0.684             | 0.700            | 0.886                                | 0.850                 |
|                                      | 20    | 0.746             | 0.676            | 0.882                                | 0.866                 |

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| Task B: SIF-suite routing (appendix) |          |                   |                  |                                      |                       |
|--------------------------------------|----------|-------------------|------------------|--------------------------------------|-----------------------|
| Model                                | Layer    | Overall<br>(base) | Overall<br>(SIF) | Overall<br>(SIF+ABTT <sub>10</sub> ) | Overall<br>(SIF+ABTT) |
| KaLM-mini                            | 21       | 0.761             | 0.739            | 0.892                                | 0.892                 |
|                                      | 22       | 0.809             | 0.719            | 0.893                                | 0.893                 |
|                                      | 23       | 0.752             | 0.759            | 0.893                                | 0.893                 |
|                                      | 24       | 0.818             | 0.706            | 0.900                                | 0.899                 |
|                                      | 25       | 0.725             | 0.748            | 0.897                                | 0.893                 |
|                                      | 26       | 0.749             | 0.772            | 0.903                                | 0.903                 |
|                                      | 27       | 0.808             | 0.760            | 0.899                                | 0.899                 |
|                                      | 28       | 0.823             | 0.774            | 0.890                                | 0.890                 |
|                                      | <b>1</b> | <b>0.585</b>      | <b>0.604</b>     | <b>0.906</b>                         | <b>0.908</b>          |
|                                      | 2        | 0.627             | 0.594            | 0.908                                | 0.908                 |
|                                      | 3        | 0.604             | 0.576            | 0.907                                | 0.907                 |
|                                      | 4        | 0.544             | 0.577            | 0.901                                | 0.901                 |
|                                      | 5        | 0.543             | 0.533            | 0.897                                | 0.897                 |
|                                      | 6        | 0.502             | 0.558            | 0.900                                | 0.900                 |
|                                      | 7        | 0.618             | 0.593            | 0.901                                | 0.901                 |
|                                      | 8        | 0.619             | 0.607            | 0.892                                | 0.878                 |
|                                      | 9        | 0.626             | 0.663            | 0.876                                | 0.876                 |
|                                      | 10       | 0.613             | 0.603            | 0.866                                | 0.866                 |
|                                      | 11       | 0.619             | 0.627            | 0.874                                | 0.874                 |
|                                      | 12       | 0.653             | 0.642            | 0.874                                | 0.874                 |
|                                      | 13       | 0.674             | 0.666            | 0.864                                | 0.864                 |
|                                      | 14       | 0.688             | 0.648            | 0.861                                | 0.861                 |
|                                      | 15       | 0.691             | 0.698            | 0.866                                | 0.866                 |
|                                      | 16       | 0.695             | 0.738            | 0.859                                | 0.859                 |
|                                      | 17       | 0.767             | 0.763            | 0.882                                | 0.882                 |
|                                      | 18       | 0.796             | 0.796            | 0.879                                | 0.879                 |
|                                      | 19       | 0.814             | 0.804            | 0.878                                | 0.878                 |
|                                      | 20       | 0.803             | 0.821            | 0.876                                | 0.876                 |
|                                      | 21       | 0.817             | 0.832            | 0.874                                | 0.876                 |
|                                      | 22       | 0.875             | 0.857            | 0.876                                | 0.876                 |
|                                      | 23       | 0.875             | 0.862            | 0.880                                | 0.881                 |
|                                      | 24       | 0.846             | 0.847            | 0.869                                | 0.869                 |

Table 23: Per-layer Task B top- $k$  ranking accuracy across all six models, averaged over  $M = 5$  query/reference reseeds of the train/test split (mean  $\pm$  std). `sif_abtt_optimal` applies SIF weighting plus ABTT with  $D$  tuned per layer on the train split. This table covers SIF+ABTT; five-seed ABTT-only values for LaTa, Qwen3-0.6B and KaLM-mini, at their train-selected layers, are in Table 26 (Appendix I). Overall assignment accuracy coincides with Acc@1 in the mseed pipeline, so we report the existing/new decomposition instead. Rows in bold mark the layer reported per model, the row printed in Table 19. For each model the reported layer is chosen on the train split: the layer with the highest training-set directory accuracy at rank 1 for SIF+ABTT in the single-seed run, the layer behind the SIF+ABTT cell of Table 8, listed in Table 11.  $D$  is tuned per layer on the train split. This is not always the layer with the highest five-seed mean.

| Task B: Top-K Ranking (appendix, 5-seed mean $\pm$ std, SIF+ABTT) |           |                                     |                                     |                                     |                                     |                                     |                                     |
|-------------------------------------------------------------------|-----------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| Model                                                             | Layer     | Acc@1<br>(base)                     | Acc@1<br>(SIF+ABTT)                 | Existing<br>(base)                  | Existing<br>(SIF+ABTT)              | New<br>(base)                       | New<br>(SIF+ABTT)                   |
| LaTa                                                              | 1         | 0.731 $\pm$ 0.010                   | 0.889 $\pm$ 0.010                   | 0.563 $\pm$ 0.007                   | 0.844 $\pm$ 0.011                   | 0.949 $\pm$ 0.011                   | 0.948 $\pm$ 0.010                   |
|                                                                   | 2         | 0.429 $\pm$ 0.011                   | 0.888 $\pm$ 0.007                   | 0.049 $\pm$ 0.006                   | 0.841 $\pm$ 0.007                   | 0.920 $\pm$ 0.008                   | 0.948 $\pm$ 0.007                   |
|                                                                   | 3         | 0.393 $\pm$ 0.011                   | 0.891 $\pm$ 0.010                   | 0.047 $\pm$ 0.006                   | 0.847 $\pm$ 0.010                   | 0.840 $\pm$ 0.020                   | 0.947 $\pm$ 0.012                   |
|                                                                   | 4         | 0.331 $\pm$ 0.019                   | 0.889 $\pm$ 0.009                   | 0.048 $\pm$ 0.009                   | 0.860 $\pm$ 0.009                   | 0.696 $\pm$ 0.030                   | 0.925 $\pm$ 0.013                   |
|                                                                   | 5         | 0.326 $\pm$ 0.021                   | 0.891 $\pm$ 0.009                   | 0.038 $\pm$ 0.005                   | 0.858 $\pm$ 0.008                   | 0.698 $\pm$ 0.028                   | 0.934 $\pm$ 0.014                   |
|                                                                   | 6         | 0.322 $\pm$ 0.018                   | 0.891 $\pm$ 0.009                   | 0.038 $\pm$ 0.007                   | 0.863 $\pm$ 0.010                   | 0.688 $\pm$ 0.026                   | 0.927 $\pm$ 0.013                   |
|                                                                   | 7         | 0.328 $\pm$ 0.020                   | 0.896 $\pm$ 0.008                   | 0.045 $\pm$ 0.007                   | 0.848 $\pm$ 0.009                   | 0.693 $\pm$ 0.028                   | 0.957 $\pm$ 0.010                   |
|                                                                   | 8         | 0.325 $\pm$ 0.022                   | 0.884 $\pm$ 0.008                   | 0.046 $\pm$ 0.010                   | 0.834 $\pm$ 0.008                   | 0.685 $\pm$ 0.032                   | 0.949 $\pm$ 0.011                   |
|                                                                   | 9         | 0.332 $\pm$ 0.019                   | 0.886 $\pm$ 0.011                   | 0.045 $\pm$ 0.010                   | 0.848 $\pm$ 0.009                   | 0.701 $\pm$ 0.024                   | 0.935 $\pm$ 0.015                   |
|                                                                   | 10        | 0.346 $\pm$ 0.017                   | 0.889 $\pm$ 0.004                   | 0.045 $\pm$ 0.010                   | 0.846 $\pm$ 0.008                   | 0.735 $\pm$ 0.021                   | 0.944 $\pm$ 0.012                   |
|                                                                   | 11        | 0.361 $\pm$ 0.018                   | 0.882 $\pm$ 0.009                   | 0.045 $\pm$ 0.009                   | 0.862 $\pm$ 0.010                   | 0.767 $\pm$ 0.029                   | 0.907 $\pm$ 0.014                   |
|                                                                   | <b>12</b> | <b>0.707 <math>\pm</math> 0.014</b> | <b>0.889 <math>\pm</math> 0.005</b> | <b>0.511 <math>\pm</math> 0.016</b> | <b>0.834 <math>\pm</math> 0.005</b> | <b>0.961 <math>\pm</math> 0.008</b> | <b>0.961 <math>\pm</math> 0.010</b> |
| PhilTa                                                            | <b>1</b>  | <b>0.705 <math>\pm</math> 0.015</b> | <b>0.898 <math>\pm</math> 0.008</b> | <b>0.531 <math>\pm</math> 0.016</b> | <b>0.858 <math>\pm</math> 0.010</b> | <b>0.931 <math>\pm</math> 0.012</b> | <b>0.951 <math>\pm</math> 0.011</b> |

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| Task B: Top-K Ranking (appendix, 5-seed mean $\pm$ std, SIF+ABTT) |       |                                     |                                     |                                     |                                     |                                     |                                     |
|-------------------------------------------------------------------|-------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| Model                                                             | Layer | Acc@1<br>(base)                     | Acc@1<br>(SIF+ABTT)                 | Existing<br>(base)                  | Existing<br>(SIF+ABTT)              | New<br>(base)                       | New<br>(SIF+ABTT)                   |
|                                                                   | 2     | 0.626 $\pm$ 0.019                   | 0.888 $\pm$ 0.010                   | 0.410 $\pm$ 0.017                   | 0.866 $\pm$ 0.010                   | 0.906 $\pm$ 0.018                   | 0.917 $\pm$ 0.017                   |
|                                                                   | 3     | 0.440 $\pm$ 0.013                   | 0.893 $\pm$ 0.009                   | 0.071 $\pm$ 0.014                   | 0.858 $\pm$ 0.010                   | 0.917 $\pm$ 0.013                   | 0.939 $\pm$ 0.011                   |
|                                                                   | 4     | 0.421 $\pm$ 0.016                   | 0.891 $\pm$ 0.009                   | 0.027 $\pm$ 0.007                   | 0.858 $\pm$ 0.008                   | 0.928 $\pm$ 0.015                   | 0.935 $\pm$ 0.013                   |
|                                                                   | 5     | 0.375 $\pm$ 0.016                   | 0.895 $\pm$ 0.010                   | 0.042 $\pm$ 0.013                   | 0.852 $\pm$ 0.010                   | 0.804 $\pm$ 0.018                   | 0.950 $\pm$ 0.014                   |
|                                                                   | 6     | 0.337 $\pm$ 0.015                   | 0.897 $\pm$ 0.009                   | 0.060 $\pm$ 0.012                   | 0.861 $\pm$ 0.010                   | 0.696 $\pm$ 0.026                   | 0.944 $\pm$ 0.013                   |
|                                                                   | 7     | 0.361 $\pm$ 0.017                   | 0.900 $\pm$ 0.009                   | 0.056 $\pm$ 0.012                   | 0.860 $\pm$ 0.008                   | 0.755 $\pm$ 0.020                   | 0.952 $\pm$ 0.013                   |
|                                                                   | 8     | 0.389 $\pm$ 0.016                   | 0.891 $\pm$ 0.010                   | 0.058 $\pm$ 0.014                   | 0.859 $\pm$ 0.009                   | 0.818 $\pm$ 0.018                   | 0.932 $\pm$ 0.015                   |
|                                                                   | 9     | 0.371 $\pm$ 0.016                   | 0.893 $\pm$ 0.008                   | 0.066 $\pm$ 0.014                   | 0.861 $\pm$ 0.009                   | 0.766 $\pm$ 0.027                   | 0.934 $\pm$ 0.016                   |
|                                                                   | 10    | 0.377 $\pm$ 0.014                   | 0.892 $\pm$ 0.009                   | 0.066 $\pm$ 0.014                   | 0.855 $\pm$ 0.005                   | 0.778 $\pm$ 0.023                   | 0.940 $\pm$ 0.017                   |
|                                                                   | 11    | 0.394 $\pm$ 0.014                   | 0.888 $\pm$ 0.008                   | 0.059 $\pm$ 0.010                   | 0.851 $\pm$ 0.010                   | 0.827 $\pm$ 0.021                   | 0.935 $\pm$ 0.013                   |
|                                                                   | 12    | 0.566 $\pm$ 0.013                   | 0.897 $\pm$ 0.010                   | 0.282 $\pm$ 0.014                   | 0.863 $\pm$ 0.010                   | 0.934 $\pm$ 0.007                   | 0.941 $\pm$ 0.018                   |
| mT5-base                                                          | 1     | 0.487 $\pm$ 0.012                   | 0.899 $\pm$ 0.005                   | 0.100 $\pm$ 0.009                   | 0.856 $\pm$ 0.007                   | 0.987 $\pm$ 0.007                   | 0.954 $\pm$ 0.011                   |
|                                                                   | 2     | <b>0.476 <math>\pm</math> 0.012</b> | <b>0.893 <math>\pm</math> 0.005</b> | <b>0.083 <math>\pm</math> 0.008</b> | <b>0.851 <math>\pm</math> 0.012</b> | <b>0.984 <math>\pm</math> 0.008</b> | <b>0.947 <math>\pm</math> 0.013</b> |
|                                                                   | 3     | 0.483 $\pm$ 0.014                   | 0.883 $\pm$ 0.011                   | 0.124 $\pm$ 0.009                   | 0.817 $\pm$ 0.014                   | 0.947 $\pm$ 0.008                   | 0.970 $\pm$ 0.007                   |
|                                                                   | 4     | 0.500 $\pm$ 0.015                   | 0.852 $\pm$ 0.007                   | 0.206 $\pm$ 0.006                   | 0.772 $\pm$ 0.010                   | 0.879 $\pm$ 0.020                   | 0.956 $\pm$ 0.010                   |
|                                                                   | 5     | 0.371 $\pm$ 0.014                   | 0.825 $\pm$ 0.008                   | 0.155 $\pm$ 0.006                   | 0.724 $\pm$ 0.007                   | 0.651 $\pm$ 0.023                   | 0.956 $\pm$ 0.006                   |
|                                                                   | 6     | 0.361 $\pm$ 0.012                   | 0.810 $\pm$ 0.005                   | 0.263 $\pm$ 0.021                   | 0.765 $\pm$ 0.012                   | 0.488 $\pm$ 0.018                   | 0.868 $\pm$ 0.016                   |
|                                                                   | 7     | 0.370 $\pm$ 0.010                   | 0.827 $\pm$ 0.008                   | 0.255 $\pm$ 0.014                   | 0.760 $\pm$ 0.011                   | 0.519 $\pm$ 0.017                   | 0.914 $\pm$ 0.013                   |
|                                                                   | 8     | 0.382 $\pm$ 0.008                   | 0.824 $\pm$ 0.012                   | 0.244 $\pm$ 0.007                   | 0.773 $\pm$ 0.012                   | 0.561 $\pm$ 0.017                   | 0.891 $\pm$ 0.016                   |
|                                                                   | 9     | 0.428 $\pm$ 0.014                   | 0.838 $\pm$ 0.010                   | 0.171 $\pm$ 0.008                   | 0.783 $\pm$ 0.013                   | 0.759 $\pm$ 0.024                   | 0.909 $\pm$ 0.006                   |
|                                                                   | 10    | 0.421 $\pm$ 0.010                   | 0.821 $\pm$ 0.017                   | 0.140 $\pm$ 0.011                   | 0.767 $\pm$ 0.015                   | 0.783 $\pm$ 0.021                   | 0.892 $\pm$ 0.019                   |
|                                                                   | 11    | 0.420 $\pm$ 0.015                   | 0.803 $\pm$ 0.012                   | 0.131 $\pm$ 0.012                   | 0.721 $\pm$ 0.013                   | 0.792 $\pm$ 0.016                   | 0.910 $\pm$ 0.015                   |
|                                                                   | 12    | 0.507 $\pm$ 0.009                   | 0.841 $\pm$ 0.009                   | 0.169 $\pm$ 0.007                   | 0.772 $\pm$ 0.016                   | 0.942 $\pm$ 0.011                   | 0.931 $\pm$ 0.011                   |
| LaBSE                                                             | 1     | 0.502 $\pm$ 0.012                   | 0.858 $\pm$ 0.007                   | 0.170 $\pm$ 0.003                   | 0.838 $\pm$ 0.007                   | 0.929 $\pm$ 0.016                   | 0.885 $\pm$ 0.009                   |
|                                                                   | 2     | 0.523 $\pm$ 0.011                   | 0.860 $\pm$ 0.004                   | 0.274 $\pm$ 0.005                   | 0.817 $\pm$ 0.003                   | 0.844 $\pm$ 0.022                   | 0.915 $\pm$ 0.008                   |
|                                                                   | 3     | 0.535 $\pm$ 0.010                   | 0.847 $\pm$ 0.006                   | 0.218 $\pm$ 0.009                   | 0.831 $\pm$ 0.009                   | 0.945 $\pm$ 0.008                   | 0.867 $\pm$ 0.007                   |
|                                                                   | 4     | 0.569 $\pm$ 0.008                   | 0.850 $\pm$ 0.006                   | 0.299 $\pm$ 0.004                   | 0.837 $\pm$ 0.006                   | 0.917 $\pm$ 0.014                   | 0.866 $\pm$ 0.011                   |
|                                                                   | 5     | 0.551 $\pm$ 0.012                   | 0.845 $\pm$ 0.005                   | 0.255 $\pm$ 0.012                   | 0.794 $\pm$ 0.005                   | 0.933 $\pm$ 0.013                   | 0.911 $\pm$ 0.010                   |
|                                                                   | 6     | 0.661 $\pm$ 0.011                   | 0.859 $\pm$ 0.004                   | 0.461 $\pm$ 0.019                   | 0.817 $\pm$ 0.008                   | 0.919 $\pm$ 0.018                   | 0.914 $\pm$ 0.015                   |
|                                                                   | 7     | 0.596 $\pm$ 0.010                   | 0.849 $\pm$ 0.004                   | 0.333 $\pm$ 0.016                   | 0.808 $\pm$ 0.006                   | 0.935 $\pm$ 0.016                   | 0.902 $\pm$ 0.010                   |
|                                                                   | 8     | 0.572 $\pm$ 0.012                   | 0.847 $\pm$ 0.006                   | 0.314 $\pm$ 0.019                   | 0.802 $\pm$ 0.009                   | 0.904 $\pm$ 0.016                   | 0.905 $\pm$ 0.008                   |
|                                                                   | 9     | 0.631 $\pm$ 0.008                   | 0.854 $\pm$ 0.007                   | 0.381 $\pm$ 0.015                   | 0.828 $\pm$ 0.002                   | 0.954 $\pm$ 0.008                   | 0.887 $\pm$ 0.012                   |
|                                                                   | 10    | 0.712 $\pm$ 0.016                   | 0.858 $\pm$ 0.005                   | 0.551 $\pm$ 0.019                   | 0.836 $\pm$ 0.005                   | 0.920 $\pm$ 0.015                   | 0.887 $\pm$ 0.010                   |
|                                                                   | 11    | <b>0.820 <math>\pm</math> 0.010</b> | <b>0.906 <math>\pm</math> 0.010</b> | <b>0.731 <math>\pm</math> 0.012</b> | <b>0.875 <math>\pm</math> 0.013</b> | <b>0.935 <math>\pm</math> 0.012</b> | <b>0.944 <math>\pm</math> 0.017</b> |
|                                                                   | 12    | 0.793 $\pm$ 0.015                   | 0.884 $\pm$ 0.006                   | 0.670 $\pm$ 0.019                   | 0.850 $\pm$ 0.012                   | 0.952 $\pm$ 0.013                   | 0.927 $\pm$ 0.015                   |
| Qwen3-0.6B                                                        | 1     | 0.550 $\pm$ 0.008                   | 0.894 $\pm$ 0.005                   | 0.223 $\pm$ 0.006                   | 0.840 $\pm$ 0.010                   | 0.972 $\pm$ 0.003                   | 0.965 $\pm$ 0.011                   |
|                                                                   | 2     | 0.572 $\pm$ 0.015                   | 0.882 $\pm$ 0.004                   | 0.363 $\pm$ 0.014                   | 0.828 $\pm$ 0.005                   | 0.841 $\pm$ 0.032                   | 0.952 $\pm$ 0.007                   |
|                                                                   | 3     | 0.541 $\pm$ 0.007                   | 0.889 $\pm$ 0.006                   | 0.216 $\pm$ 0.007                   | 0.827 $\pm$ 0.008                   | 0.962 $\pm$ 0.007                   | 0.968 $\pm$ 0.011                   |
|                                                                   | 4     | 0.580 $\pm$ 0.012                   | 0.891 $\pm$ 0.004                   | 0.353 $\pm$ 0.008                   | 0.845 $\pm$ 0.007                   | 0.874 $\pm$ 0.028                   | 0.950 $\pm$ 0.006                   |
|                                                                   | 5     | <b>0.487 <math>\pm</math> 0.017</b> | <b>0.893 <math>\pm</math> 0.007</b> | <b>0.133 <math>\pm</math> 0.012</b> | <b>0.839 <math>\pm</math> 0.011</b> | <b>0.944 <math>\pm</math> 0.014</b> | <b>0.964 <math>\pm</math> 0.009</b> |
|                                                                   | 6     | 0.568 $\pm$ 0.012                   | 0.897 $\pm$ 0.005                   | 0.350 $\pm$ 0.010                   | 0.857 $\pm$ 0.008                   | 0.850 $\pm$ 0.023                   | 0.948 $\pm$ 0.012                   |
|                                                                   | 7     | 0.541 $\pm$ 0.007                   | 0.902 $\pm$ 0.006                   | 0.211 $\pm$ 0.009                   | 0.862 $\pm$ 0.008                   | 0.967 $\pm$ 0.012                   | 0.953 $\pm$ 0.012                   |
|                                                                   | 8     | 0.609 $\pm$ 0.009                   | 0.903 $\pm$ 0.006                   | 0.362 $\pm$ 0.013                   | 0.861 $\pm$ 0.010                   | 0.929 $\pm$ 0.017                   | 0.956 $\pm$ 0.005                   |
|                                                                   | 9     | 0.612 $\pm$ 0.013                   | 0.900 $\pm$ 0.004                   | 0.393 $\pm$ 0.016                   | 0.868 $\pm$ 0.008                   | 0.894 $\pm$ 0.021                   | 0.943 $\pm$ 0.006                   |
|                                                                   | 10    | 0.638 $\pm$ 0.013                   | 0.890 $\pm$ 0.009                   | 0.452 $\pm$ 0.009                   | 0.847 $\pm$ 0.014                   | 0.878 $\pm$ 0.022                   | 0.944 $\pm$ 0.010                   |
|                                                                   | 11    | 0.578 $\pm$ 0.006                   | 0.885 $\pm$ 0.005                   | 0.257 $\pm$ 0.010                   | 0.838 $\pm$ 0.014                   | 0.991 $\pm$ 0.005                   | 0.947 $\pm$ 0.012                   |
|                                                                   | 12    | 0.627 $\pm$ 0.009                   | 0.892 $\pm$ 0.005                   | 0.366 $\pm$ 0.016                   | 0.844 $\pm$ 0.010                   | 0.964 $\pm$ 0.013                   | 0.955 $\pm$ 0.007                   |
|                                                                   | 13    | 0.639 $\pm$ 0.010                   | 0.887 $\pm$ 0.006                   | 0.383 $\pm$ 0.014                   | 0.849 $\pm$ 0.011                   | 0.970 $\pm$ 0.015                   | 0.937 $\pm$ 0.011                   |
|                                                                   | 14    | 0.646 $\pm$ 0.009                   | 0.885 $\pm$ 0.006                   | 0.402 $\pm$ 0.017                   | 0.852 $\pm$ 0.009                   | 0.960 $\pm$ 0.016                   | 0.927 $\pm$ 0.007                   |
|                                                                   | 15    | 0.648 $\pm$ 0.012                   | 0.885 $\pm$ 0.007                   | 0.410 $\pm$ 0.017                   | 0.852 $\pm$ 0.010                   | 0.954 $\pm$ 0.017                   | 0.929 $\pm$ 0.018                   |
|                                                                   | 16    | 0.691 $\pm$ 0.016                   | 0.873 $\pm$ 0.005                   | 0.546 $\pm$ 0.017                   | 0.837 $\pm$ 0.009                   | 0.878 $\pm$ 0.017                   | 0.920 $\pm$ 0.016                   |
|                                                                   | 17    | 0.658 $\pm$ 0.010                   | 0.862 $\pm$ 0.006                   | 0.420 $\pm$ 0.018                   | 0.807 $\pm$ 0.012                   | 0.966 $\pm$ 0.013                   | 0.933 $\pm$ 0.013                   |
|                                                                   | 18    | 0.709 $\pm$ 0.012                   | 0.868 $\pm$ 0.002                   | 0.503 $\pm$ 0.018                   | 0.833 $\pm$ 0.013                   | 0.975 $\pm$ 0.006                   | 0.914 $\pm$ 0.017                   |
|                                                                   | 19    | 0.691 $\pm$ 0.010                   | 0.874 $\pm$ 0.002                   | 0.462 $\pm$ 0.018                   | 0.860 $\pm$ 0.007                   | 0.986 $\pm$ 0.006                   | 0.893 $\pm$ 0.012                   |
|                                                                   | 20    | 0.743 $\pm$ 0.020                   | 0.860 $\pm$ 0.004                   | 0.588 $\pm$ 0.023                   | 0.797 $\pm$ 0.011                   | 0.943 $\pm$ 0.018                   | 0.941 $\pm$ 0.010                   |
|                                                                   | 21    | 0.758 $\pm$ 0.017                   | 0.879 $\pm$ 0.006                   | 0.585 $\pm$ 0.021                   | 0.836 $\pm$ 0.008                   | 0.981 $\pm$ 0.010                   | 0.935 $\pm$ 0.010                   |
|                                                                   | 22    | 0.799 $\pm$ 0.014                   | 0.882 $\pm$ 0.006                   | 0.671 $\pm$ 0.019                   | 0.855 $\pm$ 0.002                   | 0.965 $\pm$ 0.006                   | 0.918 $\pm$ 0.015                   |
|                                                                   | 23    | 0.757 $\pm$ 0.012                   | 0.887 $\pm$ 0.006                   | 0.577 $\pm$ 0.014                   | 0.857 $\pm$ 0.007                   | 0.989 $\pm$ 0.004                   | 0.927 $\pm$ 0.011                   |
|                                                                   | 24    | 0.810 $\pm$ 0.014                   | 0.893 $\pm$ 0.005                   | 0.695 $\pm$ 0.021                   | 0.861 $\pm$ 0.004                   | 0.959 $\pm$ 0.003                   | 0.935 $\pm$ 0.011                   |
|                                                                   | 25    | 0.732 $\pm$ 0.011                   | 0.890 $\pm$ 0.009                   | 0.527 $\pm$ 0.016                   | 0.866 $\pm$ 0.006                   | 0.997 $\pm$ 0.003                   | 0.920 $\pm$ 0.014                   |
|                                                                   | 26    | 0.749 $\pm$ 0.013                   | 0.895 $\pm$ 0.008                   | 0.562 $\pm$ 0.017                   | 0.858 $\pm$ 0.004                   | 0.991 $\pm$ 0.003                   | 0.943 $\pm$ 0.015                   |
|                                                                   | 27    | 0.812 $\pm$ 0.004                   | 0.881 $\pm$ 0.005                   | 0.741 $\pm$ 0.008                   | 0.824 $\pm$ 0.006                   | 0.903 $\pm$ 0.011                   | 0.954 $\pm$ 0.011                   |

(continued on next page)

| Task B: Top-K Ranking (appendix, 5-seed mean $\pm$ std, SIF+ABTT) |       |                                     |                                     |                                     |                                     |                                     |                                     |
|-------------------------------------------------------------------|-------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| Model                                                             | Layer | Acc@1<br>(base)                     | Acc@1<br>(SIF+ABTT)                 | Existing<br>(base)                  | Existing<br>(SIF+ABTT)              | New<br>(base)                       | New<br>(SIF+ABTT)                   |
|                                                                   | 28    | 0.824 $\pm$ 0.010                   | 0.871 $\pm$ 0.006                   | 0.736 $\pm$ 0.015                   | 0.802 $\pm$ 0.010                   | 0.939 $\pm$ 0.005                   | 0.961 $\pm$ 0.007                   |
| KaLM-mini                                                         | 1     | <b>0.596 <math>\pm</math> 0.012</b> | <b>0.893 <math>\pm</math> 0.005</b> | <b>0.344 <math>\pm</math> 0.003</b> | <b>0.846 <math>\pm</math> 0.005</b> | <b>0.922 <math>\pm</math> 0.022</b> | <b>0.953 <math>\pm</math> 0.006</b> |
|                                                                   | 2     | 0.620 $\pm$ 0.018                   | 0.893 $\pm$ 0.003                   | 0.422 $\pm$ 0.014                   | 0.845 $\pm$ 0.005                   | 0.876 $\pm$ 0.023                   | 0.955 $\pm$ 0.008                   |
|                                                                   | 3     | 0.605 $\pm$ 0.011                   | 0.893 $\pm$ 0.004                   | 0.379 $\pm$ 0.008                   | 0.843 $\pm$ 0.000                   | 0.896 $\pm$ 0.018                   | 0.959 $\pm$ 0.010                   |
|                                                                   | 4     | 0.562 $\pm$ 0.015                   | 0.889 $\pm$ 0.005                   | 0.265 $\pm$ 0.009                   | 0.842 $\pm$ 0.008                   | 0.945 $\pm$ 0.016                   | 0.949 $\pm$ 0.008                   |
|                                                                   | 5     | 0.550 $\pm$ 0.013                   | 0.890 $\pm$ 0.004                   | 0.270 $\pm$ 0.005                   | 0.841 $\pm$ 0.004                   | 0.910 $\pm$ 0.019                   | 0.952 $\pm$ 0.009                   |
|                                                                   | 6     | 0.522 $\pm$ 0.016                   | 0.893 $\pm$ 0.004                   | 0.203 $\pm$ 0.012                   | 0.850 $\pm$ 0.005                   | 0.934 $\pm$ 0.014                   | 0.949 $\pm$ 0.014                   |
|                                                                   | 7     | 0.604 $\pm$ 0.018                   | 0.893 $\pm$ 0.003                   | 0.393 $\pm$ 0.018                   | 0.840 $\pm$ 0.008                   | 0.876 $\pm$ 0.023                   | 0.962 $\pm$ 0.008                   |
|                                                                   | 8     | 0.607 $\pm$ 0.014                   | 0.866 $\pm$ 0.003                   | 0.388 $\pm$ 0.011                   | 0.798 $\pm$ 0.008                   | 0.891 $\pm$ 0.017                   | 0.954 $\pm$ 0.007                   |
|                                                                   | 9     | 0.614 $\pm$ 0.011                   | 0.874 $\pm$ 0.005                   | 0.387 $\pm$ 0.008                   | 0.821 $\pm$ 0.008                   | 0.908 $\pm$ 0.013                   | 0.943 $\pm$ 0.008                   |
|                                                                   | 10    | 0.602 $\pm$ 0.013                   | 0.859 $\pm$ 0.008                   | 0.360 $\pm$ 0.005                   | 0.774 $\pm$ 0.010                   | 0.916 $\pm$ 0.017                   | 0.969 $\pm$ 0.010                   |
|                                                                   | 11    | 0.601 $\pm$ 0.012                   | 0.864 $\pm$ 0.008                   | 0.361 $\pm$ 0.007                   | 0.792 $\pm$ 0.010                   | 0.911 $\pm$ 0.020                   | 0.956 $\pm$ 0.008                   |
|                                                                   | 12    | 0.640 $\pm$ 0.010                   | 0.868 $\pm$ 0.006                   | 0.416 $\pm$ 0.006                   | 0.801 $\pm$ 0.007                   | 0.929 $\pm$ 0.019                   | 0.954 $\pm$ 0.006                   |
|                                                                   | 13    | 0.645 $\pm$ 0.011                   | 0.862 $\pm$ 0.005                   | 0.436 $\pm$ 0.005                   | 0.793 $\pm$ 0.004                   | 0.915 $\pm$ 0.018                   | 0.951 $\pm$ 0.012                   |
|                                                                   | 14    | 0.654 $\pm$ 0.010                   | 0.859 $\pm$ 0.006                   | 0.479 $\pm$ 0.004                   | 0.795 $\pm$ 0.008                   | 0.880 $\pm$ 0.017                   | 0.941 $\pm$ 0.011                   |
|                                                                   | 15    | 0.661 $\pm$ 0.009                   | 0.870 $\pm$ 0.006                   | 0.460 $\pm$ 0.009                   | 0.814 $\pm$ 0.009                   | 0.921 $\pm$ 0.014                   | 0.941 $\pm$ 0.006                   |
|                                                                   | 16    | 0.685 $\pm$ 0.008                   | 0.870 $\pm$ 0.007                   | 0.477 $\pm$ 0.009                   | 0.824 $\pm$ 0.008                   | 0.953 $\pm$ 0.015                   | 0.930 $\pm$ 0.008                   |
|                                                                   | 17    | 0.757 $\pm$ 0.006                   | 0.886 $\pm$ 0.010                   | 0.612 $\pm$ 0.016                   | 0.841 $\pm$ 0.005                   | 0.944 $\pm$ 0.010                   | 0.943 $\pm$ 0.016                   |
|                                                                   | 18    | 0.781 $\pm$ 0.008                   | 0.880 $\pm$ 0.008                   | 0.636 $\pm$ 0.013                   | 0.840 $\pm$ 0.002                   | 0.967 $\pm$ 0.007                   | 0.931 $\pm$ 0.016                   |
|                                                                   | 19    | 0.805 $\pm$ 0.012                   | 0.877 $\pm$ 0.008                   | 0.673 $\pm$ 0.020                   | 0.831 $\pm$ 0.007                   | 0.976 $\pm$ 0.003                   | 0.935 $\pm$ 0.011                   |
|                                                                   | 20    | 0.794 $\pm$ 0.010                   | 0.878 $\pm$ 0.006                   | 0.650 $\pm$ 0.018                   | 0.831 $\pm$ 0.003                   | 0.979 $\pm$ 0.004                   | 0.937 $\pm$ 0.011                   |
|                                                                   | 21    | 0.805 $\pm$ 0.010                   | 0.871 $\pm$ 0.007                   | 0.673 $\pm$ 0.016                   | 0.831 $\pm$ 0.004                   | 0.975 $\pm$ 0.006                   | 0.924 $\pm$ 0.012                   |
|                                                                   | 22    | 0.871 $\pm$ 0.006                   | 0.873 $\pm$ 0.006                   | 0.811 $\pm$ 0.005                   | 0.813 $\pm$ 0.007                   | 0.949 $\pm$ 0.006                   | 0.951 $\pm$ 0.007                   |
|                                                                   | 23    | 0.873 $\pm$ 0.007                   | 0.885 $\pm$ 0.005                   | 0.811 $\pm$ 0.006                   | 0.835 $\pm$ 0.007                   | 0.953 $\pm$ 0.007                   | 0.950 $\pm$ 0.006                   |
|                                                                   | 24    | 0.848 $\pm$ 0.009                   | 0.856 $\pm$ 0.004                   | 0.749 $\pm$ 0.017                   | 0.807 $\pm$ 0.006                   | 0.975 $\pm$ 0.007                   | 0.918 $\pm$ 0.009                   |

## H ABTT Component Ablations

This appendix gives the two tables behind the  $D$  ablation (H1) and the subspace split (E3) of Section 5. Table 24 gives Task A test AUROC against the number of removed components at each model’s lowest-AUROC baseline layer, with medians over groups of layers; Figure 3 plots the same worst-layer curves. Table 25 ranks passages within the removed and the retained subspace, with two dimension-matched controls inside the retained one. Every quantity is fit on training embeddings only.

## I Reference Systems: the Fine-Tuning Ceilings

Tables 7 and 8 carry four reference rows below the rule: one supervised fine-tuning ceiling per fine-tuned model, and the character 3–5-gram TF-IDF reference. Table 26 gives the three fine-tuning ceilings with the pre-trained rows they are compared against. Every model we fine-tuned is reported, on whichever side of its own zero-shot row it lands.

Three encoders are fine-tuned under one recipe, LaTa, Qwen3-0.6B and KaLM-mini. Fine-tuning only the Latin-pretrained model would leave the ceiling confounded with Latin pretraining, so the second model is a multilingual decoder-only encoder with no Latin-specific pretraining, Qwen3-0.6B. The third answers the remaining question, whether supervision can clear the best cell the correction produces anywhere in the paper: KaLM-mini holds it, at 91.7 assignment accuracy and at 89.4 DirAcc@1 tied with Qwen3-0.6B. All three are trained with symmetric InfoNCE over positive pairs and in-batch negatives (no two pairs from one directory in a batch), temperature 0.05, AdamW at learning rate  $2 \times 10^{-5}$  with weight decay 0.01, linear warmup over the first 10 percent of steps then linear decay, gradient clipping at 1.0, 16 pairs per batch, up to 8 epochs with early stopping after 3 epochs without dev improvement, and seed 42. All three are mean-pooled with the same token filter as their zero-shot rows, so the fine-tuned vectors live in the space those rows are measured in; Qwen3-0.6B and KaLM-mini add gradient checkpointing, which recomputes activations rather than storing them and leaves the gradients unchanged. The dev slice holds out 28 of the 190 train directories with at least two files (71 files), the same directories for every model, so each trains on 499 of the 565 positive train pairs, and the checkpoint

is chosen by dev DirAcc@1 with dev AUROC as the tie-break. Accuracy differences at or below one dev file are treated as ties, because one file out of 71 is the pool’s whole resolution; without that, a model whose pre-trained encoder already routes all 71 dev files can never select a trained checkpoint. A 71-file pool is the weak link in this protocol: one borderline file flipped between two otherwise identical Qwen3-0.6B runs at a training loss of  $6 \times 10^{-4}$ , which is GPU non-determinism rather than a difference in the model, and it is worth 1.4 points of the selection criterion. LaTa selects epoch 7, the last epoch run before early stopping fired, and its ABTT sweep selects  $D = 10$ , the top of the grid, at every layer, so the tuned and fixed ABTT variants coincide there. Qwen3-0.6B selects epoch 3 of 6 run, and its sweep is a genuine one, selecting the top of the grid at 12 of 28 layers and  $D = 2$  at the layer its Task B row reports. KaLM-mini selects epoch 1 of 4 run, the only model of the three whose pre-trained encoder leaves the dev pool room to improve, and its sweep selects the top of the grid at 19 of 24 layers including the layer its Task B row reports. All three rows are ceilings at this training budget rather than asymptotes: training loss reaches  $4 \times 10^{-5}$  on Qwen3-0.6B by epoch 4 and  $3 \times 10^{-5}$  on KaLM-mini, so the 499 pairs are memorized and more epochs buy nothing.

**Fine-tuned LaTa by layer.** Table 27 gives fine-tuned LaTa at every layer, with no post-hoc correction, beside the pre-trained model. Fine-tuning lifts the last layer from 0.938 to 0.984 AUROC and leaves layers 2 to 11 at 0.499 to 0.569. Their top-PC share stays at 0.759 to 0.945, against 0.771 to 0.952 before fine-tuning, so the collapsed geometry stays in place.

**Lexical reference systems.** Table 28 gives the three lexical reference systems (BM25, character 3–5-gram TF-IDF, and Levenshtein similarity) on both tasks; the character  $n$ -gram row is the reference quoted in Section 7.

**The parameter-free repair matches contrastive fine-tuning on LaTa routing, but supervision clears it on Qwen3-0.6B and KaLM-mini.** Fine-tuning LaTa on the positive train pairs (499 after a directory-level dev carve; setup above) gives an uncorrected Task A AUROC of 0.984, above every zero-shot baseline row (0.838 to 0.972), and 83.4 assignment accuracy; with ABTT on top it reaches 87.8 assignment accuracy and 85.2 DirAcc@1, be-

| Layers                                    | $n$ | raw   | Removed components $D$ |       |       |       |       |       |       |       |       |       |       |
|-------------------------------------------|-----|-------|------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                           |     |       | 0                      | 1     | 2     | 3     | 5     | 7     | 10    | 15    | 20    | 30    | 50    |
| <i>Worst baseline layer of each model</i> |     |       |                        |       |       |       |       |       |       |       |       |       |       |
| LaTa (6)                                  | 1   | 0.496 | 0.482                  | 0.671 | 0.940 | 0.973 | 0.973 | 0.977 | 0.964 | 0.973 | 0.966 | 0.981 | 0.986 |
| PhilTa (10)                               | 1   | 0.538 | 0.501                  | 0.602 | 0.888 | 0.964 | 0.977 | 0.982 | 0.982 | 0.975 | 0.977 | 0.973 | 0.969 |
| mT5-base (5)                              | 1   | 0.654 | 0.670                  | 0.835 | 0.892 | 0.914 | 0.951 | 0.971 | 0.978 | 0.972 | 0.965 | 0.965 | 0.967 |
| LaBSE (1)                                 | 1   | 0.806 | 0.867                  | 0.886 | 0.915 | 0.929 | 0.962 | 0.967 | 0.977 | 0.977 | 0.977 | 0.977 | 0.979 |
| Qwen3-0.6B (1)                            | 1   | 0.858 | 0.954                  | 0.959 | 0.962 | 0.970 | 0.972 | 0.979 | 0.978 | 0.980 | 0.981 | 0.976 | 0.984 |
| KaLM-mini (5)                             | 1   | 0.861 | 0.917                  | 0.926 | 0.969 | 0.969 | 0.983 | 0.982 | 0.983 | 0.978 | 0.974 | 0.974 | 0.976 |
| <i>Median over model-layers</i>           |     |       |                        |       |       |       |       |       |       |       |       |       |       |
| Collapsed T5                              | 26  | 0.541 | 0.503                  | 0.713 | 0.902 | 0.964 | 0.973 | 0.977 | 0.977 | 0.974 | 0.975 | 0.974 | 0.971 |
| LaTa                                      | 10  | 0.502 | 0.488                  | 0.713 | 0.956 | 0.973 | 0.972 | 0.976 | 0.970 | 0.971 | 0.969 | 0.980 | 0.983 |
| PhilTa                                    | 9   | 0.542 | 0.502                  | 0.602 | 0.872 | 0.963 | 0.980 | 0.981 | 0.981 | 0.975 | 0.978 | 0.973 | 0.970 |
| mT5-base                                  | 7   | 0.657 | 0.678                  | 0.865 | 0.894 | 0.934 | 0.955 | 0.974 | 0.978 | 0.975 | 0.974 | 0.965 | 0.962 |
| Other T5                                  | 10  | 0.874 | 0.936                  | 0.952 | 0.967 | 0.973 | 0.979 | 0.976 | 0.973 | 0.972 | 0.972 | 0.973 | 0.971 |
| Embedding-trained                         | 64  | 0.903 | 0.944                  | 0.966 | 0.971 | 0.976 | 0.979 | 0.980 | 0.978 | 0.977 | 0.975 | 0.974 | 0.973 |
| All                                       | 100 | 0.885 | 0.936                  | 0.958 | 0.968 | 0.973 | 0.978 | 0.979 | 0.977 | 0.975 | 0.974 | 0.974 | 0.972 |

Table 24: Task A test AUROC against the number  $D$  of top principal components that ABTT removes, fit on training embeddings only. Raw: mean-pooled vectors with no correction;  $D = 0$ : centering on the training mean alone. The top block reads each model at its lowest-AUROC baseline layer (layer in parentheses); the bottom block gives medians over the 26 collapsed T5 layers (baseline AUROC below 0.70), also split by model, the other 10 T5 layers, the 64 layers of the embedding-trained models, and all 100 model-layers. The paper’s selection grid for  $D$  stops at 10.

| Model                                                        | $\ell$ | $D$ | Var. | Full vector |       | Removed subspace |            |            | Retained subspace |           |       |
|--------------------------------------------------------------|--------|-----|------|-------------|-------|------------------|------------|------------|-------------------|-----------|-------|
|                                                              |        |     |      | Raw         | Cent. | PC1              | PCs 2– $D$ | PCs 1– $D$ | Next $D$          | Rand. $D$ | All   |
| <i>Pre-trained, train-selected layer and <math>D</math></i>  |        |     |      |             |       |                  |            |            |                   |           |       |
| LaTa                                                         | 12     | 10  | 0.35 | 0.938       | 0.972 | 0.670            | 0.825      | 0.868      | 0.907             | 0.861     | 0.971 |
| PhilTa                                                       | 9      | 10  | 0.99 | 0.539       | 0.501 | 0.462            | 0.545      | 0.487      | 0.922             | 0.873     | 0.983 |
| mT5-base                                                     | 2      | 10  | 0.57 | 0.822       | 0.910 | 0.610            | 0.735      | 0.751      | 0.896             | 0.866     | 0.975 |
| LaBSE                                                        | 11     | 10  | 0.48 | 0.960       | 0.958 | 0.647            | 0.932      | 0.909      | 0.958             | 0.902     | 0.987 |
| Qwen3-0.6B                                                   | 2      | 10  | 0.44 | 0.863       | 0.940 | 0.607            | 0.792      | 0.798      | 0.906             | 0.882     | 0.973 |
| KaLM-mini                                                    | 1      | 10  | 0.41 | 0.887       | 0.958 | 0.518            | 0.846      | 0.819      | 0.910             | 0.880     | 0.981 |
| Collapsed T5, median (26)                                    | –      | –   | 0.99 | 0.541       | 0.503 | 0.463            | 0.601      | 0.489      | 0.919             | 0.874     | 0.977 |
| <i>Fine-tuning contrast, same layer, <math>D = 10</math></i> |        |     |      |             |       |                  |            |            |                   |           |       |
| LaTa, pre-trained                                            | 12     | 10  | 0.35 | 0.938       | 0.972 | 0.670            | 0.825      | 0.868      | 0.907             | 0.861     | 0.971 |
| LaTa, fine-tuned                                             | 12     | 10  | 0.33 | 0.984       | 0.985 | 0.697            | 0.935      | 0.953      | 0.900             | 0.874     | 0.970 |
| Qwen3-0.6B, pre-trained                                      | 27     | 10  | 0.41 | 0.967       | 0.969 | 0.643            | 0.913      | 0.917      | 0.909             | 0.884     | 0.982 |
| Qwen3-0.6B, fine-tuned                                       | 27     | 10  | 0.31 | 0.993       | 0.996 | 0.684            | 0.980      | 0.983      | 0.969             | 0.927     | 0.992 |
| KaLM-mini, pre-trained                                       | 24     | 10  | 0.32 | 0.956       | 0.979 | 0.663            | 0.909      | 0.924      | 0.909             | 0.855     | 0.981 |
| KaLM-mini, fine-tuned                                        | 24     | 10  | 0.23 | 0.997       | 0.996 | 0.699            | 0.977      | 0.978      | 0.967             | 0.918     | 0.994 |

Table 25: Task A test AUROC of cosine ranking within parts of the pooled vector. ABTT removes the top  $D$  principal components of the centered training embeddings; Var. is their share of the centered training variance. Raw: uncorrected vector; Cent.: centered on the training mean; PC1: projection onto the first component alone, where cosine reduces to a sign match; PCs 2- $D$  and PCs 1- $D$ : projections onto the rest of the removed subspace and onto all of it. Retained subspace, a dimension-matched control: Next  $D$ , PCs  $D+1$  to  $2D$ ; Rand.  $D$ , random  $D$ -dimensional projections of the retained vectors (mean over five seeds); All, the ABTT output. Top block: each pre-trained model at its train-selected ABTT layer  $\ell$  with  $D$  chosen on training DirAcc@1, and the median over the 26 collapsed T5 layers (baseline AUROC below 0.70), each at its own layer and  $D$ . Bottom block: each contrastively fine-tuned encoder at its train-selected ABTT layer, next to its pre-trained model at the same layer, both with  $D = 10$ . All components are fit on training embeddings only.

low every zero-shot ABTT cell (88.5 to 91.7 and 86.1 to 89.4). ABTT also lowers that row’s AUROC from 0.984 to 0.970 in point estimate (within the bootstrap interval of Appendix J) while widening its cosine gap from 0.387 to 0.548, consistent with fine-tuning already removing part of the common component: the repair does not help every

row, and it matches what the available supervision buys on routing at this budget. The same recipe lands elsewhere on the other two encoders we fine-tuned: Qwen3-0.6B reaches 92.1 assignment accuracy and 91.4 DirAcc@1 with ABTT, and KaLM-mini 93.4 and 92.5, both above every zero-shot ABTT cell in point estimate; on the single split

| System                          | Task A |            | Task B       |             |                     |
|---------------------------------|--------|------------|--------------|-------------|---------------------|
|                                 | AUROC  | Cosine gap | Assign. acc. | Dir. acc.@1 | Dir. acc.@1 5 seeds |
| LaTa (pre-trained)              | 0.938  | 0.237      | 73.8         | 72.1        | 73.1 $\pm$ 1.0      |
| LaTa (pre-trained) + ABTT       | 0.971  | 0.525      | 88.5         | 86.1        | 87.7 $\pm$ 0.4      |
| LaTa (fine-tuned)               | 0.984  | 0.387      | 83.4         | 81.6        | 83.4 $\pm$ 0.9      |
| LaTa (fine-tuned) + ABTT        | 0.970  | 0.548      | 87.8         | 85.2        | 87.7 $\pm$ 0.8      |
| Qwen3-0.6B (pre-trained)        | 0.966  | 0.024      | 82.3         | 80.3        | 82.4 $\pm$ 1.0      |
| Qwen3-0.6B (pre-trained) + ABTT | 0.973  | 0.553      | 91.5         | 89.4        | 90.5 $\pm$ 0.4      |
| Qwen3-0.6B (fine-tuned)         | 0.996  | 0.684      | 91.4         | 90.8        | 91.7 $\pm$ 0.6      |
| Qwen3-0.6B (fine-tuned) + ABTT  | 0.994  | 0.716      | 92.1         | 91.4        | 92.3 $\pm$ 0.2      |
| KaLM-mini (pre-trained)         | 0.972  | 0.056      | 87.5         | 85.9        | 87.3 $\pm$ 0.7      |
| KaLM-mini (pre-trained) + ABTT  | 0.981  | 0.568      | 91.7         | 89.4        | 90.9 $\pm$ 0.6      |
| KaLM-mini (fine-tuned)          | 0.997  | 0.636      | 92.4         | 91.7        | 92.6 $\pm$ 0.8      |
| KaLM-mini (fine-tuned) + ABTT   | 0.994  | 0.651      | 93.4         | 92.5        | 93.0 $\pm$ 0.5      |

Table 26: Supervised fine-tuning reference ceiling on LaTa, Qwen3-0.6B and KaLM-mini. Fine-tuning raises Task A AUROC for every model, and ABTT on top of it lowers AUROC again in point estimate; on five-seed routing, the fine-tuned encoder with ABTT is level with its pre-trained ABTT row for LaTa and above it for Qwen3-0.6B and KaLM-mini. The fine-tuned encoders are trained contrastively on 499 of the 565 positive train pairs (in-batch negatives, symmetric InfoNCE), with a dev slice carved out of train by directory for model selection; the test split is untouched until this evaluation. Task A columns are test AUROC and cosine gap at the layer chosen by train AUROC; Task B columns are test assignment accuracy and directory accuracy at rank 1, in percent, at the layer chosen by train directory accuracy, with  $\tau$  learned on train. The selected layers are listed in Table 11. The last column repeats directory accuracy at rank 1 at the same layer as a mean  $\pm$  standard deviation over five random reassignments of query and reference files. ABTT rows for LaTa sweep  $D$  per layer on train and select  $D = 10$  everywhere, which is the top of the grid  $\{1, 2, 3, 5, 7, 10\}$ , so the tuned and fixed variants coincide. The selected checkpoint for LaTa is epoch 7, the terminal epoch of the sweep under the 8-epoch budget, so this is a ceiling at this training budget rather than an asymptote. ABTT rows for Qwen3-0.6B sweep  $D$  per layer on train over the grid  $\{1, 2, 3, 5, 7, 10\}$ ; 12 of 28 layer rows select the top of the grid. The selected checkpoint for Qwen3-0.6B is epoch 3 of 6 run under the 8-epoch budget, so this is a ceiling at this training budget rather than an asymptote. ABTT rows for KaLM-mini sweep  $D$  per layer on train over the grid  $\{1, 2, 3, 5, 7, 10\}$ ; 19 of 24 layer rows select the top of the grid. The selected checkpoint for KaLM-mini is epoch 1 of 4 run under the 8-epoch budget, so this is a ceiling at this training budget rather than an asymptote. This is a reference ceiling, not a proposed method: it consumes supervision the zero-shot pipeline never sees.

only KaLM-mini’s DirAcc@1 lead over its zero-shot ABTT cell excludes zero (+3.1, 1.2 to 5.2; Appendix J). Over five Task B query and reference reseeds the fine-tuned encoder with ABTT beats its zero-shot ABTT row by 1.8 points on Qwen3-0.6B ( $0.923 \pm 0.002$  against  $0.905 \pm 0.004$ ) and by 2.1 points on KaLM-mini ( $0.930 \pm 0.005$  against  $0.909 \pm 0.006$ ), while on LaTa the two are 0.877 and 0.877. So the match is a statement about LaTa, not about post-processing in general; it does not survive on the model with the best zero-shot cell in Table 8. All three ceilings are also flattered: 206 of the 535 test queries whose directory already exists (38.5 percent) sit in a directory that supplied a training pair, and witnesses inside a directory are near-duplicates, so read each margin as an upper bound.

## J Bootstrap Intervals and Routing Checks

**Intervals.** A bootstrap replicate draws the 514 test directories with replacement ( $B = 10,000$ , seed 233). A drawn directory keeps every file and pair: a directory drawn  $c$  times weights its files and within-directory pairs by  $c$  and its pairs with a directory drawn  $c'$  times by  $cc'$ , and no pair is formed between two copies of one directory. The layer, threshold, ABTT components,  $D$ , and SIF weights stay fit on the training split, and every score is computed once on the full test set, so a replicate only reweights test pairs and files. Intervals are 95% percentiles; a paired difference recomputes both cells on each replicate. Before resampling, every cell was recomputed with the paper’s evaluation code and matched its printed value. Table 29 gives every cell and Table 30 the contrasts quoted in Section 7.

| Layer | AUROC |       | Top-PC share |       | Eff. rank |        |
|-------|-------|-------|--------------|-------|-----------|--------|
|       | PT    | FT    | PT           | FT    | PT        | FT     |
| 1     | 0.934 | 0.936 | 0.050        | 0.050 | 134.33    | 134.44 |
| 2     | 0.563 | 0.569 | 0.771        | 0.759 | 4.61      | 4.92   |
| 3     | 0.513 | 0.512 | 0.935        | 0.929 | 1.59      | 1.64   |
| 4     | 0.500 | 0.499 | 0.952        | 0.945 | 1.38      | 1.43   |
| 5     | 0.496 | 0.500 | 0.943        | 0.931 | 1.41      | 1.48   |
| 6     | 0.496 | 0.500 | 0.934        | 0.921 | 1.44      | 1.51   |
| 7     | 0.498 | 0.501 | 0.937        | 0.923 | 1.43      | 1.50   |
| 8     | 0.502 | 0.505 | 0.950        | 0.940 | 1.36      | 1.42   |
| 9     | 0.503 | 0.507 | 0.943        | 0.931 | 1.40      | 1.48   |
| 10    | 0.505 | 0.510 | 0.942        | 0.930 | 1.43      | 1.50   |
| 11    | 0.508 | 0.512 | 0.939        | 0.927 | 1.47      | 1.55   |
| 12    | 0.938 | 0.984 | 0.081        | 0.069 | 132.22    | 137.68 |

Table 27: LaTa before (PT) and after (FT) contrastive fine-tuning on training pairs, per layer, with no post-hoc correction. AUROC: Task A test pairwise AUROC of cosine on mean-pooled vectors. Top-PC share and effective rank: label-free geometry of the centered training vectors, which include the passages of the fine-tuning pairs. Fine-tuning lifts the last layer and leaves the middle layers collapsed.

**Routing checks.** Table 31 reports, at the Task B cells, the AUROC of each test file’s maximum cosine for existing against new files (no threshold), the gap between the best test threshold and the training threshold, the skewness of the 10-occurrence distribution (Radovanović et al., 2010), centering alone ( $D = 0$ ), and assignment accuracy with  $\tau$  refit as the exact best-F1 cut over all training pair scores.

**Threshold grid.** We keep the published grid of 200 evenly spaced thresholds in  $[0, 1]$ . Every finer grid optimizes the same training objective, the best F1 over same- against different-directory training pairs, and differs only in resolution. With the layer and  $D$  fixed, a grid of 10,001 values moves 45 of the 66 printed routing numbers, by a median of 3 files (0.35 points); ABTT, SIF+ABTT, and fine-tuned cells move by at most 0.8 points, and Base and SIF cells by up to 2.6 (mT5-base’s baseline). The exact best-F1 cut moves 48 of them. Refitting would trade one set of differences of one to a few files for another and changes no conclusion of Section 7.

## K Token Attribution at Retrieval-Selected Layers

This appendix compares local explanations of the retrieval score before and after ABTT: methods, metrics, setup, and results. Appendix L gives the secondary metrics and sweeps, and Appendix M gives one qualitative example.

| Baseline        | Task A |       | Task B  |       |
|-----------------|--------|-------|---------|-------|
|                 | AUROC  | Gap   | Assign. | Dir@1 |
| BM25 (word)     | 0.974  | 0.243 | 62.5    | 56.3  |
| TF-IDF char 3–5 | 0.987  | 0.555 | 91.8    | 89.9  |
| Levenshtein     | 0.952  | 0.362 | 87.8    | 86.2  |

Table 28: Lexical baselines on the same 50/50 split, the same held-out test set and the same evaluation code as the embedding models, so the numbers are directly comparable to the headline Task A and Task B tables. Task A is measured over the test  $n \times n$  score matrix: AUROC treats same-directory test pairs as positives, and Gap is the mean same-directory minus the mean different-directory score. Task B is reported in percent: Assign. is the existing-versus-new decision against a threshold  $\tau$  fitted on train, and Dir@1 additionally requires the correct directory. TF-IDF char 3–5 is the strongest of the three at 89.9 percent Dir@1, which puts plain surface-form overlap within reach of the best embedding configurations on this corpus of hand-copied witnesses. BM25 separates pairs well but routes poorly: a raw BM25 score grows with query length, so each file’s maximum score tracks document length as much as relatedness, and the train-fitted  $\tau$  transfers badly. BM25 uses  $k_1 = 1.5$ ,  $b = 0.75$  with document frequencies, vocabulary and average document length taken from train files only; the character TF-IDF vectoriser is likewise fitted on train only, and normalised Levenshtein has nothing to fit. Every score matrix is min-max rescaled using the train block; the map is monotone, so AUROC is unchanged, while the threshold  $\tau$  is fitted on a fixed grid afterwards, which matters for BM25 (unrescaled,  $\tau$  pins at the grid ceiling) and is the identity for char TF-IDF, whose train scores already span  $[0, 1]$ . Under the 5-seed query-versus-reference Task B protocol, in which only the query/reference partition of the test half is redrawn, directory accuracy at rank 1 is BM25 (word)  $68.3 \pm 7.1$ ; TF-IDF char 3–5  $91.1 \pm 0.5$ ; Levenshtein  $87.2 \pm 0.3$ .

### K.1 Local Attribution Methods

We evaluate local explanations for a fixed query-candidate retrieval score. The design goal is a review aid: attributions are shown as highlights connecting matching tokens across the two witnesses, so a scholar judging a proposed match can see which words the score rests on. Our metrics test faithfulness, whether a highlight tracks what the score depends on, not plausibility, whether it points at the words a philologist would compare (Jacovi and Goldberg, 2020). Integrated gradients (Sundararajan et al., 2017) attributes the pairwise cosine to the input tokens of each side of the pair (pair-matrix view: Figure 9).

Our learned-mask attribution is adapted from MaRC (Brinner and Zarri  , 2023). MaRC was

| Model                   | Setting  | Task A               |                      | Task B            |                   |
|-------------------------|----------|----------------------|----------------------|-------------------|-------------------|
|                         |          | AUROC                | Cosine gap           | Assignment acc.   | DirAcc@1          |
| LaTa                    | Base     | 0.938 [0.910, 0.963] | 0.237 [0.222, 0.253] | 73.8 [70.2, 77.3] | 72.1 [68.4, 75.8] |
|                         | SIF      | 0.948 [0.915, 0.974] | 0.366 [0.342, 0.390] | 83.2 [80.1, 86.3] | 81.8 [78.5, 85.1] |
|                         | ABTT     | 0.971 [0.948, 0.991] | 0.525 [0.491, 0.559] | 88.5 [85.8, 90.9] | 86.1 [83.2, 88.9] |
|                         | SIF+ABTT | 0.966 [0.938, 0.989] | 0.580 [0.541, 0.617] | 90.2 [87.6, 92.6] | 88.2 [85.4, 90.9] |
| PhilTa                  | Base     | 0.939 [0.918, 0.958] | 0.231 [0.217, 0.247] | 70.9 [66.9, 74.9] | 69.3 [65.2, 73.5] |
|                         | SIF      | 0.955 [0.935, 0.972] | 0.319 [0.300, 0.339] | 79.0 [75.6, 82.4] | 77.7 [74.2, 81.3] |
|                         | ABTT     | 0.983 [0.969, 0.993] | 0.540 [0.508, 0.571] | 91.1 [88.9, 93.2] | 88.3 [85.6, 90.9] |
|                         | SIF+ABTT | 0.982 [0.969, 0.993] | 0.573 [0.538, 0.608] | 91.4 [89.1, 93.6] | 88.9 [86.2, 91.5] |
| mT5-base                | Base     | 0.838 [0.810, 0.866] | 0.081 [0.072, 0.092] | 47.2 [43.1, 51.6] | 46.6 [42.5, 51.0] |
|                         | SIF      | 0.845 [0.816, 0.876] | 0.036 [0.031, 0.042] | 54.5 [50.3, 58.8] | 54.1 [49.8, 58.4] |
|                         | ABTT     | 0.975 [0.954, 0.992] | 0.536 [0.498, 0.574] | 90.3 [87.9, 92.7] | 88.5 [85.7, 91.2] |
|                         | SIF+ABTT | 0.974 [0.950, 0.993] | 0.519 [0.480, 0.559] | 90.3 [88.0, 92.6] | 87.5 [84.8, 90.2] |
| LaBSE                   | Base     | 0.956 [0.932, 0.977] | 0.183 [0.171, 0.195] | 83.3 [80.2, 86.4] | 81.5 [78.1, 84.7] |
|                         | SIF      | 0.956 [0.931, 0.977] | 0.184 [0.172, 0.197] | 83.4 [80.2, 86.5] | 81.9 [78.6, 85.2] |
|                         | ABTT     | 0.987 [0.977, 0.995] | 0.611 [0.575, 0.646] | 90.8 [88.5, 92.9] | 88.5 [85.8, 90.9] |
|                         | SIF+ABTT | 0.976 [0.960, 0.989] | 0.580 [0.543, 0.617] | 89.6 [87.2, 91.9] | 86.9 [84.2, 89.5] |
| Qwen3-0.6B              | Base     | 0.966 [0.950, 0.981] | 0.024 [0.023, 0.026] | 82.3 [79.1, 85.4] | 80.3 [76.9, 83.6] |
|                         | SIF      | 0.942 [0.919, 0.963] | 0.053 [0.048, 0.060] | 76.0 [72.5, 79.4] | 74.2 [70.6, 77.8] |
|                         | ABTT     | 0.973 [0.948, 0.990] | 0.553 [0.513, 0.590] | 91.5 [89.1, 93.7] | 89.4 [86.7, 92.0] |
|                         | SIF+ABTT | 0.972 [0.947, 0.991] | 0.540 [0.499, 0.580] | 90.8 [88.3, 93.1] | 88.6 [85.8, 91.2] |
| KaLM-mini               | Base     | 0.972 [0.955, 0.986] | 0.056 [0.053, 0.060] | 87.5 [84.8, 90.1] | 85.9 [82.9, 88.7] |
|                         | SIF      | 0.964 [0.944, 0.982] | 0.056 [0.051, 0.062] | 85.7 [82.8, 88.5] | 84.0 [80.9, 87.1] |
|                         | ABTT     | 0.981 [0.963, 0.993] | 0.568 [0.529, 0.605] | 91.7 [89.5, 93.8] | 89.4 [86.7, 91.9] |
|                         | SIF+ABTT | 0.981 [0.962, 0.995] | 0.535 [0.496, 0.573] | 90.8 [88.4, 93.0] | 88.0 [85.2, 90.7] |
| LaTa (fine-tuned)       | Base     | 0.984 [0.973, 0.992] | 0.387 [0.368, 0.406] | 83.4 [80.2, 86.5] | 81.6 [78.1, 84.9] |
|                         | ABTT     | 0.970 [0.947, 0.990] | 0.548 [0.511, 0.583] | 87.8 [85.2, 90.2] | 85.2 [82.3, 87.9] |
| Qwen3-0.6B (fine-tuned) | Base     | 0.996 [0.993, 0.999] | 0.684 [0.653, 0.714] | 91.4 [88.7, 93.8] | 90.8 [88.1, 93.3] |
|                         | ABTT     | 0.994 [0.989, 0.998] | 0.716 [0.680, 0.751] | 92.1 [89.6, 94.3] | 91.4 [88.8, 93.7] |
| KaLM-mini (fine-tuned)  | Base     | 0.997 [0.993, 0.999] | 0.636 [0.605, 0.666] | 92.4 [90.1, 94.6] | 91.7 [89.3, 94.0] |
|                         | ABTT     | 0.994 [0.988, 0.998] | 0.651 [0.612, 0.687] | 93.4 [91.2, 95.4] | 92.5 [90.3, 94.7] |
| BM25 (word)             | –        | 0.974 [0.956, 0.987] | 0.243 [0.222, 0.266] | 62.5 [58.2, 66.6] | 56.3 [52.0, 60.5] |
| TF-IDF char 3–5         | –        | 0.987 [0.976, 0.996] | 0.555 [0.519, 0.591] | 91.8 [89.7, 93.8] | 89.9 [87.3, 92.3] |
| Levenshtein             | –        | 0.952 [0.927, 0.975] | 0.362 [0.332, 0.393] | 87.8 [85.0, 90.4] | 86.2 [83.2, 89.2] |

Table 29: Every headline cell of Tables 7 and 8 with its 95% percentile intervals from a directory-level bootstrap ( $B = 10,000$  resamples of the 514 test directories with replacement, seed 233). A replicate keeps every file and pair of a drawn directory; a directory drawn twice counts twice, and pairs between its two copies are not formed. The threshold  $\tau$ , the ABTT components and  $D$ , the SIF weights and the layer stay fit on the fixed training split; only the test evaluation is resampled, so the intervals cover test-set sampling and not the choice of split. Each cell is read at its train-selected layer (Table 11); Task B in percent. The lexical rows are the reference systems of Table 28.

proposed for classification, where a soft input mask preserves a target class score while sparsity and smoothness regularizers encourage concise rationales. We keep the per-instance mask optimization and replace the classification target with pairwise retrieval cosine: the mask is optimized on one side of a query-candidate pair while the partner representation stays fixed, and swapping the sides gives candidate-side attribution.

Let  $\mathbf{E}(\mathbf{x}_q) \in \mathbb{R}^{T_q \times d_{\text{emb}}}$  denote the input-embedding lookup,  $E_{>0}$  the encoder with the lookup removed, and  $\mathbf{z}_c^*$  the candidate representation held fixed during optimization. We parameterize  $\mathbf{m} = \sigma(\boldsymbol{\zeta})$  with free logits  $\boldsymbol{\zeta} \in \mathbb{R}^{T_q}$  and write  $\mathcal{I}$  for content-token positions. For the ABTT variant, define  $s_{\text{ABTT}}(\mathbf{a}, \mathbf{b}) = \cos(\text{ABTT}(\mathbf{a}, \mathbf{b}))$ .

The retrieval-adapted MaRC loss is

$$\mathcal{L}(\mathbf{m}) = (s_{\text{ABTT}}(\tilde{\mathbf{z}}_q(\mathbf{m}), \mathbf{z}_c^*) - s_{\text{ABTT}}(\mathbf{z}_q, \mathbf{z}_c^*))^2 + \lambda \sum_{i \in \mathcal{I}} \frac{m_i}{|\mathcal{I}|} + \gamma \sum_{i, i+1 \in \mathcal{I}} \frac{|m_{i+1} - m_i|}{|\mathcal{I}| - 1}, \quad (1)$$

where  $\tilde{\mathbf{z}}_q(\mathbf{m}) = \text{pool}(E_{>0}(\text{diag}(\mathbf{m}) \mathbf{E}(\mathbf{x}_q)), \mathbf{a}_q)$ . The baseline variant uses raw cosine in place of the ABTT-corrected cosine in Equation 1. Logits outside  $\mathcal{I}$  are frozen at  $-\infty$ .

## K.2 Attribution Metrics

We report two threshold-free metrics:  $\rho_{\text{LOO}}$ , the Spearman correlation between attribution magnitude and the leave-one-out drop in the same retrieval score, and the DeLAUC gap, the random-order minus attribution-order deletion-curve area. A shuffled-attribution control recomputes each of six candidate metrics ( $\rho_{\text{LOO}}$ , Kendall  $\tau_b$ , the DeLAUC and InsAUC gaps, AOPC comprehensive-

| Contrast                         | Model      | $\Delta$ AUROC          | $\Delta$ Assignment acc. | $\Delta$ DirAcc@1  |
|----------------------------------|------------|-------------------------|--------------------------|--------------------|
| ABTT — Base                      | LaTa       | +0.034 [0.017, 0.054]   | +14.7 [11.1, 18.3]       | +14.0 [10.4, 17.6] |
|                                  | PhilTa     | +0.043 [0.029, 0.060]   | +20.3 [16.2, 24.5]       | +19.0 [15.0, 23.1] |
|                                  | mT5-base   | +0.137 [0.109, 0.166]   | +43.1 [38.6, 47.6]       | +41.8 [37.2, 46.4] |
|                                  | LaBSE      | +0.030 [0.014, 0.049]   | +7.5 [4.6, 10.4]         | +7.0 [4.1, 10.0]   |
|                                  | Qwen3-0.6B | +0.006 [−0.010, 0.019]  | +9.2 [6.3, 12.2]         | +9.1 [6.1, 12.2]   |
|                                  | KaLM-mini  | +0.009 [−0.002, 0.021]  | +4.2 [1.7, 6.7]          | +3.5 [1.0, 6.0]    |
| Fine-tuned: ABTT — Base          | LaTa       | −0.014 [−0.032, 0.001]  | +4.3 [1.7, 7.1]          | +3.6 [1.0, 6.3]    |
|                                  | Qwen3-0.6B | −0.002 [−0.005, 0.001]  | +0.7 [−0.7, 2.1]         | +0.6 [−0.8, 2.0]   |
|                                  | KaLM-mini  | −0.003 [−0.007, 0.001]  | +0.9 [−0.3, 2.2]         | +0.8 [−0.5, 2.1]   |
| Fine-tuned ABTT — zero-shot ABTT | LaTa       | −0.002 [−0.010, 0.005]  | −0.7 [−2.6, 1.2]         | −0.9 [−2.8, 1.0]   |
|                                  | Qwen3-0.6B | +0.022 [0.006, 0.044]   | +0.6 [−1.6, 2.9]         | +2.0 [0.0, 4.0]    |
|                                  | KaLM-mini  | +0.013 [0.002, 0.029]   | +1.6 [−0.3, 3.6]         | +3.1 [1.2, 5.2]    |
| TF-IDF char 3–5 — ABTT           | LaTa       | +0.016 [0.003, 0.030]   | +3.4 [1.1, 5.7]          | +3.7 [1.4, 6.2]    |
|                                  | PhilTa     | +0.004 [−0.002, 0.011]  | +0.7 [−1.0, 2.4]         | +1.5 [−0.2, 3.3]   |
|                                  | mT5-base   | +0.012 [0.000, 0.027]   | +1.5 [−0.6, 3.7]         | +1.4 [−0.7, 3.6]   |
|                                  | LaBSE      | 0.000 [−0.004, 0.005]   | +1.0 [−1.0, 3.1]         | +1.4 [−0.8, 3.6]   |
|                                  | Qwen3-0.6B | +0.014 [0.002, 0.031]   | +0.3 [−1.6, 2.3]         | +0.5 [−1.4, 2.5]   |
|                                  | KaLM-mini  | +0.006 [−0.002, 0.016]  | +0.1 [−1.6, 1.9]         | +0.5 [−1.3, 2.3]   |
| Centering — Base (ABTT layer)    | LaTa       | +0.034 [0.021, 0.049]   | −2.2 [−5.2, 0.9]         | +6.8 [4.6, 9.1]    |
|                                  | PhilTa     | −0.038 [−0.060, −0.016] | +12.9 [9.0, 16.9]        | +12.4 [8.5, 16.3]  |
|                                  | mT5-base   | +0.088 [0.066, 0.109]   | +20.5 [16.0, 24.8]       | +15.6 [11.3, 19.9] |
|                                  | LaBSE      | −0.002 [−0.009, 0.004]  | −4.7 [−7.4, −2.0]        | −4.7 [−7.3, −2.0]  |
|                                  | Qwen3-0.6B | +0.076 [0.061, 0.092]   | +19.2 [15.4, 23.2]       | +16.7 [12.7, 20.7] |
|                                  | KaLM-mini  | +0.070 [0.052, 0.089]   | +19.3 [15.5, 23.2]       | +16.9 [13.1, 20.7] |
| ABTT — centering (ABTT layer)    | LaTa       | −0.001 [−0.016, 0.012]  | +40.4 [35.8, 44.8]       | +49.2 [44.1, 53.9] |
|                                  | PhilTa     | +0.482 [0.460, 0.504]   | +7.3 [4.9, 10.0]         | +6.6 [4.1, 9.3]    |
|                                  | mT5-base   | +0.065 [0.050, 0.081]   | +24.5 [20.7, 28.3]       | +27.7 [23.7, 31.7] |
|                                  | LaBSE      | +0.029 [0.013, 0.048]   | +12.1 [8.8, 15.5]        | +11.7 [8.4, 15.0]  |
|                                  | Qwen3-0.6B | +0.033 [0.018, 0.050]   | +25.9 [22.0, 29.9]       | +27.7 [23.5, 32.1] |
|                                  | KaLM-mini  | +0.023 [0.008, 0.040]   | +12.0 [9.1, 15.1]        | +13.8 [10.7, 17.0] |
| Spread over six models           | Base       | 0.134 [0.110, 0.158]    | 40.3 [35.7, 44.7]        | 39.3 [34.6, 43.7]  |
|                                  | ABTT       | 0.015 [0.006, 0.034]    | 3.3 [1.8, 5.8]           | 3.3 [1.8, 6.0]     |

Table 30: Paired differences between headline cells, with 95% intervals from the same directory-level bootstrap replicates as Table 29 ( $B = 10,000$ , 514 test directories, seed 233). Both cells of a contrast are recomputed on each replicate, so the interval reflects their correlation. Each cell sits at its own train-selected layer. The two centering contrasts compare baseline, centering alone ( $D = 0$ ) and ABTT at the layer of the ABTT cell, the Task A ABTT layer for  $\Delta$  AUROC and the Task B ABTT layer for the two routing columns. Spread: the largest minus the smallest of the six zero-shot models. Task B in points. Differences are computed at full precision, so they can differ by one unit in the last digit from the difference of the rounded cells. The threshold  $\tau$ , the ABTT components and  $D$ , the SIF weights and the layer stay fit on the fixed training split; only the test evaluation is resampled, so the intervals cover test-set sampling and not the choice of split.

ness and sufficiency) on five permutations of the attribution’s own scores in all twelve model-method-variant cells; a metric qualifies for the headline table (Table 32) only if the real attribution beats the permutation mean in every cell at the operational layers. ERASER-style sufficiency and comprehensiveness (DeYoung et al., 2020) and MinFrac, defined for retrieval cosine in Appendix L, stay out of the headline table because their verdicts depend on the threshold and on the erasure operator. Every curve metric ranks tokens by  $|a|$ , ties broken by position; Appendix L reports them and states both conventions and their consequences.

### K.3 Setup and Layer Selection

MaRC runs up to 200 Adam steps per pair (learning rate 0.1, stopping after step 50 once the masked cosine is within 0.01 of the full one) from  $\zeta_0 = \log 9$  with  $\lambda = 0.01$  and  $\gamma = 0.001$ ; integrated gradients uses 40 integration steps. Attribution

layers follow a predeclared train-only rule, a variant of the Task B rule of Section 3: the earliest layer within 0.5 points of the best training-set DirAcc@1 under ABTT. It selects LaTa layer 7, PhilTa layer 1, and mT5-base layer 1 for attribution, with  $D = 10$  ABTT components; the most anisotropic layers on the test split, LaTa 8, PhilTa 6, and mT5-base 5, are kept as mechanism checks (Appendix B). The rule locks these layers before any attribution metric is computed.

### K.4 Results

Table 32 compares baseline and ABTT attributions on 200 positive query-candidate pairs per model for LaTa, PhilTa, and mT5-base. ABTT repairs the scores being explained at these layers: LaTa layer 7 moves from AUROC 0.498 to 0.962 and DirAcc@1 30.2 to 86.8, PhilTa layer 1 from 0.939 to 0.977 and 69.3 to 88.3, and mT5-base layer 1 from 0.822 to 0.979 and 45.1 to 88.5.

| Model      | Setting            | L  | Cos. SD | Exist./new AUROC     | Assign. | Oracle gap         | $\tau$ | Exact | Skew |
|------------|--------------------|----|---------|----------------------|---------|--------------------|--------|-------|------|
| LaTa       | Base               | 1  | 0.112   | 0.869 [0.840, 0.896] | 73.8    | +5.6 [2.8, 9.4]    | 0.844  | 74.6  | 2.81 |
|            | Base @ ABTT L      | 8  | 0.632   | 0.621 [0.583, 0.658] | 50.2    | +16.4 [11.9, 21.9] | 0.995  | 44.1  | 1.59 |
|            | Centering @ ABTT L | 8  | 0.901   | 0.604 [0.564, 0.644] | 48.0    | +14.3 [9.4, 21.8]  | 0.995  | 42.8  | 1.97 |
|            | ABTT               | 8  | 0.100   | 0.943 [0.925, 0.960] | 88.5    | +0.6 [0.2, 1.7]    | 0.472  | 88.5  | 1.03 |
| PhilTa     | Base               | 1  | 0.111   | 0.843 [0.813, 0.873] | 70.9    | +7.0 [3.7, 11.5]   | 0.894  | 71.0  | 3.15 |
|            | Centering @ ABTT L | 1  | 0.126   | 0.906 [0.882, 0.928] | 83.8    | +1.2 [0.6, 2.5]    | 0.548  | 84.4  | 1.78 |
|            | ABTT               | 1  | 0.086   | 0.960 [0.944, 0.974] | 91.1    | +0.7 [0.2, 1.8]    | 0.417  | 91.1  | 0.72 |
| mT5-base   | Base               | 12 | 0.104   | 0.685 [0.642, 0.726] | 47.2    | +19.6 [14.3, 26.8] | 0.970  | 49.5  | 2.59 |
|            | Base @ ABTT L      | 1  | 0.065   | 0.696 [0.654, 0.737] | 45.3    | +23.1 [16.5, 30.2] | 0.995  | 52.2  | 3.83 |
|            | Centering @ ABTT L | 1  | 0.242   | 0.765 [0.729, 0.801] | 65.9    | +5.6 [2.0, 10.5]   | 0.688  | 66.0  | 3.29 |
|            | ABTT               | 1  | 0.081   | 0.952 [0.935, 0.967] | 90.3    | +0.9 [0.2, 2.4]    | 0.437  | 91.1  | 1.13 |
| LaBSE      | Base               | 11 | 0.039   | 0.922 [0.901, 0.942] | 83.3    | +1.0 [0.1, 3.5]    | 0.975  | 82.5  | 2.00 |
|            | Centering @ ABTT L | 11 | 0.209   | 0.891 [0.866, 0.914] | 78.7    | +2.8 [0.7, 6.3]    | 0.704  | 79.0  | 1.84 |
|            | ABTT               | 11 | 0.102   | 0.960 [0.944, 0.973] | 90.8    | +0.6 [0.1, 1.6]    | 0.452  | 90.8  | 0.58 |
| Qwen3-0.6B | Base               | 28 | 0.043   | 0.900 [0.874, 0.923] | 82.3    | +1.0 [0.2, 2.6]    | 0.945  | 82.6  | 2.51 |
|            | Base @ ABTT L      | 5  | 0.034   | 0.708 [0.666, 0.748] | 46.4    | +23.4 [17.7, 30.4] | 0.990  | 53.0  | 2.49 |
|            | Centering @ ABTT L | 5  | 0.171   | 0.802 [0.767, 0.835] | 65.6    | +9.6 [5.9, 14.5]   | 0.653  | 66.6  | 1.77 |
|            | ABTT               | 5  | 0.086   | 0.954 [0.935, 0.971] | 91.5    | +0.7 [0.2, 1.7]    | 0.417  | 91.5  | 1.07 |
| KaLM-mini  | Base               | 23 | 0.021   | 0.931 [0.910, 0.950] | 87.5    | 0.0 [0.0, 1.1]     | 0.965  | 86.2  | 1.97 |
|            | Base @ ABTT L      | 3  | 0.054   | 0.756 [0.719, 0.792] | 60.4    | +11.3 [6.7, 17.0]  | 0.975  | 58.3  | 3.58 |
|            | Centering @ ABTT L | 3  | 0.135   | 0.881 [0.853, 0.906] | 79.7    | +2.7 [1.2, 4.8]    | 0.558  | 79.5  | 1.75 |
|            | ABTT               | 3  | 0.085   | 0.958 [0.941, 0.973] | 91.7    | +0.2 [0.0, 1.6]    | 0.397  | 91.7  | 0.87 |

Table 31: Routing checks at the Task B cells (Table 8) of the six zero-shot models: the Base and ABTT cells at their train-selected layers (L), and baseline and centering alone ( $D = 0$ , subtract the training mean) at the ABTT cell’s layer (a second baseline row only where that layer differs). Cos. SD: standard deviation of all test pairwise cosines. Exist./new AUROC: threshold-free AUROC of each test file’s maximum cosine for existing against new files. Assign.: assignment accuracy at the train-fit  $\tau$  (the printed cell). Oracle gap: best assignment accuracy over every test threshold minus Assign. Exact: assignment accuracy with  $\tau$  re-fit as the exact best-F1 cut over all training pair scores instead of the 200-point grid. Skew: skewness of the  $k$ -occurrence distribution  $N_{10}$  over test files (Radovanović et al., 2010). Brackets: 95% directory-bootstrap intervals ( $B = 10,000$ ). Holding layer and  $D$  fixed, the fine grid (step  $10^{-4}$ ) moves 45 and the exact cut 48 of the 66 printed Task B numbers; re-selecting layer and  $D$  under the new grid as well moves 42 (fine) and 44 (exact) of 60.

| Model    | Method | $\rho_{LOO}$ |              | DeLAUC gap   |              |
|----------|--------|--------------|--------------|--------------|--------------|
|          |        | base         | ABTT         | base         | ABTT         |
| LaTa     | IG     | 0.003        | <b>0.370</b> | <b>0.859</b> | 0.181        |
|          | MaRC   | 0.083        | <b>0.538</b> | <b>0.524</b> | 0.327        |
| PhilTa   | IG     | 0.329        | <b>0.614</b> | 0.113        | <b>0.403</b> |
|          | MaRC   | <b>0.469</b> | 0.445        | 0.201        | <b>0.313</b> |
| mT5-base | IG     | 0.143        | <b>0.686</b> | 0.061        | <b>0.561</b> |
|          | MaRC   | 0.257        | <b>0.504</b> | 0.042        | <b>0.463</b> |

Table 32: Attribution faithfulness at the predeclared operational layers, 200 positive pairs per model, for integrated gradients (IG) and retrieval-adapted MaRC.  $\rho_{LOO}$  correlates attribution magnitude with the leave-one-out change in the cosine. DeLAUC gap is the random-order minus attribution-order deletion-curve area, positive when attribution beats chance; the reference runs from 0.692 to 0.961 here, so zero is chance. Higher is better in both; boldface marks the better variant. ABTT wins 5/6 and 4/6. Ties, within 2 standard errors of zero:  $\rho$  for PhilTa MaRC. Ratio metrics are undefined below a full-query cosine of 0.05, so the DeLAUC columns average 195 to 199 ABTT pairs against 199 to 200 baseline pairs. Cross-variant comparisons are descriptive. Secondary metrics: Table 34.

ABTT improves rank faithfulness in five of six model-view cells ( $\rho_{LOO}$  5/6, Kendall  $\tau_b$  agreeing in the same five) by 10.3 to 22.5 standard errors,

the sixth a tie at 1.5; chance-corrected deletion faithfulness improves in four of six, and both LaTa cells favour the baseline, at 2.1 and 8.5 standard errors. Comp@25% and the InsAUC gap point the same way (Table 34): Comp@25% favours ABTT in four of six cells, both LaTa cells to the baseline as for DeLAUC gap, and the InsAUC gap in five of six, LaTa/IG to the baseline; MinFrac@0.80 does not, favouring ABTT in two. Neither is headline evidence, and InsAUC gap and AOPC-Suff (identical gaps by construction) fall to  $-0.024$  on mT5-base/IG baseline, the one non-positive cell of the control (Table 35). So we claim better-ranked rationales, not a gain on every axis.

As a mechanism check we repeated the analysis on the same pairs at each model’s most anisotropic layer (LaTa 8, PhilTa 6, mT5-base 5; Table 33), where baseline AUROC is 0.502, 0.541 and 0.654 and ABTT restores 0.965, 0.981 and 0.978. ABTT improves  $\rho_{LOO}$  in all six cells (10.8 to 18.3 standard errors), but the mean gain is no larger than at the operational layers (0.34 against 0.31): the baseline explanation falls with the score, and baseline integrated gradients for LaTa and mT5-base no longer beat a shuffle of their own scores, so  $\rho_{LOO}$

| Model    | Layer   | Method | $\rho_{\text{LOO}}$ |              |                            | DeLAUC gap   |              |                            |
|----------|---------|--------|---------------------|--------------|----------------------------|--------------|--------------|----------------------------|
|          |         |        | base                | ABTT         | $\Delta$                   | base         | ABTT         | $\Delta$                   |
| LaTa     | op. 7   | IG     | 0.003               | <b>0.370</b> | +0.367 (16.0)              | <b>0.859</b> | 0.181        | −0.651 (−8.5)              |
|          |         | MaRC   | 0.083               | <b>0.538</b> | +0.455 (17.0)              | <b>0.524</b> | 0.327        | −0.177 (−2.1)              |
|          | anis. 8 | IG     | −0.018              | <b>0.357</b> | +0.375 (17.0)              | <b>0.913</b> | 0.182        | −0.711 (−10.9)             |
|          |         | MaRC   | 0.108               | <b>0.535</b> | +0.427 (16.5)              | <b>0.567</b> | 0.336        | −0.212 (−3.2)              |
| PhilTa   | op. 1   | IG     | 0.329               | <b>0.614</b> | +0.285 (17.5)              | 0.113        | <b>0.403</b> | +0.290 (28.4)              |
|          |         | MaRC   | <b>0.469</b>        | 0.445        | −0.024 (−1.5) <sup>†</sup> | 0.201        | <b>0.313</b> | +0.112 (8.0)               |
|          | anis. 6 | IG     | 0.065               | <b>0.278</b> | +0.213 (10.8)              | 0.052        | <b>0.140</b> | +0.089 (0.8) <sup>†</sup>  |
|          |         | MaRC   | 0.107               | <b>0.484</b> | +0.377 (15.3)              | 0.044        | <b>0.285</b> | +0.241 (3.6)               |
| mT5-base | op. 1   | IG     | 0.143               | <b>0.686</b> | +0.543 (22.5)              | 0.061        | <b>0.561</b> | +0.499 (24.0)              |
|          |         | MaRC   | 0.257               | <b>0.504</b> | +0.247 (10.3)              | 0.042        | <b>0.463</b> | +0.421 (19.4)              |
|          | anis. 5 | IG     | −0.026              | <b>0.272</b> | +0.298 (18.3)              | <b>0.262</b> | 0.230        | −0.037 (−0.6) <sup>†</sup> |
|          |         | MaRC   | 0.041               | <b>0.413</b> | +0.372 (17.8)              | 0.028        | <b>0.370</b> | +0.341 (7.5)               |

Table 33: Attribution faithfulness at two layer sets on the same 200 positive pairs per model: the operational layers of Table 32 (op., LaTa 7, PhilTa 1, and mT5-base 1, chosen by the train-only retrieval rule) and each model’s most anisotropic layer (anis., LaTa 8, PhilTa 6, and mT5-base 5, the largest top-PC variance share on the test split, Table 9). Methods, settings, token filter and erasure operator are those of Table 32; ABTT removes  $D = 10$  components fit on training embeddings at each layer.  $\rho_{\text{LOO}}$  correlates attribution magnitude with the leave-one-out change in the cosine; DeLAUC gap is the random-order minus attribution-order deletion-curve area, positive when attribution beats chance. Boldface marks the better variant.  $\Delta$  is the paired ABTT minus baseline mean over pairs valid under both variants, with its ratio to the paired standard error in parentheses; a dagger marks a tie, within two standard errors of zero. ABTT wins, ties and loses 5/1/0 cells on  $\rho_{\text{LOO}}$  and 4/0/2 on DeLAUC gap at the operational layers, and 6/0/0 and 2/2/2 at the most anisotropic layers. At the most anisotropic layers the real attribution does not beat a shuffle of its own scores on  $\rho_{\text{LOO}}$  in two of the twelve cells (LaTa IG baseline, mT5-base IG baseline). At the most anisotropic layers the DeLAUC columns average 195 to 200 ABTT pairs against 191 to 198 baseline pairs, since ratio metrics are undefined below a full-query cosine of 0.05.

fails the every-cell control at these layers. Chance-corrected deletion faithfulness is weaker than at the operational layers (ABTT wins two cells, ties two and loses both LaTa cells, against four wins and two losses), so the mixed attribution picture is not an artifact of the layer choice.

Masking acts on cached layer- $L$  hidden states, so every faithfulness statement is about the pooled read-out at that layer, not the encoder’s input-to-output map. Appendix L carries the threshold sweeps and Figure 8; Appendix M shows one PhilTa demonstration pair, outside the 200 evaluated pairs and at a different layer, with its integrated-gradient pair matrix before and after ABTT.

## L Secondary Attribution Metrics and Sweeps

**Ranking conventions.** Two ranking conventions hold for every curve metric and every method, variant and control alike. Tokens are ranked by attribution magnitude  $|a|$  rather than by signed importance, because the question is which tokens carry the pair cosine and both methods produce two-signed scores. Ties are broken by token position, which matters most for MaRC, whose many

exactly-zero scores are then ordered by reading order. Absolute values are therefore not comparable with a signed-ranking implementation, and Kendall  $\tau_b$  in Table 34 is the tie-corrected companion to  $\rho_{\text{LOO}}$  under the second convention.

**Secondary metrics.** Figure 8 plots the rank-faithfulness column of Table 32. Table 34 carries the metrics Table 32 does not: tie-corrected Kendall  $\tau_b$ , chance-corrected insertion faithfulness, and the three ERASER-style rationale metrics at the global threshold settings. Sufficiency keeps the top-ranked query tokens and measures how much of the full pair cosine remains, comprehensiveness removes those tokens and measures the score drop, and MinFrac reports the smallest retained query-token fraction needed to recover a fixed share of the full cosine. Four computed quantities are printed in neither table: the raw deletion and insertion areas, which appear here only as the chance-corrected gaps, and AOPC-Sufficiency and AOPC-Comprehensiveness, whose per-threshold components are in the sweeps below. Table 35 gives the per-cell shuffled-attribution gaps for the six candidate metrics, the control Appendix K quotes as a validity statement. The rest of this appendix expands those rationale metrics across Sufficiency

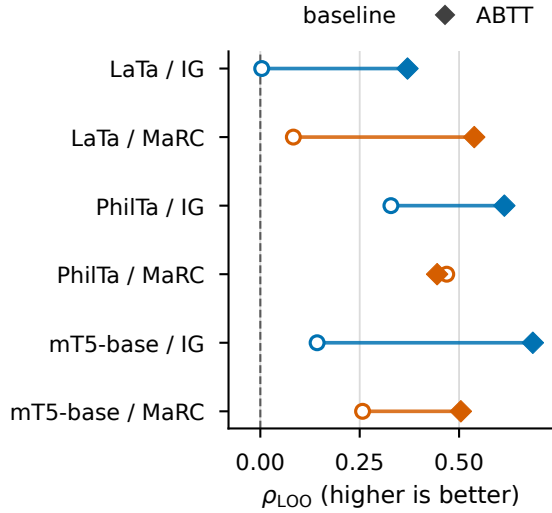


Figure 8: Leave-one-out rank correlation  $\rho_{LOO}$  at the predeclared operational attribution layers, for integrated gradients (IG) and retrieval-adapted MaRC. Each line connects the baseline and ABTT variants of one model-method cell. ABTT raises  $\rho_{LOO}$  in five of the six cells; the sixth, PhilTa MaRC, moves from 0.469 to 0.445, a tie within two standard errors (Table 32). Secondary metrics: Table 34.

and Comprehensiveness thresholds of 10%, 25%, and 50%, and MinFrac recovery thresholds of 0.70, 0.80, 0.90, and 0.95. We use one global midpoint setting rather than selecting thresholds per model or per method. Table 36 gives the sweep for integrated gradients and retrieval-adapted MaRC, and Table 37 gives it for the supplemental attribution methods. Every table in this appendix uses the same representation-level erasure operator as Table 32, so the cells are comparable across them, except Table 38, which moves the choices behind the DeLAUC gap, the erasure operator among them, one at a time.

## M Qualitative Attribution Examples

This appendix keeps the example-level views for one PhilTa retrieval pair, at PhilTa layer 6, its most anisotropic layer (the layer of the mechanism check in Appendix K), not the operational layer of Table 32; a demonstration pair chosen because ABTT reverses the baseline decision, outside the 200 evaluated pairs. Figure 9 gives the integrated-gradient pair matrix before and after ABTT. The attention map in Figure 10 is context only, since ABTT is applied after encoding. The retrieval-adapted MaRC masks in Figure 11 show the learned-mask view for the same pair.

## N Cluster Geometry

Figures 12 and 13 give the t-SNE and UMAP panels for LaTa, PhilTa, and mT5-base, and Figures 14 and 15 extend them to LaBSE, Qwen3-0.6B, and KaLM-mini. All four draw all 1,705 witnesses, color the same six largest directories, and compare baseline mean pooling with ABTT-only at  $D = 10$ ; each model sits at the layer chosen on the train split by DirAcc@1 under ABTT, the Task B ABTT layer of Table 11.

In t-SNE (van der Maaten and Hinton, 2008) and UMAP (McInnes et al., 2018) projections at the train-selected layers, ABTT-only with  $D = 10$  forms tighter same-directory groups than baseline mean pooling: on each of the three T5 encoders the t-SNE silhouette of the 1,160 winnable witnesses rises, from 0.50 to 0.57 for PhilTa, from  $-0.25$  to 0.48 for mT5-base, and from  $-0.66$  to 0.50 for LaTa. Silhouette lies in  $[-1, 1]$ , so a rise can be as large as 2; the three rises are 0.07, 0.72 and 1.16.

| Model    | Method | $\tau_{\text{LOO}}$ |              | InsAUC gap   |              | Suff@25%     |              | Comp@25%     |              | MinFrac@0.80 |              |
|----------|--------|---------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|          |        | base                | ABTT         | base         | ABTT         | base         | ABTT         | base         | ABTT         | base         | ABTT         |
| LaTa     | IG     | 0.007               | <b>0.262</b> | <b>0.284</b> | 0.061        | <b>0.990</b> | 0.804        | <b>1.242</b> | 0.177        | <b>0.111</b> | 0.272        |
|          | MaRC   | 0.068               | <b>0.400</b> | 0.088        | <b>0.138</b> | 0.731        | <b>0.917</b> | <b>0.781</b> | 0.280        | 0.184        | <b>0.177</b> |
| PhilTa   | IG     | 0.234               | <b>0.462</b> | 0.004        | <b>0.245</b> | 0.814        | <b>0.971</b> | 0.113        | <b>0.522</b> | 0.240        | <b>0.136</b> |
|          | MaRC   | <b>0.334</b>        | 0.318        | 0.063        | <b>0.193</b> | <b>0.888</b> | 0.878        | 0.093        | <b>0.363</b> | <b>0.149</b> | 0.259        |
| mT5-base | IG     | 0.101               | <b>0.515</b> | -0.024       | <b>0.399</b> | 0.933        | <b>1.126</b> | 0.012        | <b>0.642</b> | <b>0.086</b> | 0.178        |
|          | MaRC   | 0.181               | <b>0.359</b> | 0.003        | <b>0.327</b> | 0.960        | <b>1.013</b> | 0.014        | <b>0.582</b> | <b>0.031</b> | 0.246        |

Table 34: Secondary attribution metrics, on the same pairs, the same layers and the same erasure operator as Table 32. Boldface marks the better variant within a pair; higher is better everywhere except MinFrac. None of these columns is in Table 32, and each is out for its own reason. Kendall  $\tau_b$  agrees with  $\rho_{\text{LOO}}$  in 5/6 cells, but it is the tie-corrected twin of the same statistic rather than a second witness. Chance-corrected insertion faithfulness favours ABTT in 5/6 cells, and we do not report it in Table 32 because in one of the twelve cells, a baseline cell, the real attribution does not beat a permutation of its own scores (Table 35), so the measurement does not meet the validity bar we set for a headline column. The threshold-based ERASER metrics are reported for completeness: their baseline-versus-ABTT verdict depends on the threshold and on the erasure operator, which is why Table 32 uses threshold-free, chance-corrected metrics instead. The InsAUC columns average 195 to 199 ABTT pairs against the baseline’s 199 to 200, for the same small-denominator reason. Full threshold sweeps are in Tables 36 and 37.

| Model    | Method | Variant | $\rho_{\text{LOO}}$ | $\tau_{\text{LOO}}$ | DelAUC gap, AOPC-Comp | InsAUC gap, AOPC-Suff |
|----------|--------|---------|---------------------|---------------------|-----------------------|-----------------------|
| LaTa     | IG     | base    | +0.007 (0.018)      | +0.010 (0.013)      | +0.887 (0.071)        | +0.300 (0.044)        |
|          |        | ABTT    | +0.379 (0.014)      | +0.268 (0.010)      | +0.183 (0.009)        | +0.064 (0.008)        |
|          | MaRC   | base    | +0.087 (0.021)      | +0.070 (0.016)      | +0.509 (0.082)        | +0.073 (0.072)        |
|          |        | ABTT    | +0.535 (0.018)      | +0.398 (0.014)      | +0.327 (0.010)        | +0.140 (0.011)        |
| PhilTa   | IG     | base    | +0.327 (0.017)      | +0.233 (0.012)      | +0.113 (0.006)        | +0.006 (0.003)        |
|          |        | ABTT    | +0.614 (0.013)      | +0.462 (0.010)      | +0.406 (0.011)        | +0.246 (0.010)        |
|          | MaRC   | base    | +0.466 (0.016)      | +0.332 (0.012)      | +0.202 (0.005)        | +0.065 (0.004)        |
|          |        | ABTT    | +0.446 (0.019)      | +0.319 (0.014)      | +0.316 (0.015)        | +0.195 (0.016)        |
| mT5-base | IG     | base    | +0.143 (0.025)      | +0.101 (0.018)      | +0.061 (0.003)        | <b>-0.024 (0.001)</b> |
|          |        | ABTT    | +0.688 (0.013)      | +0.517 (0.011)      | +0.556 (0.020)        | +0.397 (0.026)        |
|          | MaRC   | base    | +0.256 (0.021)      | +0.180 (0.015)      | +0.042 (0.001)        | +0.003 (0.001)        |
|          |        | ABTT    | +0.502 (0.017)      | +0.357 (0.013)      | +0.460 (0.022)        | +0.323 (0.028)        |

Table 35: Shuffled-attribution control for the six candidate metrics, on the same pairs, layers and erasure operator as Table 32. Each entry is the real metric minus its mean over five permutations of the same attribution’s scores across the query tokens, with the standard error over pairs in parentheses: a positive gap means the real attribution beats a fake one drawn from its own score distribution, and a metric passes the control only when every cell is positive. DelAUC gap and AOPC-Comprehensiveness share one column, as do InsAUC gap and AOPC-Sufficiency, because the random-order reference and the constant trapezoid offset cancel in the difference, so each pair has the same gap to machine precision and one verdict. Boldface marks the one cell at or below zero: mT5-base IG baseline on InsAUC gap, at -0.024. Four further cells are positive but within two standard errors of zero: LaTa IG baseline on  $\rho_{\text{LOO}}$ , LaTa IG baseline on  $\tau_{\text{LOO}}$ , LaTa MaRC baseline on InsAUC gap and PhilTa IG baseline on InsAUC gap. The rank columns use 200 pairs per cell; the AUC columns, undefined below a full-query cosine of 0.05, average 195 to 199 ABTT pairs against 199 to 200 baseline pairs.

| Method          | $\rho_{\text{LOO}} \uparrow$ |       | Suff<br>@10% $\uparrow$ |       | Suff<br>@25% $\uparrow$ |       | Suff<br>@50% $\uparrow$ |       | Comp<br>@10% $\uparrow$ |       | Comp<br>@25% $\uparrow$ |       | Comp<br>@50% $\uparrow$ |       |
|-----------------|------------------------------|-------|-------------------------|-------|-------------------------|-------|-------------------------|-------|-------------------------|-------|-------------------------|-------|-------------------------|-------|
|                 | Base                         | ABTT  | Base                    | ABTT  | Base                    | ABTT  | Base                    | ABTT  | Base                    | ABTT  | Base                    | ABTT  | Base                    | ABTT  |
| <i>LaTa</i>     |                              |       |                         |       |                         |       |                         |       |                         |       |                         |       |                         |       |
| IG              | 0.003                        | 0.370 | 0.949                   | 0.556 | 0.990                   | 0.804 | 1.014                   | 0.969 | 1.144                   | 0.069 | 1.242                   | 0.177 | 1.203                   | 0.354 |
| MaRC            | 0.083                        | 0.538 | 0.696                   | 0.797 | 0.731                   | 0.917 | 0.807                   | 1.004 | 0.487                   | 0.112 | 0.781                   | 0.280 | 0.889                   | 0.556 |
| <i>PhilTa</i>   |                              |       |                         |       |                         |       |                         |       |                         |       |                         |       |                         |       |
| IG              | 0.329                        | 0.614 | 0.637                   | 0.844 | 0.814                   | 0.971 | 0.937                   | 1.038 | 0.042                   | 0.288 | 0.113                   | 0.522 | 0.231                   | 0.746 |
| MaRC            | 0.469                        | 0.445 | 0.786                   | 0.691 | 0.888                   | 0.878 | 0.971                   | 1.001 | 0.032                   | 0.169 | 0.093                   | 0.363 | 0.263                   | 0.625 |
| <i>mT5-base</i> |                              |       |                         |       |                         |       |                         |       |                         |       |                         |       |                         |       |
| IG              | 0.143                        | 0.686 | 0.839                   | 0.903 | 0.933                   | 1.126 | 0.976                   | 1.210 | 0.004                   | 0.312 | 0.012                   | 0.642 | 0.038                   | 1.005 |
| MaRC            | 0.257                        | 0.504 | 0.918                   | 0.791 | 0.960                   | 1.013 | 0.989                   | 1.121 | 0.004                   | 0.274 | 0.014                   | 0.582 | 0.042                   | 0.902 |

| Method          | MinFrac<br>@0.70 $\downarrow$ |       | MinFrac<br>@0.80 $\downarrow$ |       | MinFrac<br>@0.90 $\downarrow$ |       | MinFrac<br>@0.95 $\downarrow$ |       |
|-----------------|-------------------------------|-------|-------------------------------|-------|-------------------------------|-------|-------------------------------|-------|
|                 | Base                          | ABTT  | Base                          | ABTT  | Base                          | ABTT  | Base                          | ABTT  |
| <i>LaTa</i>     |                               |       |                               |       |                               |       |                               |       |
| IG              | 0.082                         | 0.196 | 0.111                         | 0.272 | 0.159                         | 0.388 | 0.197                         | 0.475 |
| MaRC            | 0.139                         | 0.101 | 0.184                         | 0.177 | 0.270                         | 0.328 | 0.317                         | 0.430 |
| <i>PhilTa</i>   |                               |       |                               |       |                               |       |                               |       |
| IG              | 0.156                         | 0.084 | 0.240                         | 0.136 | 0.411                         | 0.221 | 0.548                         | 0.290 |
| MaRC            | 0.076                         | 0.199 | 0.149                         | 0.259 | 0.299                         | 0.350 | 0.435                         | 0.406 |
| <i>mT5-base</i> |                               |       |                               |       |                               |       |                               |       |
| IG              | 0.045                         | 0.126 | 0.086                         | 0.178 | 0.184                         | 0.252 | 0.323                         | 0.304 |
| MaRC            | 0.016                         | 0.175 | 0.031                         | 0.246 | 0.085                         | 0.347 | 0.199                         | 0.412 |

Table 36: Appendix attribution sweep for the two paper-facing attribution views, IG and retrieval-adapted MARC, across the three main-paper models. Direction arrows are printed in the column headers. Sufficiency and Comprehensiveness are ratio metrics at  $\{10\%, 25\%, 50\%\}$ ; MinFrac is the minimum retained query-token fraction needed to recover  $\tau \in \{0.70, 0.80, 0.90, 0.95\}$  of the full cosine (lower is better). Ratio metrics require  $|S_v(q_{\text{full}}, c)| \geq 0.05$ ;  $\rho_{\text{LOO}}$  excludes  $|\Delta| < 10^{-6}$ . The completeness report contains no missing metric cells.

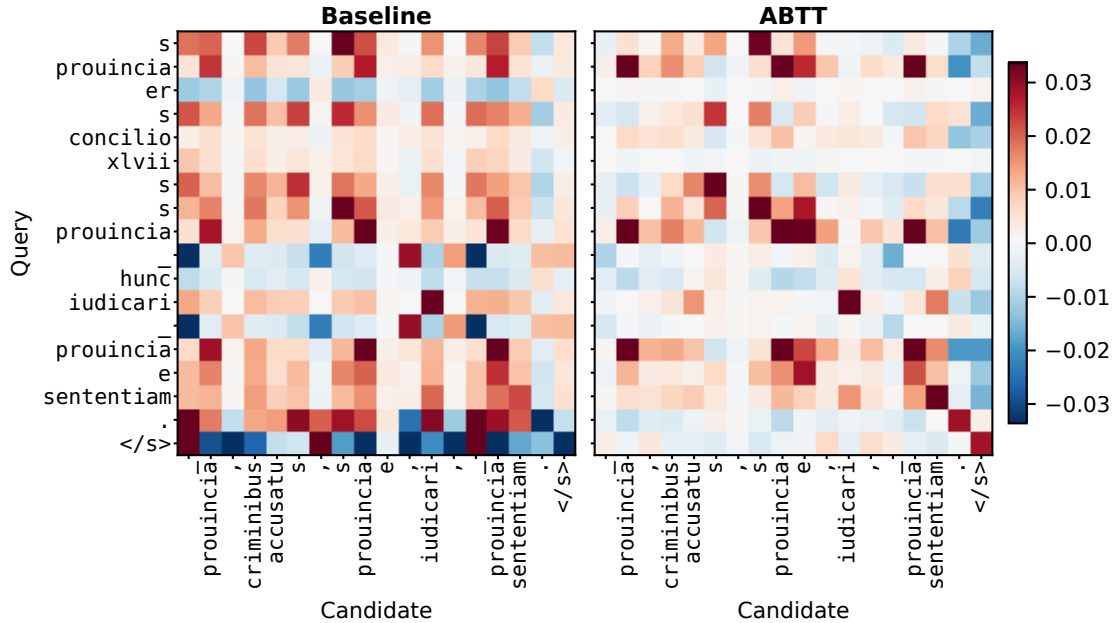


Figure 9: Derived pairwise heatmap for one PhilTa retrieval pair, at PhilTa layer 6, its most anisotropic layer (the layer of the mechanism check in Appendix K), not the operational layer of Table 32; a demonstration pair chosen because ABTT reverses the baseline decision, outside the 200 evaluated pairs. The left panel shows the baseline representation and the right panel shows the same example after ABTT. Both axes use original decoded tokens: the 18 query and 18 candidate tokens with the largest integrated-gradient magnitude under either representation. Cell values combine token-token cosine similarity with single-sequence integrated-gradient scores.

| Method          | $\rho_{\text{LOO}} \uparrow$ |        | Suff<br>@10% $\uparrow$ |        | Suff<br>@25% $\uparrow$ |        | Suff<br>@50% $\uparrow$ |       | Comp<br>@10% $\uparrow$ |        | Comp<br>@25% $\uparrow$ |        | Comp<br>@50% $\uparrow$ |        |
|-----------------|------------------------------|--------|-------------------------|--------|-------------------------|--------|-------------------------|-------|-------------------------|--------|-------------------------|--------|-------------------------|--------|
|                 | Base                         | ABTT   | Base                    | ABTT   | Base                    | ABTT   | Base                    | ABTT  | Base                    | ABTT   | Base                    | ABTT   | Base                    | ABTT   |
| <i>LaTa</i>     |                              |        |                         |        |                         |        |                         |       |                         |        |                         |        |                         |        |
| BERTScore       | 0.282                        | 0.496  | -0.019                  | 0.565  | -0.032                  | 0.788  | 0.016                   | 0.960 | 0.292                   | 0.074  | 0.298                   | 0.192  | 0.282                   | 0.426  |
| OT              | 0.031                        | 0.382  | 0.929                   | 0.581  | 0.961                   | 0.820  | 0.979                   | 0.974 | 0.872                   | 0.071  | 0.882                   | 0.179  | 0.827                   | 0.361  |
| Att-weighted    | -0.014                       | 0.001  | -0.185                  | 0.265  | -0.216                  | 0.459  | 0.341                   | 0.709 | 0.015                   | 0.046  | 0.022                   | 0.135  | 0.182                   | 0.255  |
| Att-standalone  | 0.144                        | 0.575  | 0.788                   | 0.721  | 0.910                   | 0.905  | 0.962                   | 1.010 | 0.737                   | 0.108  | 0.954                   | 0.275  | 1.074                   | 0.553  |
| DLA             | 0.075                        | 0.601  | 0.973                   | 0.704  | 1.009                   | 0.893  | 1.035                   | 0.989 | 1.200                   | 0.123  | 1.250                   | 0.292  | 1.251                   | 0.555  |
| <i>random</i>   | 0.004                        | -0.002 | 0.265                   | 0.547  | 0.519                   | 0.736  | 0.819                   | 0.871 | 0.026                   | 0.021  | 0.030                   | 0.053  | 0.177                   | 0.136  |
| <i>inverse</i>  | -0.003                       | -0.370 | -0.202                  | 0.276  | -0.179                  | 0.436  | -0.205                  | 0.653 | -0.007                  | -0.017 | -0.013                  | -0.022 | -0.015                  | 0.035  |
| <i>PhilTa</i>   |                              |        |                         |        |                         |        |                         |       |                         |        |                         |        |                         |        |
| BERTScore       | 0.561                        | 0.646  | 0.695                   | 0.841  | 0.874                   | 1.016  | 0.984                   | 1.105 | 0.047                   | 0.231  | 0.113                   | 0.442  | 0.253                   | 0.686  |
| OT              | 0.340                        | 0.616  | 0.641                   | 0.846  | 0.818                   | 0.971  | 0.940                   | 1.040 | 0.043                   | 0.289  | 0.114                   | 0.523  | 0.233                   | 0.749  |
| Att-weighted    | 0.111                        | 0.041  | 0.589                   | 0.355  | 0.807                   | 0.592  | 0.923                   | 0.775 | 0.013                   | 0.050  | 0.051                   | 0.156  | 0.145                   | 0.311  |
| Att-standalone  | 0.287                        | 0.217  | 0.666                   | 0.486  | 0.822                   | 0.705  | 0.937                   | 0.888 | 0.022                   | 0.095  | 0.073                   | 0.240  | 0.228                   | 0.458  |
| DLA             | 0.531                        | 0.599  | 0.729                   | 0.872  | 0.862                   | 0.970  | 0.959                   | 1.006 | 0.068                   | 0.311  | 0.159                   | 0.567  | 0.337                   | 0.776  |
| <i>random</i>   | -0.004                       | -0.002 | 0.650                   | 0.400  | 0.815                   | 0.603  | 0.922                   | 0.792 | 0.011                   | 0.034  | 0.031                   | 0.093  | 0.084                   | 0.220  |
| <i>inverse</i>  | -0.329                       | -0.614 | 0.514                   | 0.046  | 0.636                   | 0.109  | 0.774                   | 0.255 | 0.002                   | -0.030 | 0.015                   | -0.049 | 0.065                   | -0.038 |
| <i>mT5-base</i> |                              |        |                         |        |                         |        |                         |       |                         |        |                         |        |                         |        |
| BERTScore       | 0.351                        | 0.569  | 0.906                   | 0.777  | 0.968                   | 1.059  | 0.991                   | 1.159 | 0.004                   | 0.227  | 0.012                   | 0.486  | 0.036                   | 0.790  |
| OT              | 0.151                        | 0.678  | 0.840                   | 0.908  | 0.933                   | 1.127  | 0.976                   | 1.205 | 0.004                   | 0.313  | 0.012                   | 0.641  | 0.039                   | 0.996  |
| Att-weighted    | 0.242                        | 0.564  | 0.937                   | 0.962  | 0.974                   | 1.067  | 0.990                   | 1.075 | 0.004                   | 0.319  | 0.011                   | 0.589  | 0.035                   | 0.845  |
| Att-standalone  | 0.145                        | 0.268  | 0.886                   | 0.616  | 0.960                   | 0.760  | 0.986                   | 0.857 | 0.004                   | 0.221  | 0.011                   | 0.412  | 0.035                   | 0.587  |
| DLA             | 0.237                        | 0.552  | 0.901                   | 0.934  | 0.950                   | 1.051  | 0.979                   | 1.032 | 0.004                   | 0.341  | 0.014                   | 0.622  | 0.046                   | 0.825  |
| <i>random</i>   | 0.002                        | 0.002  | 0.917                   | 0.350  | 0.968                   | 0.534  | 0.989                   | 0.741 | 0.001                   | 0.048  | 0.004                   | 0.119  | 0.011                   | 0.268  |
| <i>inverse</i>  | -0.143                       | -0.686 | 0.687                   | -0.201 | 0.884                   | -0.186 | 0.962                   | 0.002 | 0.002                   | -0.092 | 0.007                   | -0.168 | 0.025                   | -0.210 |

| Method          | MinFrac<br>@0.70 $\downarrow$ |       | MinFrac<br>@0.80 $\downarrow$ |       | MinFrac<br>@0.90 $\downarrow$ |       | MinFrac<br>@0.95 $\downarrow$ |       |
|-----------------|-------------------------------|-------|-------------------------------|-------|-------------------------------|-------|-------------------------------|-------|
|                 | Base                          | ABTT  | Base                          | ABTT  | Base                          | ABTT  | Base                          | ABTT  |
| <i>LaTa</i>     |                               |       |                               |       |                               |       |                               |       |
| BERTScore       | 0.377                         | 0.210 | 0.403                         | 0.297 | 0.436                         | 0.433 | 0.467                         | 0.517 |
| OT              | 0.087                         | 0.184 | 0.115                         | 0.258 | 0.160                         | 0.376 | 0.197                         | 0.464 |
| Att-weighted    | 0.334                         | 0.476 | 0.347                         | 0.584 | 0.377                         | 0.705 | 0.408                         | 0.787 |
| Att-standalone  | 0.116                         | 0.132 | 0.172                         | 0.209 | 0.273                         | 0.341 | 0.322                         | 0.436 |
| DLA             | 0.086                         | 0.128 | 0.118                         | 0.208 | 0.154                         | 0.353 | 0.192                         | 0.452 |
| <i>random</i>   | 0.183                         | 0.220 | 0.219                         | 0.330 | 0.288                         | 0.514 | 0.338                         | 0.659 |
| <i>inverse</i>  | 0.615                         | 0.513 | 0.706                         | 0.645 | 0.811                         | 0.797 | 0.847                         | 0.883 |
| <i>PhilTa</i>   |                               |       |                               |       |                               |       |                               |       |
| BERTScore       | 0.130                         | 0.122 | 0.199                         | 0.178 | 0.311                         | 0.254 | 0.393                         | 0.310 |
| OT              | 0.152                         | 0.084 | 0.237                         | 0.136 | 0.401                         | 0.221 | 0.541                         | 0.289 |
| Att-weighted    | 0.187                         | 0.376 | 0.268                         | 0.485 | 0.415                         | 0.630 | 0.550                         | 0.723 |
| Att-standalone  | 0.148                         | 0.270 | 0.240                         | 0.358 | 0.405                         | 0.479 | 0.533                         | 0.560 |
| DLA             | 0.102                         | 0.069 | 0.182                         | 0.115 | 0.345                         | 0.205 | 0.486                         | 0.285 |
| <i>random</i>   | 0.150                         | 0.351 | 0.242                         | 0.478 | 0.412                         | 0.642 | 0.576                         | 0.751 |
| <i>inverse</i>  | 0.325                         | 0.859 | 0.516                         | 0.909 | 0.730                         | 0.951 | 0.837                         | 0.970 |
| <i>mT5-base</i> |                               |       |                               |       |                               |       |                               |       |
| BERTScore       | 0.030                         | 0.178 | 0.051                         | 0.244 | 0.104                         | 0.315 | 0.187                         | 0.356 |
| OT              | 0.045                         | 0.127 | 0.086                         | 0.178 | 0.183                         | 0.252 | 0.321                         | 0.305 |
| Att-weighted    | 0.015                         | 0.085 | 0.028                         | 0.137 | 0.068                         | 0.228 | 0.148                         | 0.292 |
| Att-standalone  | 0.032                         | 0.270 | 0.060                         | 0.374 | 0.124                         | 0.499 | 0.220                         | 0.586 |
| DLA             | 0.024                         | 0.083 | 0.045                         | 0.133 | 0.110                         | 0.231 | 0.245                         | 0.312 |
| <i>random</i>   | 0.028                         | 0.401 | 0.046                         | 0.537 | 0.095                         | 0.697 | 0.175                         | 0.785 |
| <i>inverse</i>  | 0.114                         | 0.869 | 0.158                         | 0.912 | 0.260                         | 0.953 | 0.408                         | 0.974 |

Table 37: Supplemental attribution sweep for additional methods and sanity baselines. *random* and *inverse* are diagnostic baselines, not paper-facing attribution methods. Direction arrows are printed in the column headers. Sufficiency and Comprehensiveness are ratio metrics at  $\{10\%, 25\%, 50\%\}$ ; MinFrac is the minimum retained query-token fraction needed to recover  $\tau \in \{0.70, 0.80, 0.90, 0.95\}$  of the full cosine (lower is better). Ratio metrics require  $|S_v(q_{\text{full}}, c)| \geq 0.05$ ;  $\rho_{\text{LOO}}$  excludes  $|\Delta| < 10^{-6}$ . The completeness report contains no missing metric cells.

| Knob                   | Setting                                | W/T/L        | base gap      | ABTT gap      | $ \Delta $ |
|------------------------|----------------------------------------|--------------|---------------|---------------|------------|
| Predeclared            | every token, drop, 5 draws, query only | <b>4/1/1</b> | 0.04 to 0.81  | 0.18 to 0.56  | 0.352      |
| Deletion step schedule | every 5% of the query                  | 4/1/1        | 0.04 to 0.81  | 0.18 to 0.56  | 0.353      |
|                        | every 10% of the query                 | 4/0/2        | 0.04 to 0.80  | 0.17 to 0.55  | 0.352      |
|                        | every 20% of the query                 | 4/0/2        | 0.03 to 0.74  | 0.16 to 0.52  | 0.337      |
| Deleted token becomes  | zero vector, denominator kept          | 3/1/2        | 0.04 to 0.81  | 0.08 to 0.27  | 0.242      |
|                        | corpus mean vector                     | 4/1/1        | 0.01 to 0.43  | 0.18 to 0.56  | 0.326      |
| Random orderings       | 1 draw                                 | 4/1/1        | 0.04 to 0.76  | 0.18 to 0.56  | 0.327      |
|                        | 20 draws                               | 4/0/2        | 0.04 to 0.83  | 0.18 to 0.56  | 0.358      |
|                        | 50 draws                               | 4/0/2        | 0.04 to 0.83  | 0.18 to 0.56  | 0.358      |
| Random-order seed      | seed 20260101                          | 4/0/2        | 0.04 to 0.83  | 0.18 to 0.56  | 0.356      |
|                        | seed 7                                 | 4/0/2        | 0.04 to 0.83  | 0.18 to 0.57  | 0.359      |
| Token filter           | none (mismatches the pooling)          | 1/1/4        | -0.21 to 0.42 | -0.13 to 0.01 | 0.170      |
|                        | no_empty (mismatches the pooling)      | 6/0/0        | 0.02 to 0.17  | 0.16 to 0.48  | 0.281      |
| Side erased            | query and candidate                    | 3/1/2        | 0.00 to 1.30  | 0.13 to 0.39  | 0.420      |
| Combined               | 10% grid + zero vector                 | 3/1/2        | 0.04 to 0.80  | 0.08 to 0.26  | 0.241      |
|                        | 10% grid + corpus mean                 | 4/1/1        | 0.01 to 0.41  | 0.17 to 0.55  | 0.316      |
|                        | 10% grid + both sides                  | 3/1/2        | 0.01 to 1.28  | 0.13 to 0.38  | 0.414      |
|                        | zero vector + both sides               | 3/0/3        | 0.00 to 1.30  | 0.05 to 0.10  | 0.365      |
|                        | corpus mean + both sides               | 4/1/1        | 0.00 to 0.39  | 0.13 to 0.39  | 0.189      |
|                        | 10% grid + zero + both sides           | 3/0/3        | 0.01 to 1.28  | 0.05 to 0.09  | 0.359      |

Table 38: Sensitivity of the chance-corrected deletion AUC gap to the choices behind it, over the same 200 positive pairs per model and the same stored hidden states as Table 32. Each row recomputes the metric with one knob moved and everything else at the predeclared setting in the first row, which draws five random orderings; Table 32 uses 20, the setting of the 20-draw row. W/T/L counts the six model-view cells in which ABTT beats, ties or loses to the uncorrected baseline, a tie being a paired difference within two standard errors of zero. The two range columns give the smallest and largest cell gap that setting produces for each variant, the spread Table 32 reports cell by cell. The last column is the mean absolute paired difference over the six cells. Across the 18 settings that pool the same vectors the components were fitted on, ABTT wins 3 to 4 of six cells and the predeclared setting gives 4. The 2 token-filter rows are diagnostics, not candidate settings: the artifacts were generated by pooling one filtered token set, so any other filter scores a vector the ABTT components were not fitted on. Cells average 194 to 200 pairs, the rest falling below the full-query cosine floor of 0.05 at which the ratio is undefined.

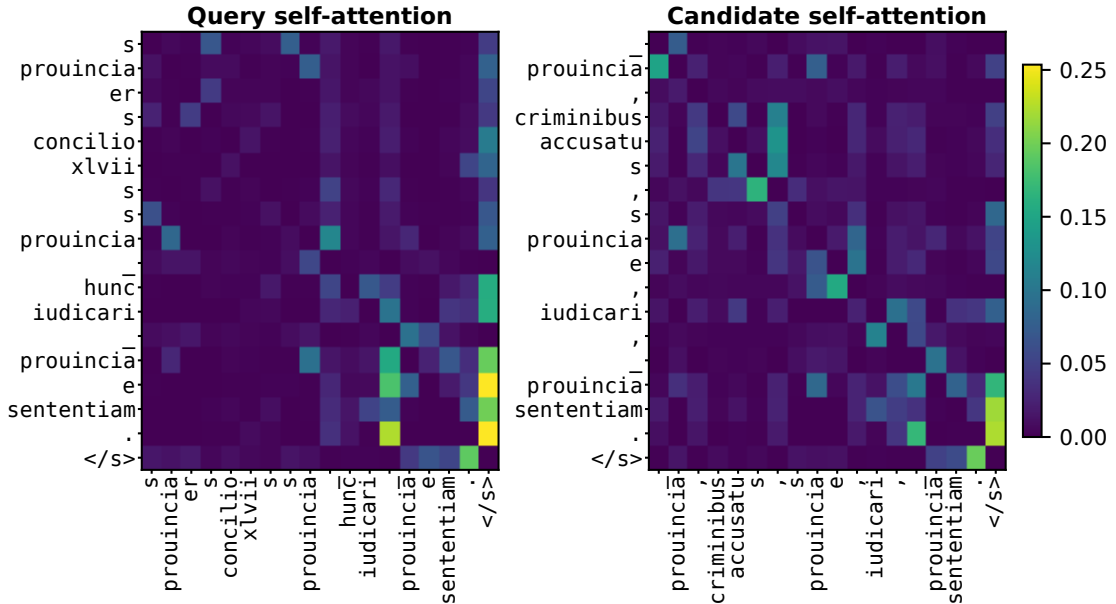


Figure 10: Companion self-attention maps at PhilTa layer 6 for the same demonstration pair as Figure 9, which is outside the 200 evaluated pairs. ABTT is applied after encoding, so attention is included only as supporting context.

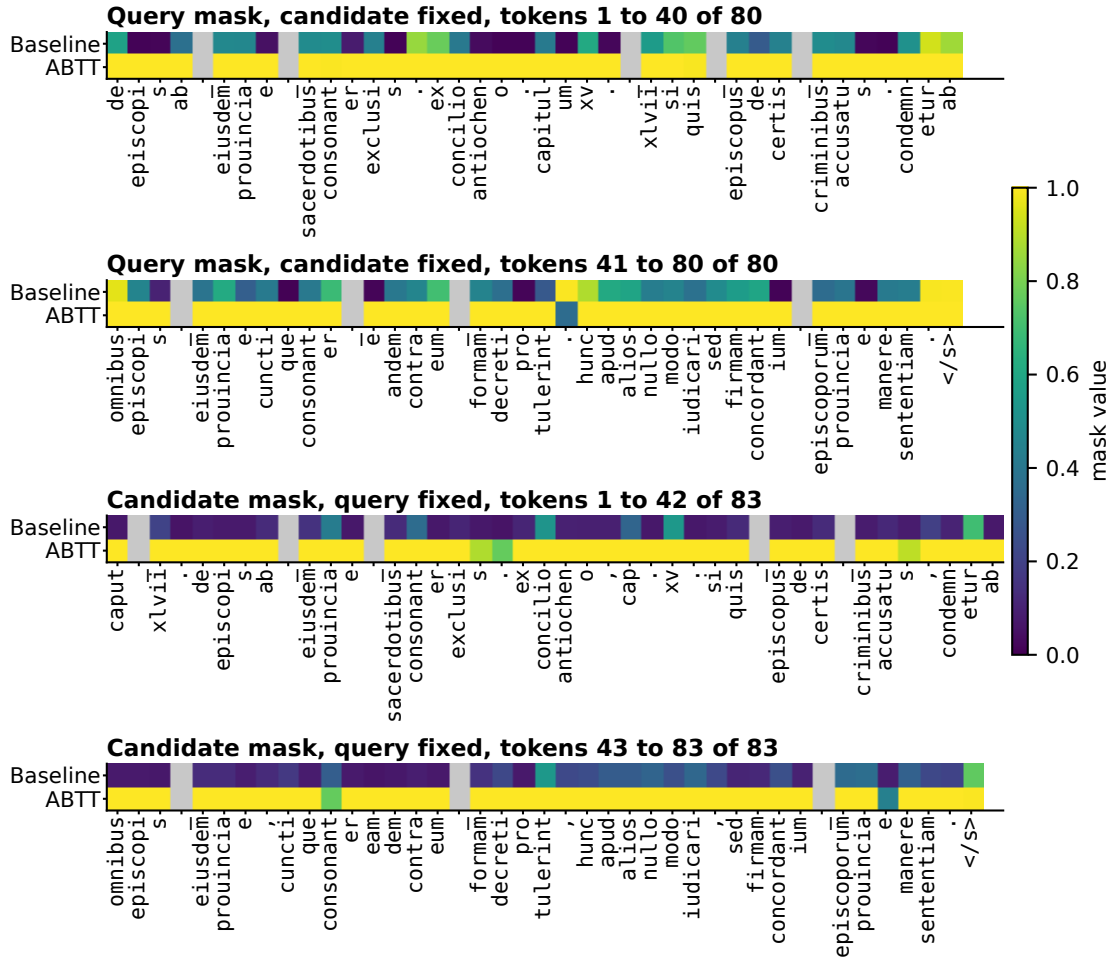


Figure 11: Retrieval-adapted MaRC masks at PhilTa layer 6 for the same demonstration pair as Figure 9, which is outside the 200 evaluated pairs. The upper two strips give the mask on the query with the candidate fixed, and the lower two the mask on the candidate with the query fixed; each sequence wraps onto two strips. In every strip the Baseline row uses the raw cosine target and the ABTT row the ABTT-corrected cosine target. Gray cells are non-content positions, frozen at zero.

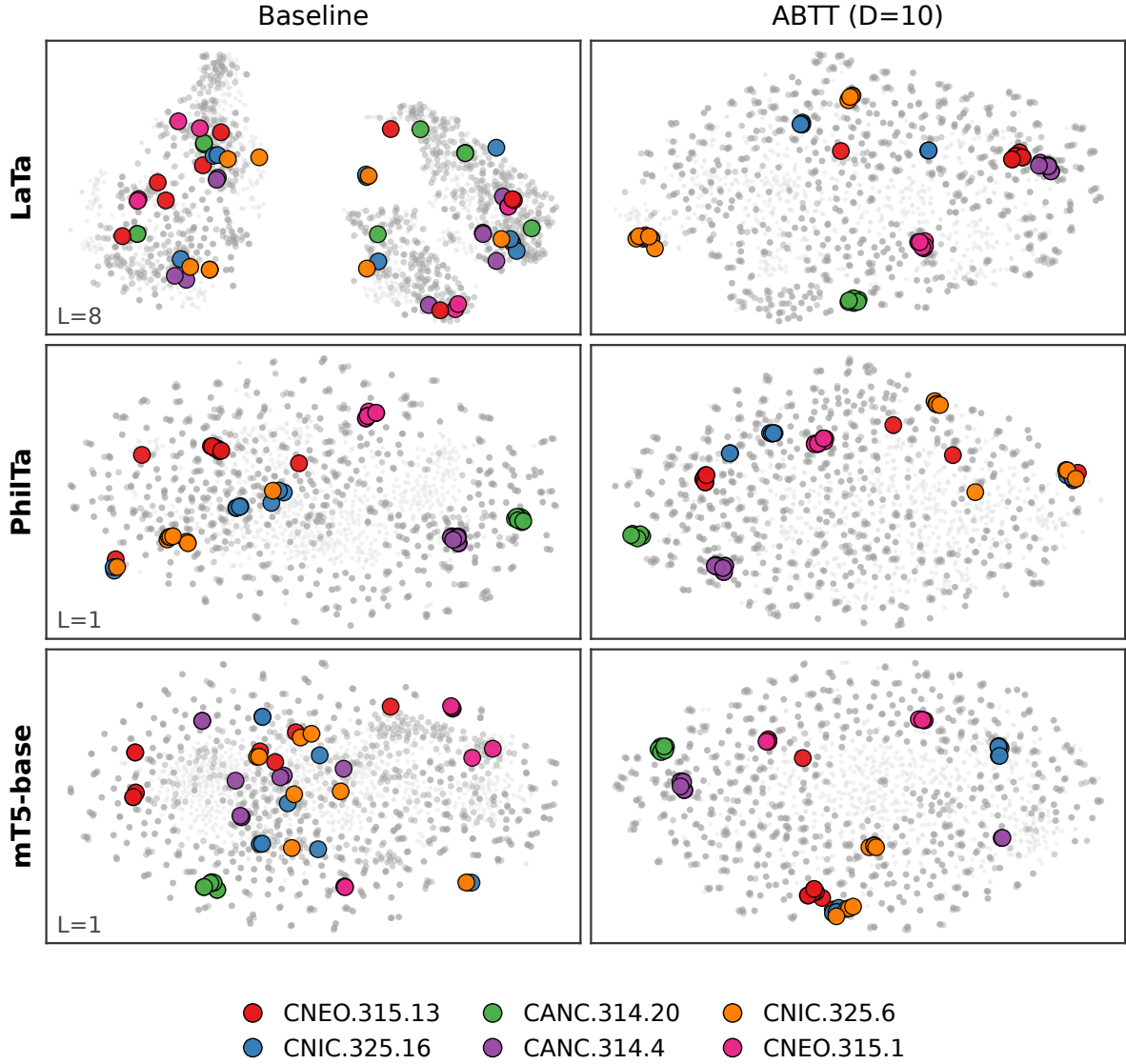


Figure 12: t-SNE projections of all 1,705 manuscript witnesses for LaTa, PhilTa, and mT5-base, each at the layer chosen on the train split: the layer with the highest training-set DirAcc@1 under ABTT, printed in each panel as  $L$  and equal to the Task B ABTT layer in Table 11. Columns compare baseline mean pooling with ABTT-only using  $D = 10$ , fit on the training embeddings. Colors mark the six largest directories; other witnesses of directories with at least two witnesses are mid gray, and singletons light gray. The silhouette scores in Appendix N are computed in these two-dimensional coordinates over the 1,160 winnable witnesses, those whose directory has at least two witnesses.

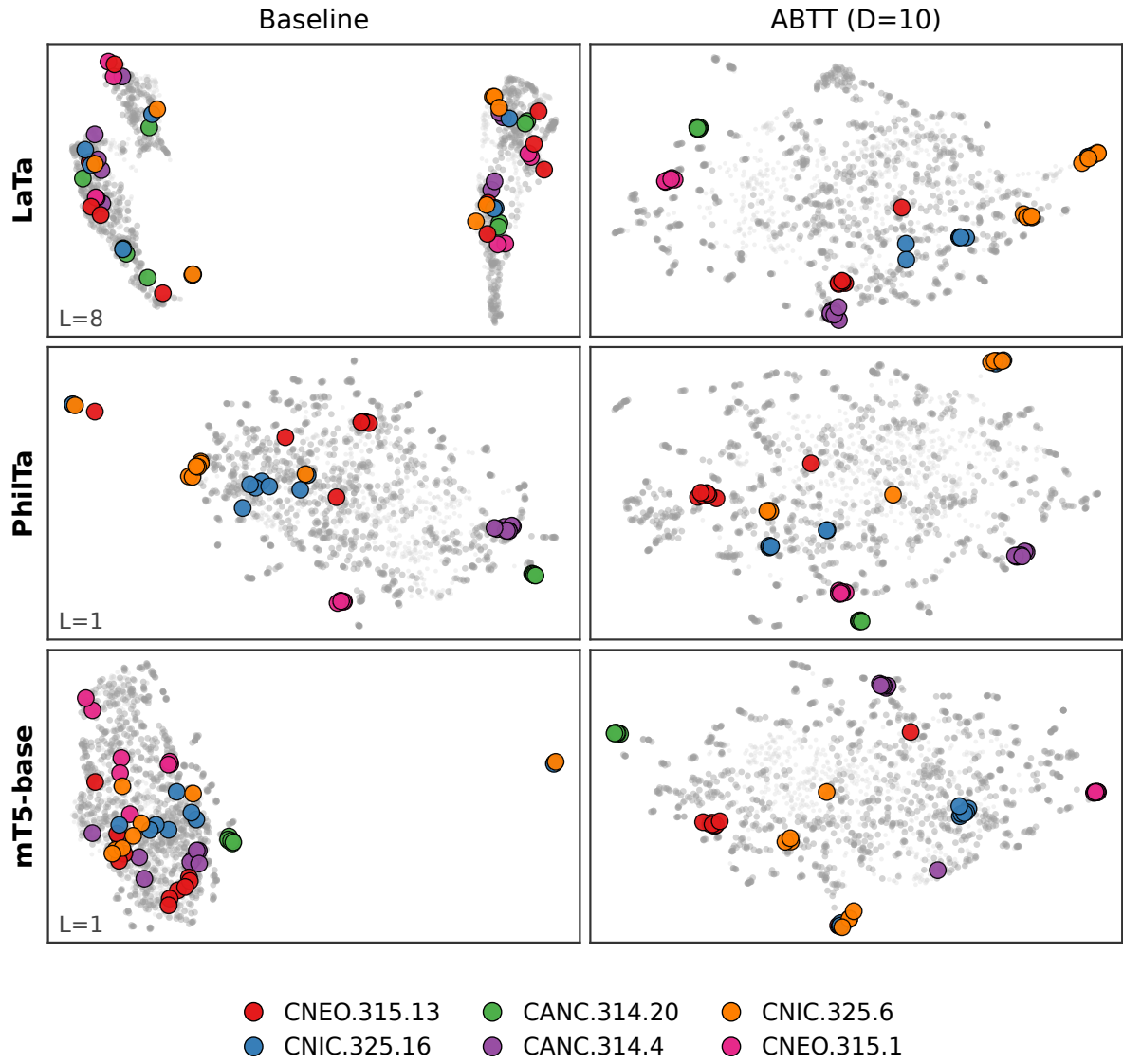


Figure 13: UMAP projections with the same configuration as Figure 12. The same cluster-tightening pattern appears under a different projection method.

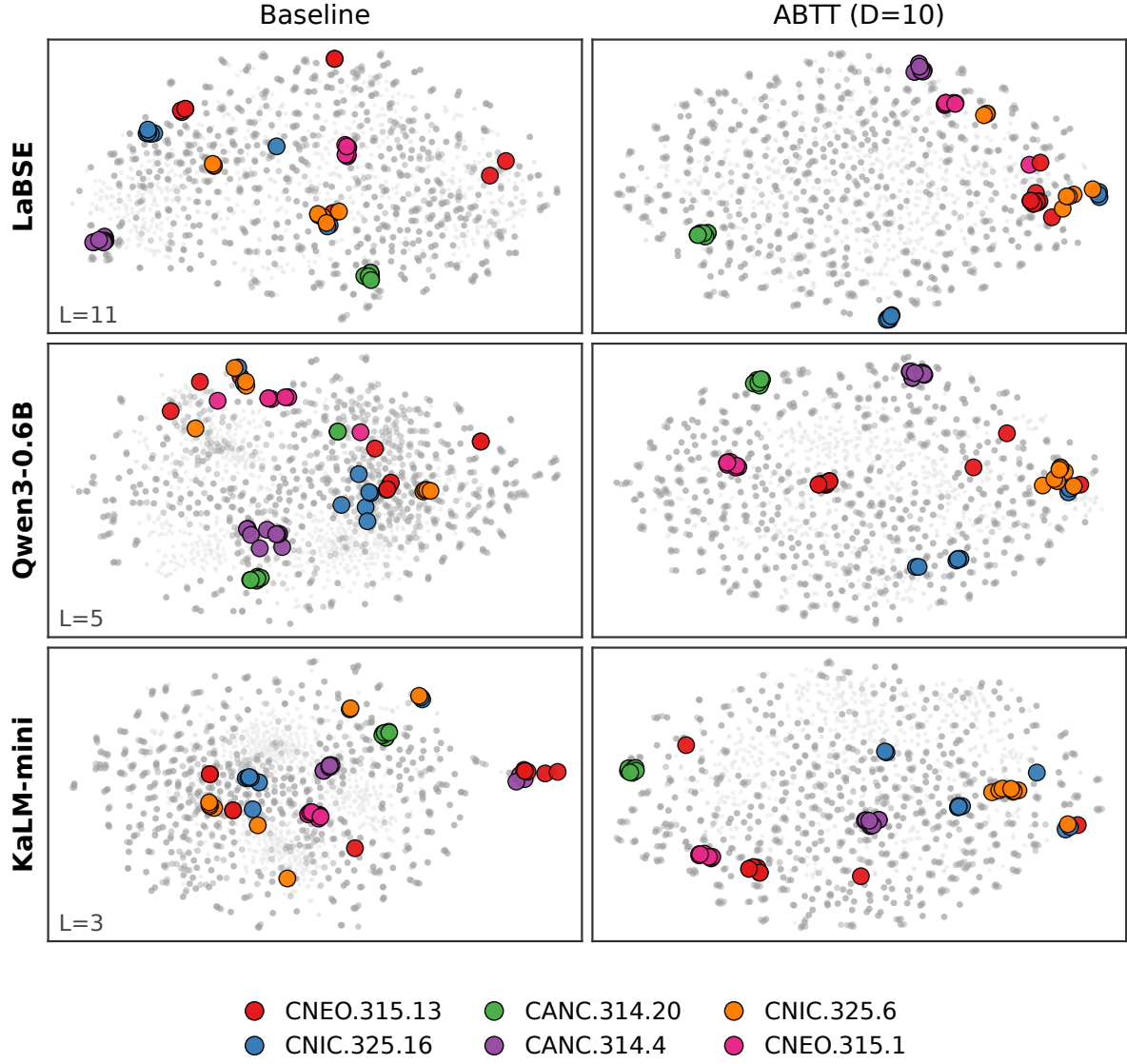


Figure 14: t-SNE projections of all 1,705 manuscript witnesses for LaBSE, Qwen3-0.6B, and KaLM-mini under the layer rule of Figure 12: the layer with the highest training-set DirAcc@1 under ABTT, printed in each panel and equal to the Task B ABTT layer in Table 11. Columns compare baseline mean pooling with ABTT-only using  $D = 10$ . Colors mark the same six largest directories as Figure 12.

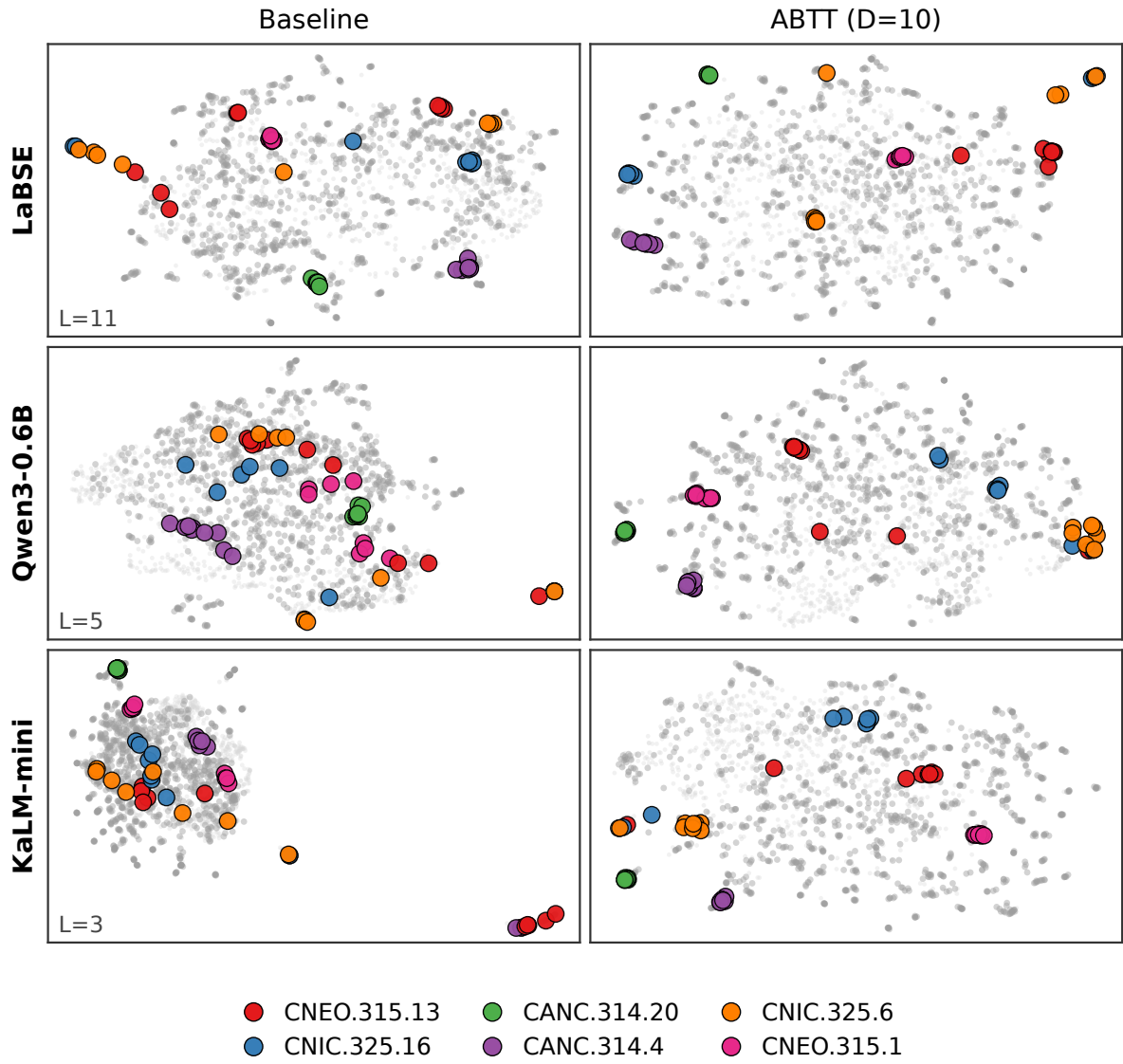


Figure 15: UMAP projections matching Figure 14. The cluster-tightening pattern appears under a second projection method.